Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
```

```
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]
```

```
# if camp == "2025_Jun":
#     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
# elif camp == "2025_Sep":
#     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
if camp == "2025_Apr":
    #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
    #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
    mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
elif camp == "2025_Jun":
    mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
elif camp == "2025_Sep":
    mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : id_channel

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : id_channel

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test $<$ 0.95 (under performance of the test)

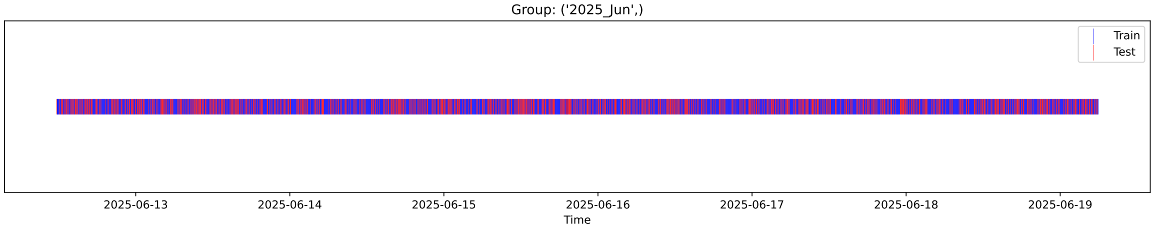
[Green] r_test $>$ 1.05 (over performance of the test)

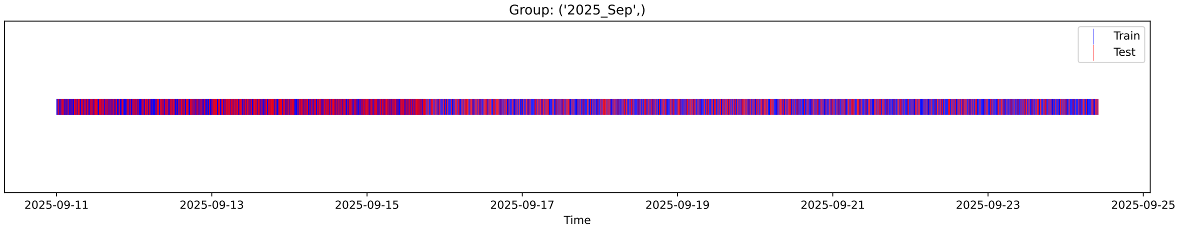
[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

External Split: Train: 5717 samples Test: 2451 samples

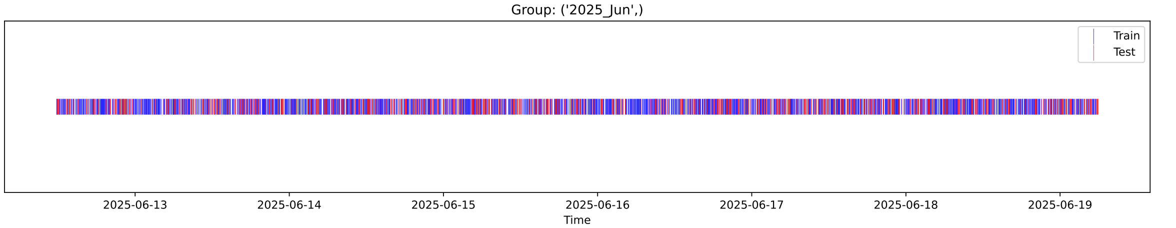


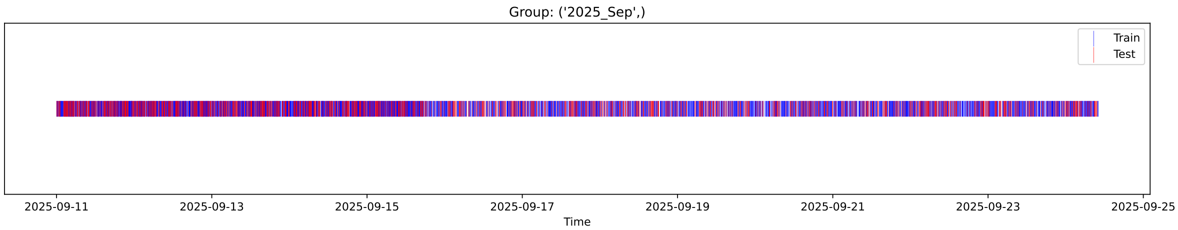


2.2.2 Internal CV Folds

Fold 1/5

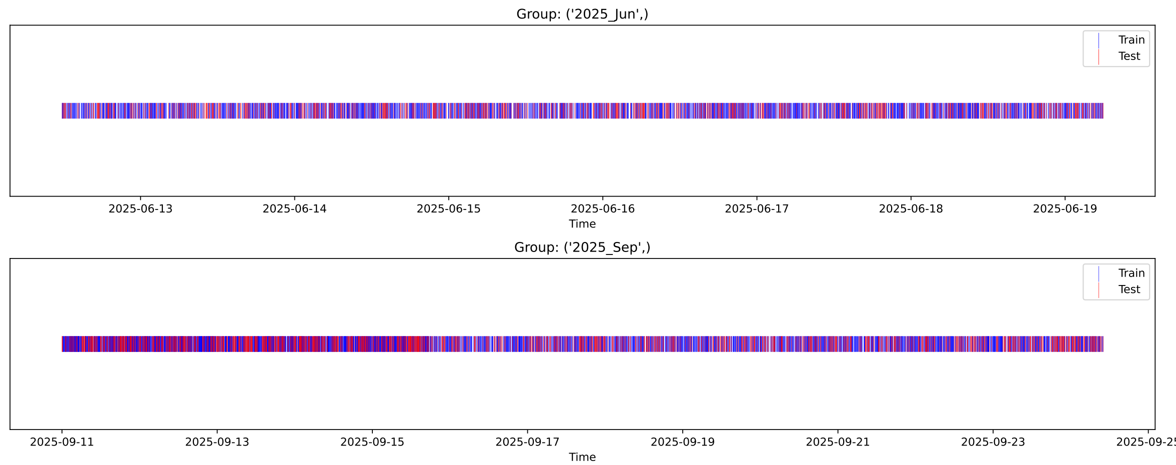
Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h





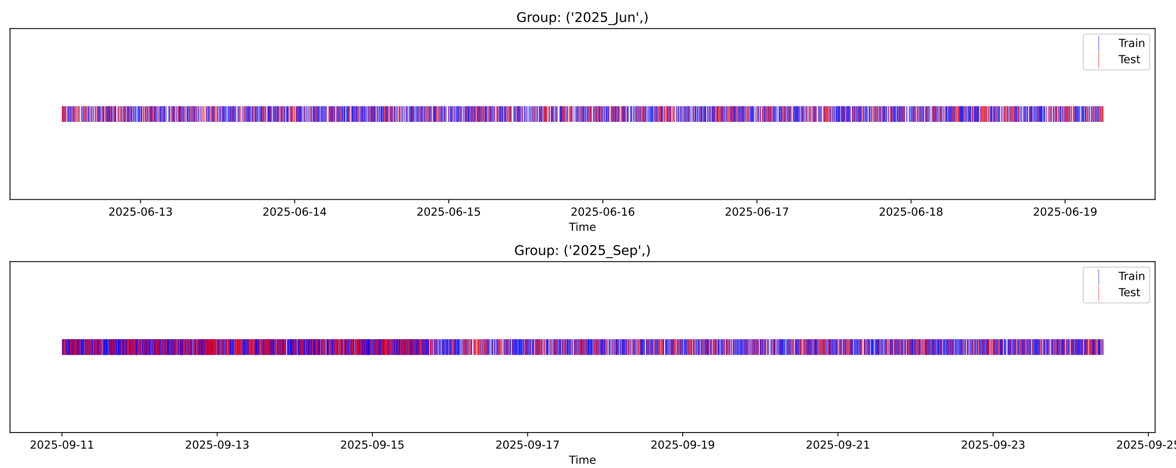
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



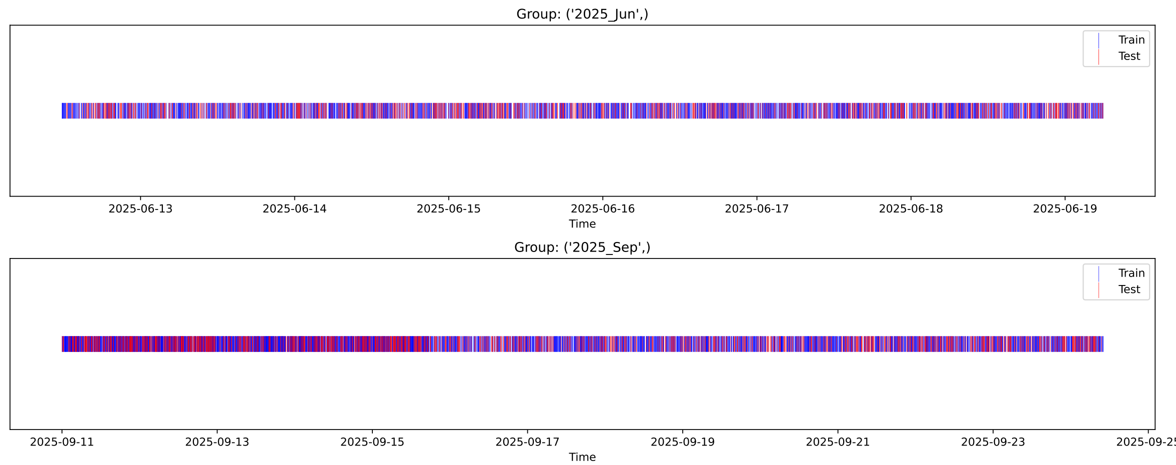
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



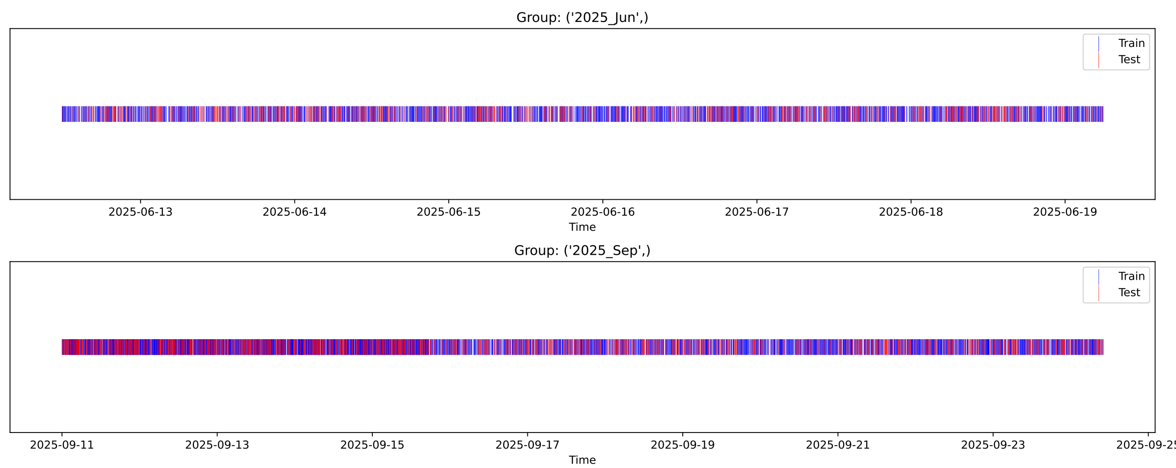
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 3

Len(group) : 1356, **Len(training)** : 950, **Len(test)** : 406

Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086 y_test: mean=2.318, std=0.740, min=1.068, max=5.447

Fold 0 — y_test: mean=2.39, std=0.76, y_train: mean=2.29, std=0.68

Fold 1 — y_test: mean=2.34, std=0.73, y_train: mean=2.31, std=0.70

Fold 2 — y_test: mean=2.33, std=0.71, y_train: mean=2.31, std=0.71

Fold 3 — y_test: mean=2.33, std=0.69, y_train: mean=2.31, std=0.71

Fold 4 — y_test: mean=2.30, std=0.73, y_train: mean=2.32, std=0.70

train scores CV: [0.62268767 0.60265136 0.58040391 0.56906438 0.60353391]

test scores CV: [0.49521601 0.53276144 0.58141377 0.60879453 0.53165288]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5957	0.55	0.0402	0.6403	0.6408	0.0908	1.1651	1.0831

Coefficients:

coef_r_CO2 = -0.0897

coef_r_NH3 = 0.0392

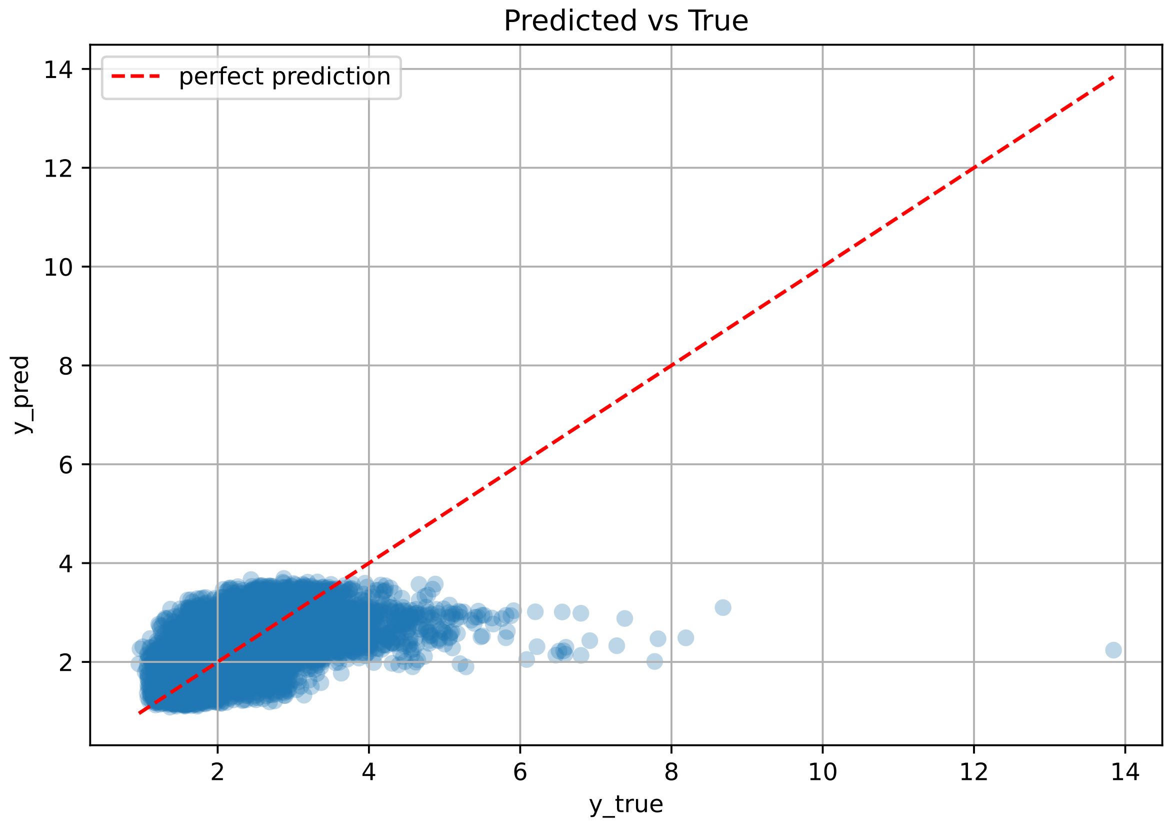
coef_r_THI = 0.4472

coef_r_temperature = -0.3735

coef_r_umidity = -0.5516

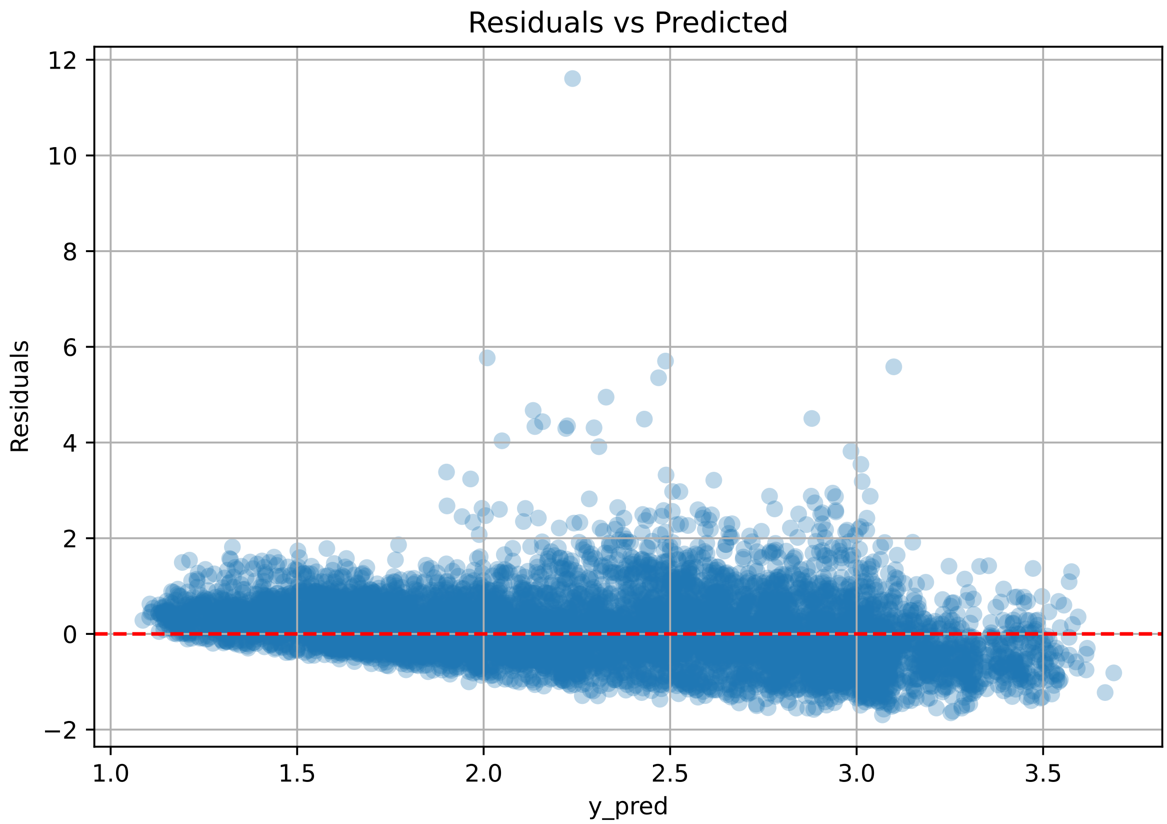
intercept = 2.3165

2.3.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14
2
4
6
8
10
12
14

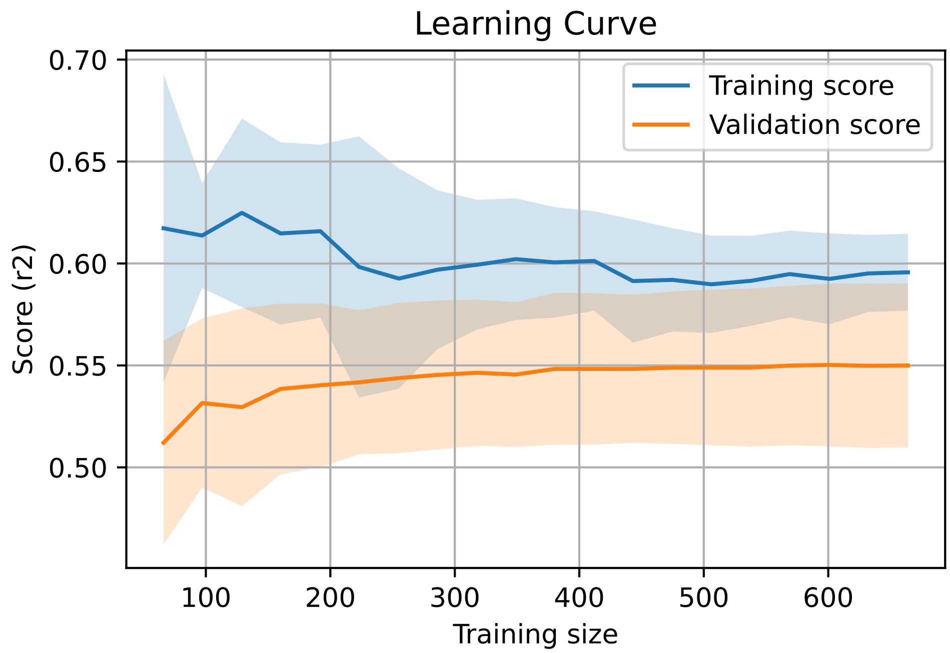
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 3

Len(group) : 1356, **Len(training)** : 950,

Len(training_real) : 665, **Len(test_of_training)** : 285, **Len(test)** : 406



2.3.5 Gridsearch — Group: 4

Len(group) : 1360, **Len(training)** : 952, **Len(test)** : 408

Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383 y_test: mean=2.202, std=0.791, min=1.062, max=7.822

Fold 0 — y_test: mean=2.22, std=0.80, y_train: mean=2.25, std=0.84

Fold 1 — y_test: mean=2.22, std=0.70, y_train: mean=2.25, std=0.88

Fold 2 — y_test: mean=2.29, std=0.86, y_train: mean=2.22, std=0.81

Fold 3 — y_test: mean=2.31, std=0.86, y_train: mean=2.21, std=0.82

Fold 4 — y_test: mean=2.22, std=0.83, y_train: mean=2.25, std=0.83

train scores CV: [0.31407089 0.27350601 0.3045029 0.31978092 0.29846023]

test scores CV: [0.23746916 0.36893865 0.26706608 0.23176705 0.28402149]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3021	0.2779	0.0494	0.2765	0.2771	-0.0008	0.9973	1.0871

Coefficients:

coef_r_CO2 = -0.1021

coef_r_NH3 = 0.1428

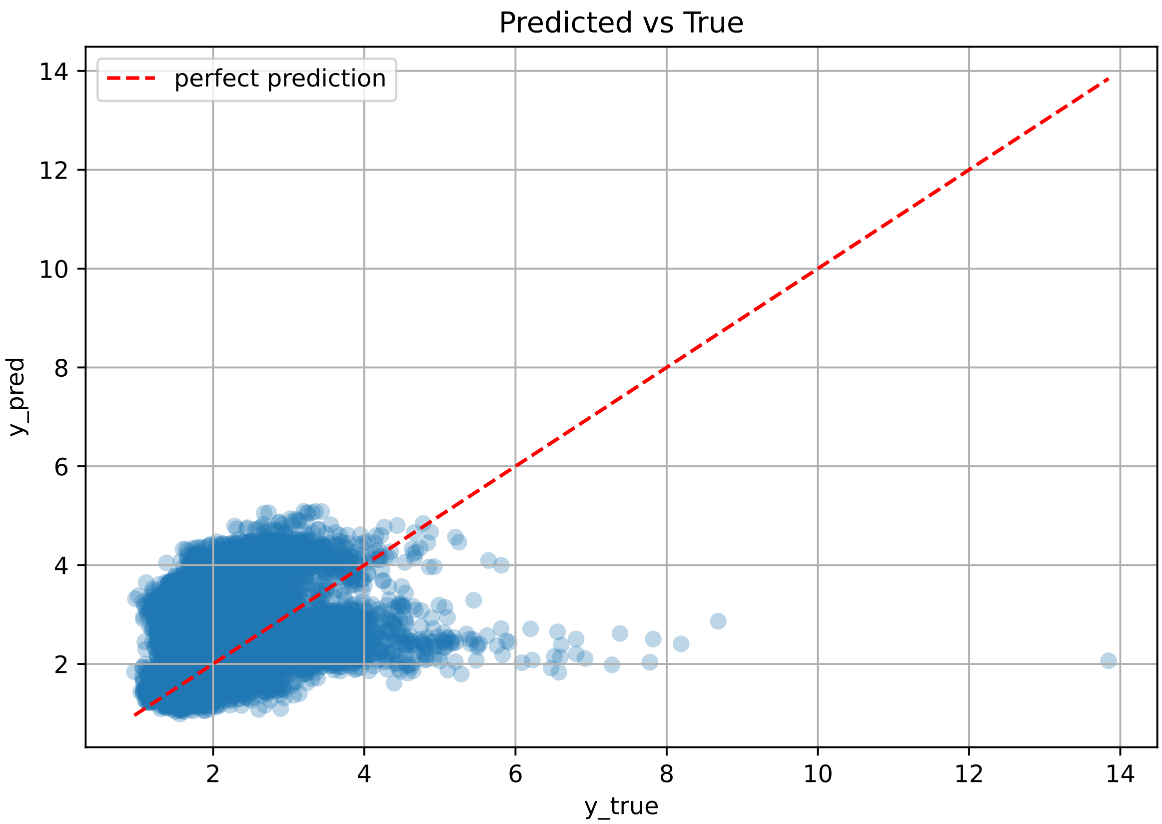
coef_r_THI = 1.0882

coef_r_temperature = -1.7734

coef_r_umidity = -1.1164

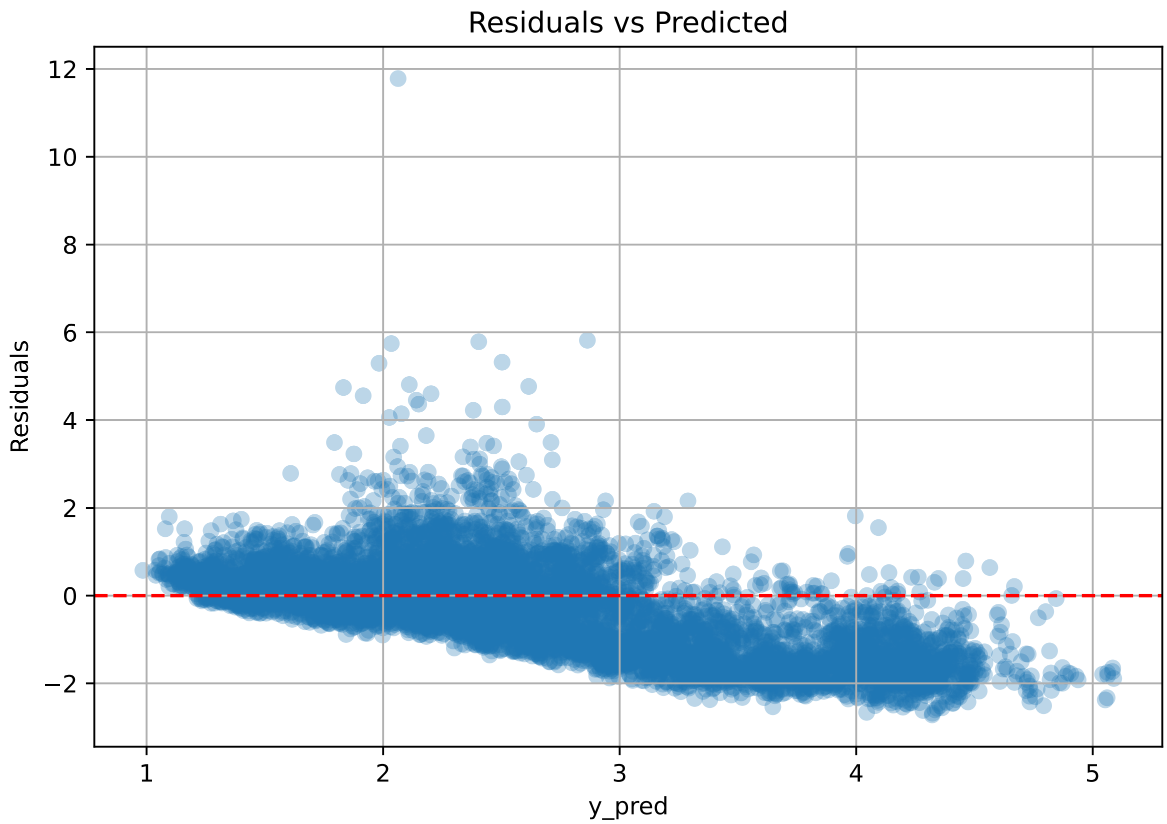
intercept = 2.2387

2.3.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

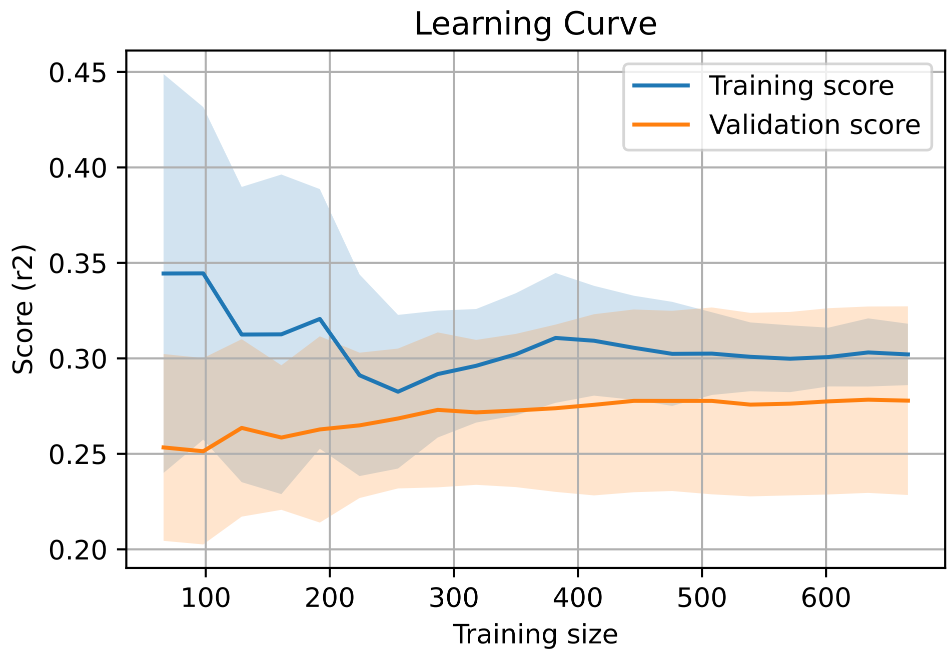
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 4

Len(group) : 1360, **Len(training)** : 952,

Len(training_real) : 667, **Len(test_of_training)** : 285, **Len(test)** : 408



2.3.9 Gridsearch — Group: 5

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813 y_test: mean=2.212, std=0.659, min=1.071, max=4.858

Fold 0 — y_test: mean=2.27, std=0.68, y_train: mean=2.22, std=0.68

Fold 1 — y_test: mean=2.25, std=0.71, y_train: mean=2.23, std=0.67

Fold 2 — y_test: mean=2.20, std=0.65, y_train: mean=2.25, std=0.69

Fold 3 — y_test: mean=2.21, std=0.65, y_train: mean=2.25, std=0.70

Fold 4 — y_test: mean=2.22, std=0.63, y_train: mean=2.24, std=0.70

train scores CV: [0.4277706 0.44021497 0.41172833 0.43714743 0.44005866]

test scores CV: [0.41299685 0.39069188 0.45040747 0.3878951 0.37549212]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4314	0.4035	0.0264	0.4566	0.4572	0.0537	1.1332	1.0691

Coefficients:

coef_r_CO2 = -0.0952

coef_r_NH3 = -0.1254

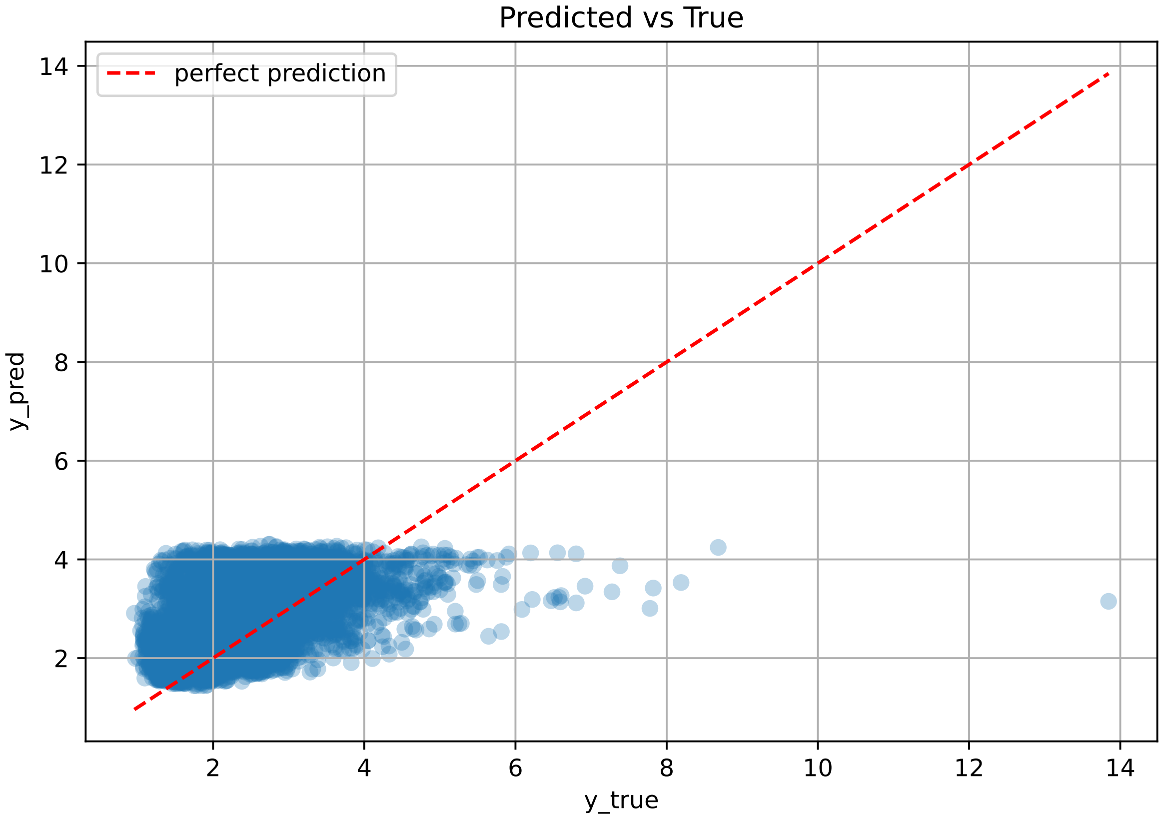
coef_r_THI = 0.0994

coef_r_temperature = 0.0362

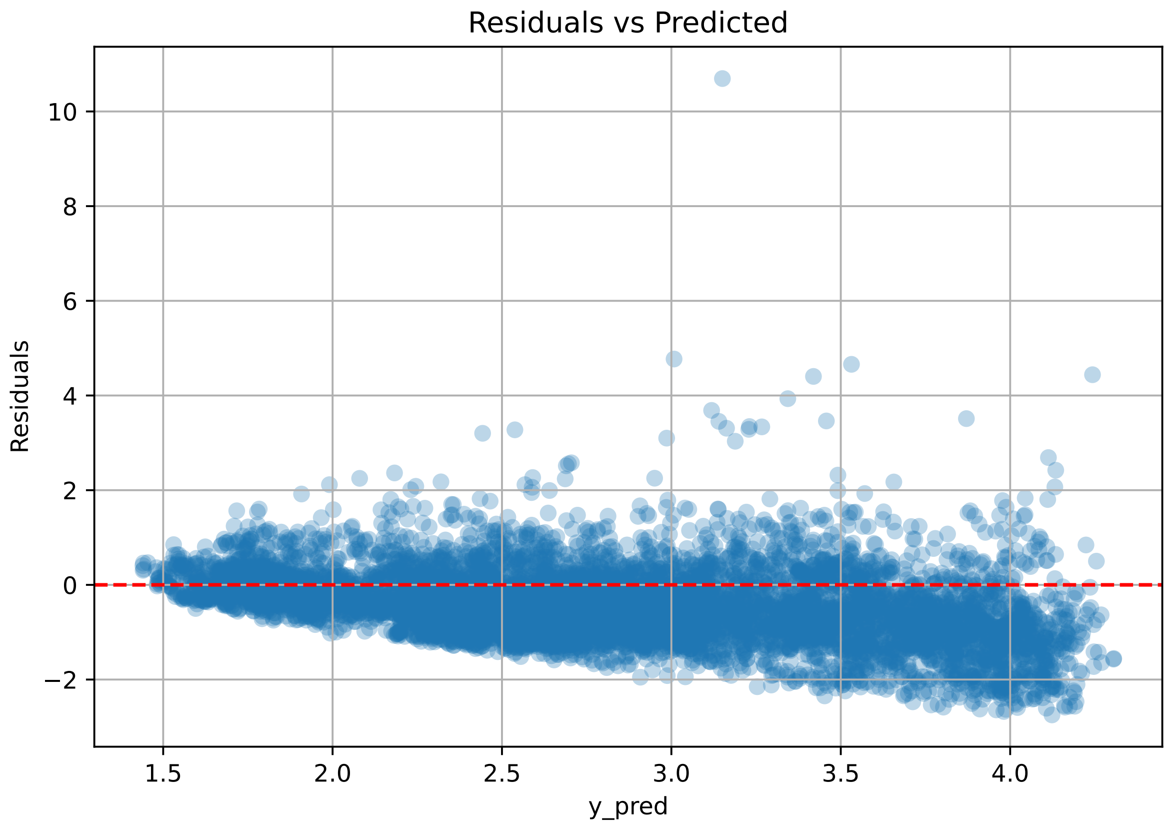
coef_r_umidity = -0.4502

intercept = 2.2379

2.3.10 Scatter



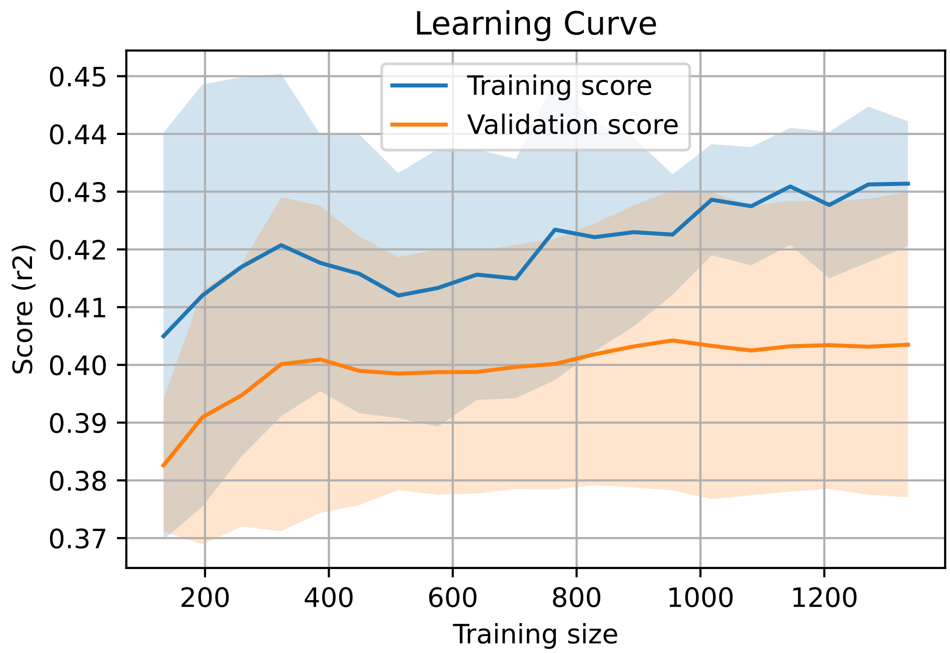
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 5

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817



2.3.13 Gridsearch — Group: 11

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682 y_test: mean=2.561, std=0.947, min=1.120, max=13.845

Fold 0 — y_test: mean=2.58, std=0.86, y_train: mean=2.52, std=0.86

Fold 1 — y_test: mean=2.56, std=0.84, y_train: mean=2.53, std=0.87

Fold 2 — y_test: mean=2.50, std=0.84, y_train: mean=2.55, std=0.87

Fold 3 — y_test: mean=2.53, std=0.87, y_train: mean=2.55, std=0.85

Fold 4 — y_test: mean=2.53, std=0.82, y_train: mean=2.54, std=0.87

train scores CV: [0.46497694 0.46580479 0.46495948 0.46844345 0.45091902]

test scores CV: [0.45757013 0.45504757 0.45666765 0.44978339 0.49647352]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.463	0.4631	0.0169	0.3942	0.3945	-0.0686	0.8519	0.9998

Coefficients:

coef_r_CO2 = 0.0009

coef_r_NH3 = 0.0485

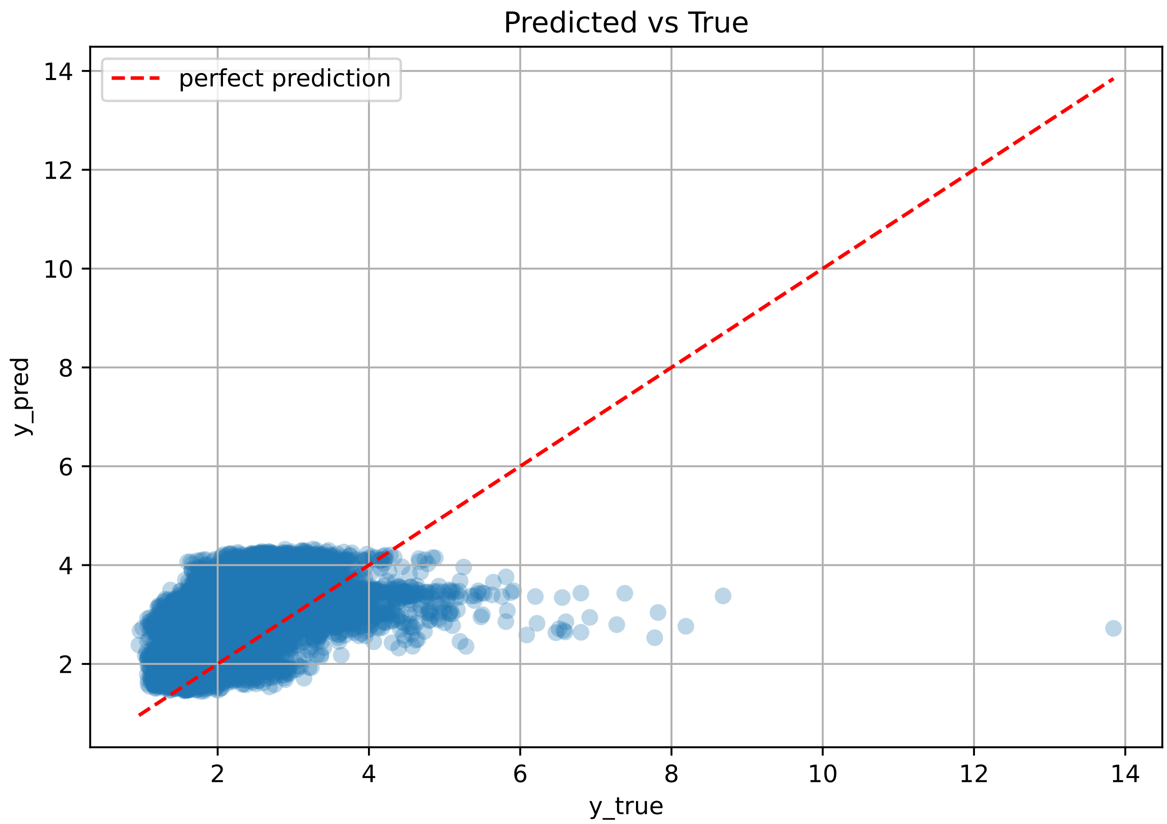
coef_r_THI = 0.8323

coef_r_temperature = -0.9731

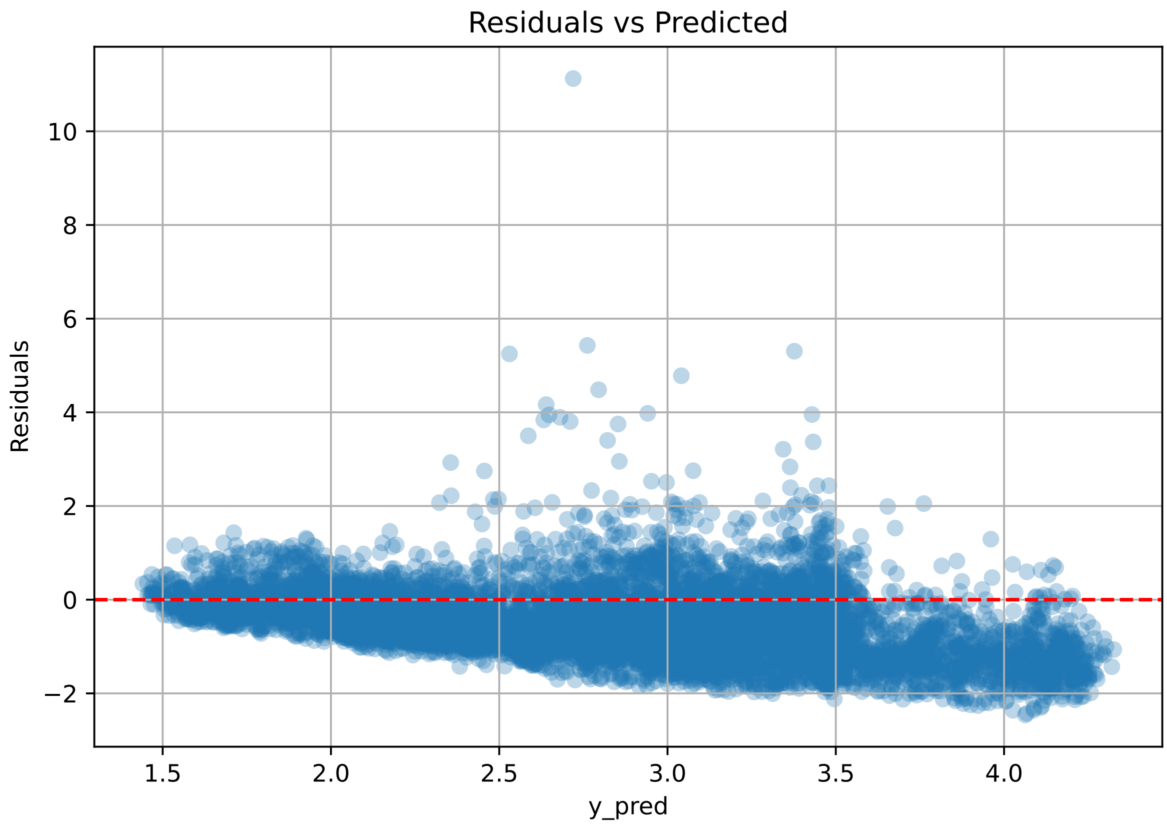
coef_r_umidity = -0.7852

intercept = 2.5398

2.3.14 Scatter



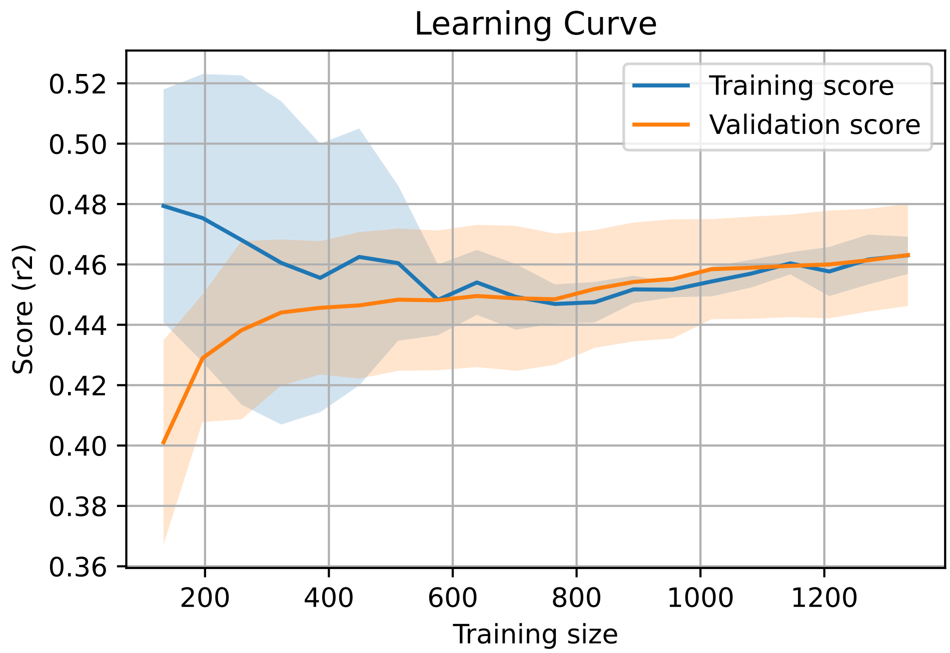
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 11

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817



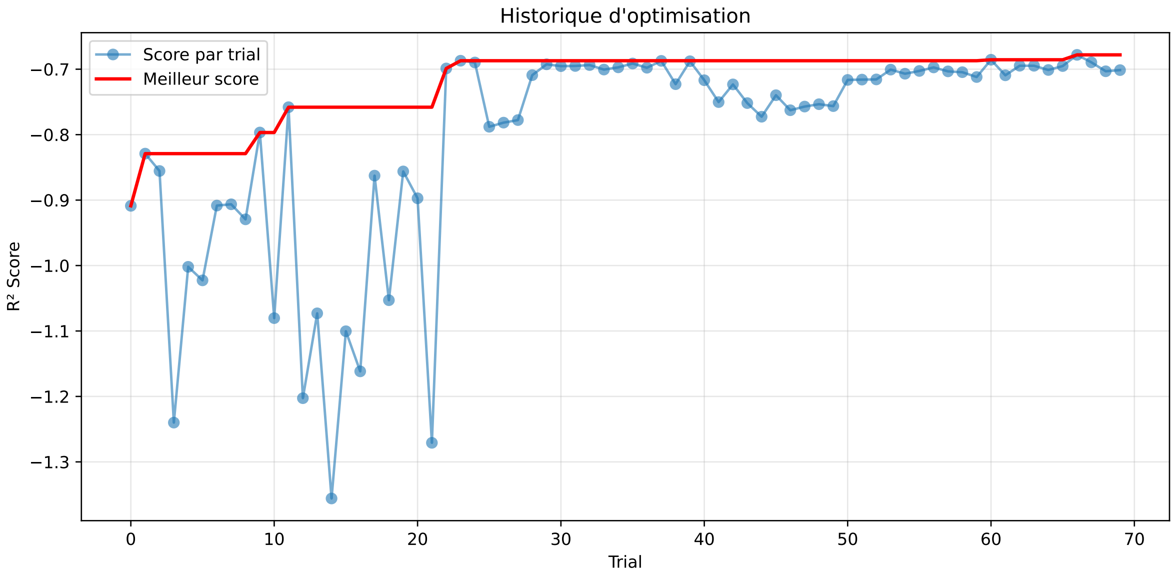
2.4 Model : LGBM

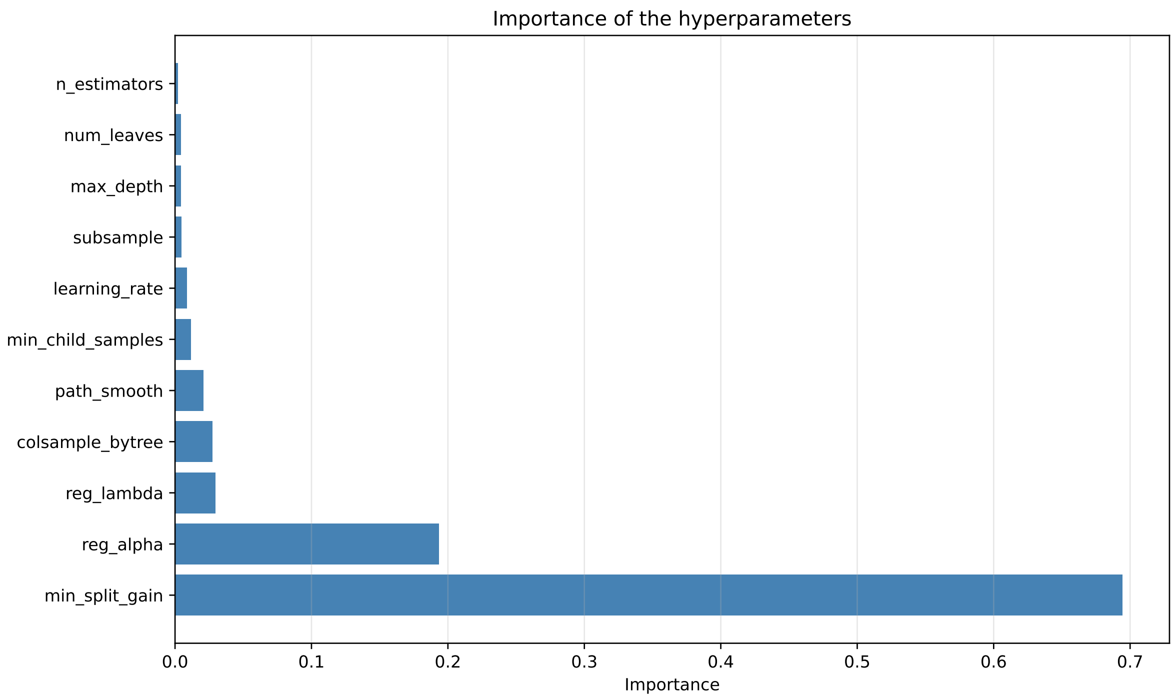
2.4.1 Gridsearch — Group: 3

Len(group) : 1356, **Len(training)** : 950, **Len(test)** : 406

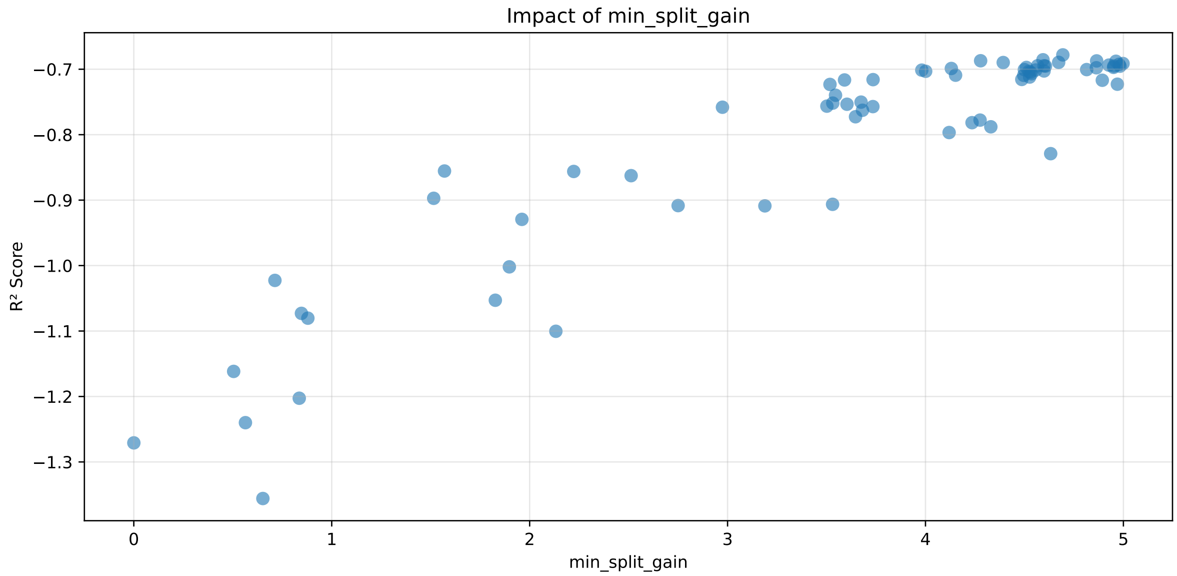
Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086 y_test: mean=2.318, std=0.740, min=1.068, max=5.447

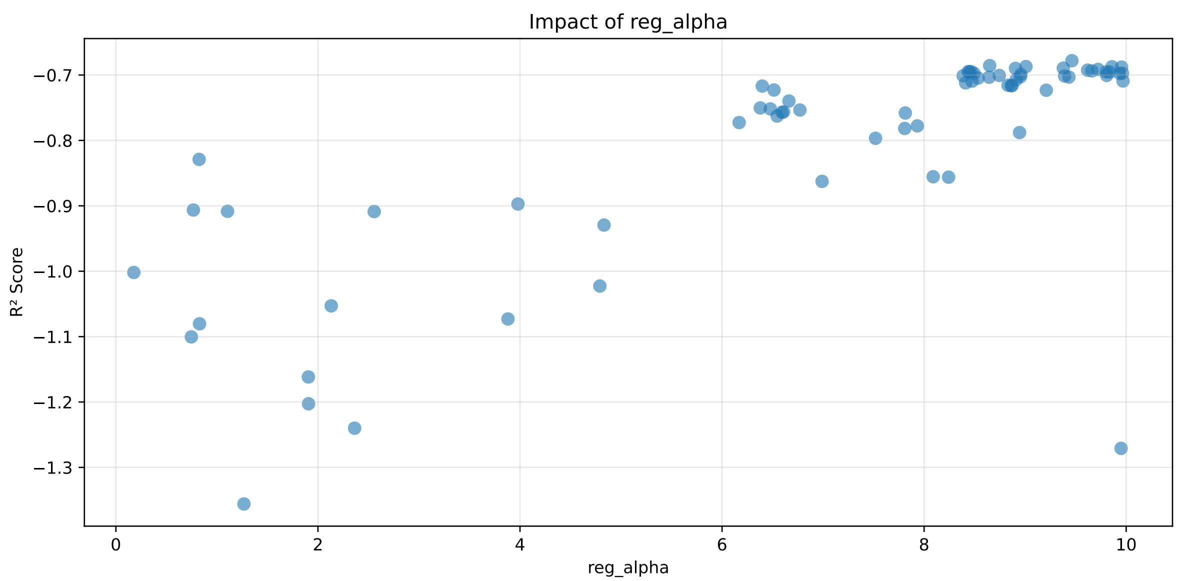
==================== TOP Importance FEATURES ==================== feature importance r_umidity 30 r_temperature 9 r_NH3 8 r_CO2 0 r_THI 0





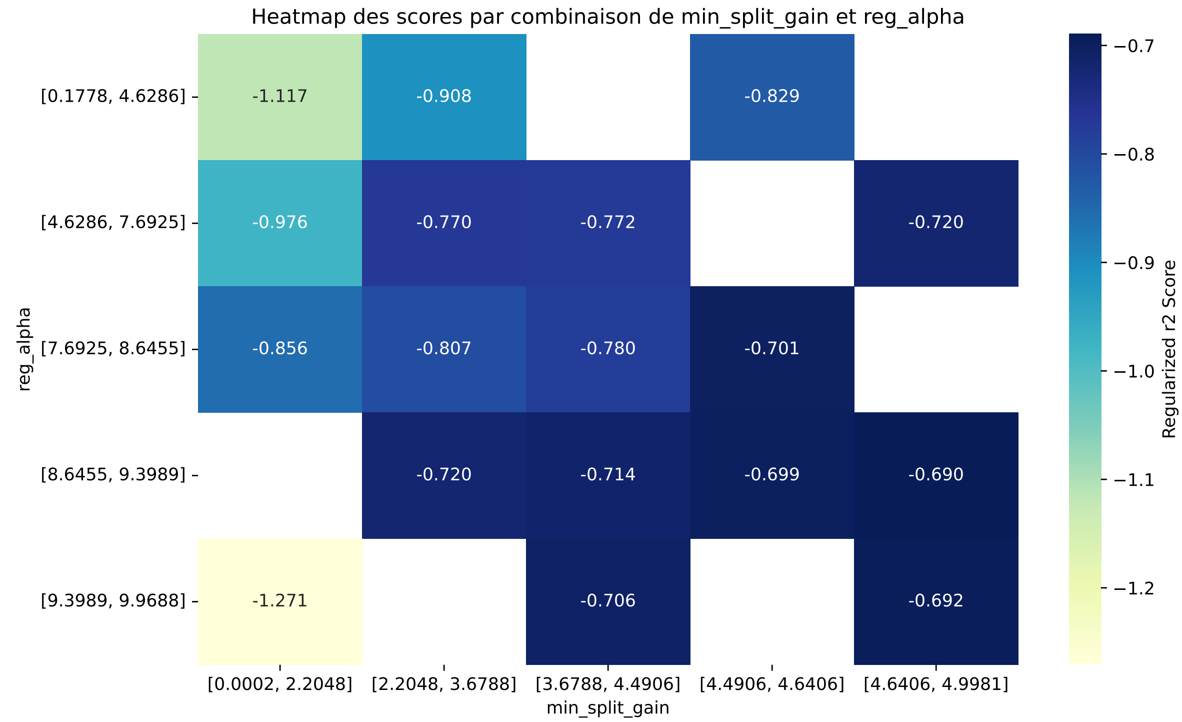
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5695	0.5101	0.581	0.0389	-0.6781	0.6101	0.0999	1.1959	1.1165
2	0.5682	0.5085	0.5806	0.0397	-0.6856	0.6108	0.1024	1.2013	1.1174
3	0.5739	0.5138	0.5856	0.041	-0.6871	0.6152	0.1013	1.1972	1.1169
4	0.5658	0.5061	0.577	0.0391	-0.6873	0.6076	0.1015	1.2006	1.1179
5	0.5661	0.5064	0.5771	0.0401	-0.6878	0.6059	0.0995	1.1965	1.1179
6	0.5665	0.5067	0.5775	0.04	-0.6895	0.608	0.1013	1.1999	1.118
7	0.5716	0.5115	0.5837	0.0406	-0.6898	0.6126	0.1011	1.1976	1.1174
8	0.5665	0.5066	0.5763	0.0404	-0.6913	0.6029	0.0964	1.1902	1.1182
9	0.5641	0.5043	0.5742	0.0398	-0.6925	0.6054	0.1011	1.2004	1.1187
10	0.5662	0.5062	0.5756	0.0403	-0.6937	0.6059	0.0997	1.197	1.1185
---	---	---	---	---	---	---	---	---	---
70	0.697	0.5992	0.6874	0.0324	-1.3559	0.6991	0.0999	1.1667	1.1631

Hyperparameters:

Rank 1: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.08698999944126816, 'reg___num_leaves': 42, 'reg___subsample': 0.6202187477976923, 'reg___colsample_bytree': 0.7690320278470953, 'reg___min_child_samples': 36, 'reg___reg_alpha': 9.462714894489533, 'reg___reg_lambda': 46.53646477433272, 'reg___min_split_gain': 4.693994566533224, 'reg___path_smooth': 1.5572743006229592}

Rank 2: {'reg___n_estimators': 200, 'reg___max_depth': 3, 'reg___learning_rate': 0.0643734132942536, 'reg___num_leaves': 39, 'reg___subsample': 0.6155281969720198, 'reg___colsample_bytree': 0.788489052087602, 'reg___min_child_samples': 41, 'reg___reg_alpha': 8.648812340131917, 'reg___reg_lambda': 46.12183288882852, 'reg___min_split_gain': 4.5937733603786, 'reg___path_smooth': 1.6625562410053987}

Rank 3: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.09012371918741617, 'reg___num_leaves': 52, 'reg___subsample': 0.7053295020656914, 'reg___colsample_bytree': 0.8859013559117079, 'reg___min_child_samples': 11, 'reg___reg_alpha': 9.00827791442368, 'reg___reg_lambda': 49.4361841992527, 'reg___min_split_gain': 4.278595536490568, 'reg___path_smooth': 4.8981662556025585}

Rank 4: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.07160168223977346, 'reg___num_leaves': 36, 'reg___subsample': 0.6724284389087918, 'reg___colsample_bytree': 0.8502919099088924, 'reg___min_child_samples': 34, 'reg___reg_alpha': 9.8597101898629, 'reg___reg_lambda': 40.20185001361375, 'reg___min_split_gain': 4.864742174636623, 'reg___path_smooth': 1.3283280491147864}

Rank 5: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.07967186710751248, 'reg___num_leaves': 36, 'reg___subsample': 0.666343000700893, 'reg___colsample_bytree': 0.8464004886907852, 'reg___min_child_samples': 33, 'reg___reg_alpha': 9.954516992727653, 'reg___reg_lambda': 42.486876473701535, 'reg___min_split_gain': 4.962356131014538, 'reg___path_smooth': 1.4635402101947341}

Rank 6: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.06505580294652163, 'reg___num_leaves': 41, 'reg___subsample': 0.5986694065105357, 'reg___colsample_bytree': 0.8713246854661865, 'reg___min_child_samples': 26, 'reg___reg_alpha': 9.377837618937946, 'reg___reg_lambda': 46.09582681219639, 'reg___min_split_gain': 4.672852963930574, 'reg___path_smooth': 1.6099933721054263}

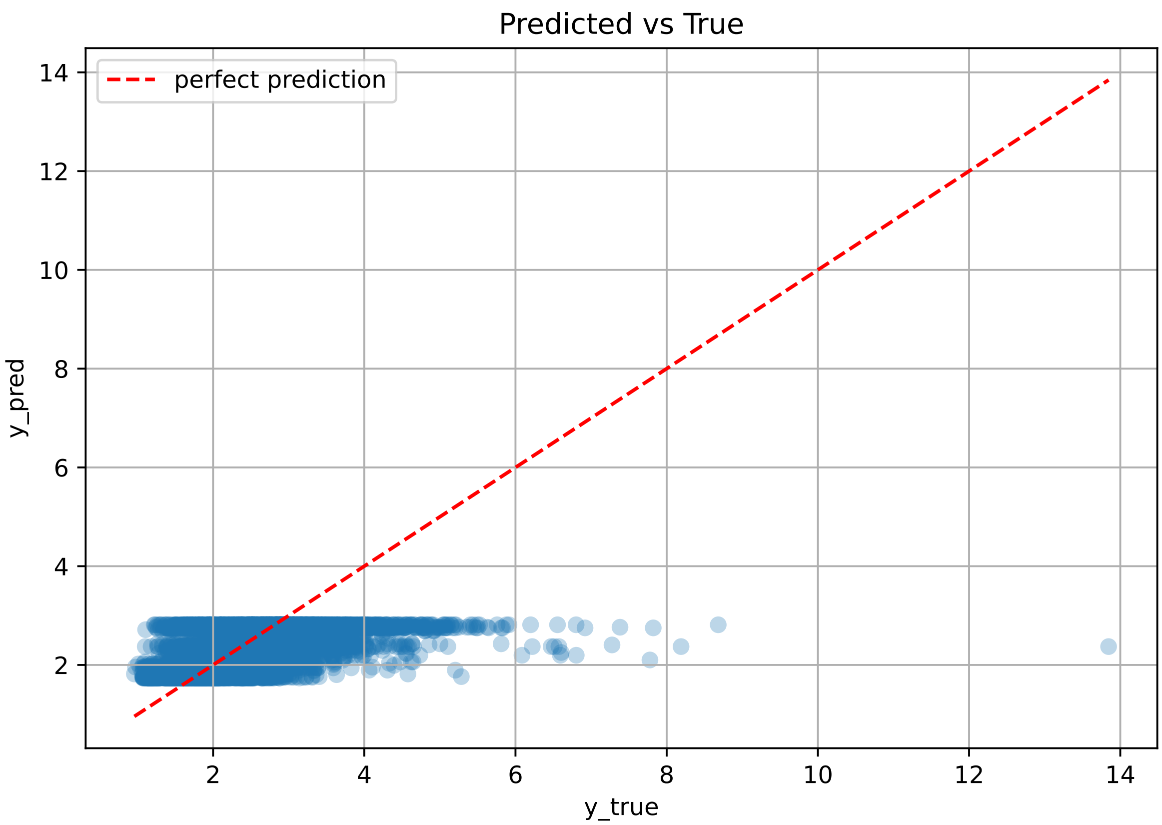
Rank 7: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.07926873935359324, 'reg___num_leaves': 54, 'reg___subsample': 0.7107568598597636, 'reg___colsample_bytree': 0.852392044350931, 'reg___min_child_samples': 36, 'reg___reg_alpha': 8.904388783550397, 'reg___reg_lambda': 46.5056597098908, 'reg___min_split_gain': 4.392723443891457, 'reg___path_smooth': 1.172162697341879}

Rank 8: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.07421247106450168, 'reg___num_leaves': 52, 'reg___subsample': 0.689027822070301, 'reg___colsample_bytree': 0.8490194342389664, 'reg___min_child_samples': 17, 'reg___reg_alpha': 9.721932711739324, 'reg___reg_lambda': 39.96660083686656, 'reg___min_split_gain': 4.99814087820008, 'reg___path_smooth': 1.2471143645210865}

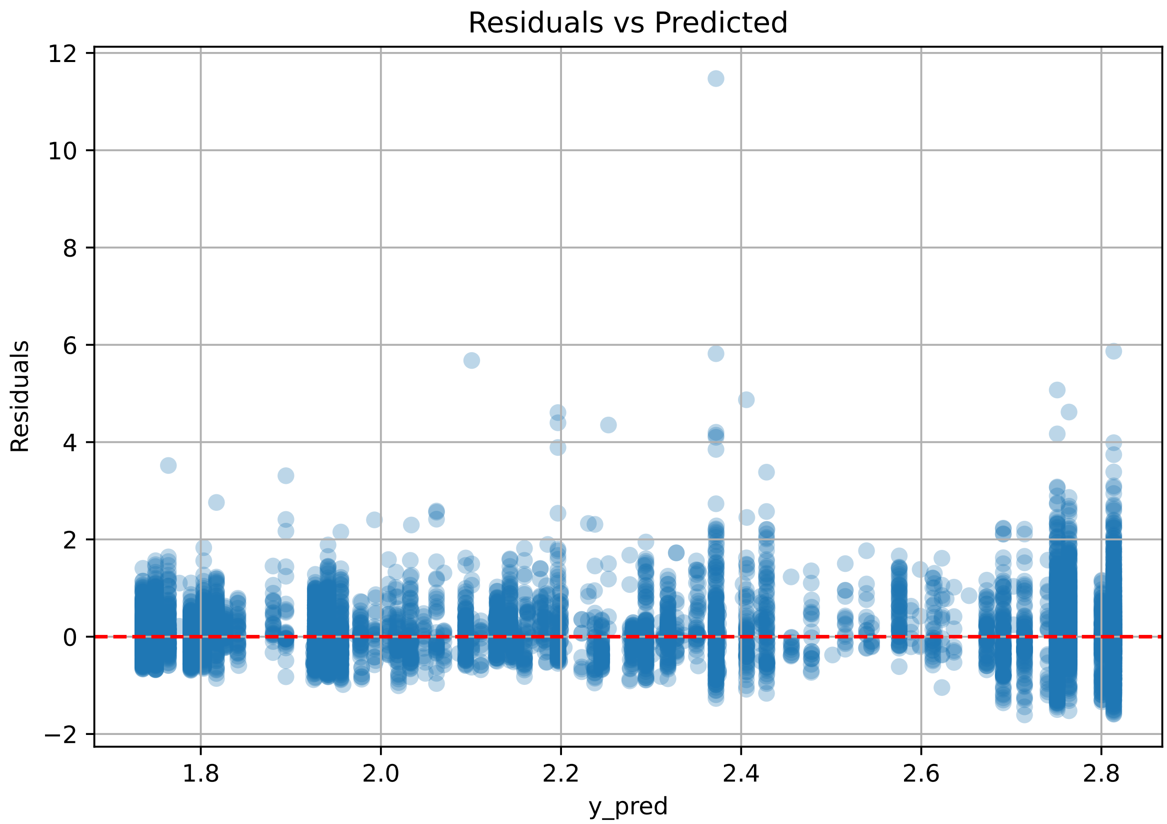
Rank 9: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.03784647012775166, 'reg___num_leaves': 51, 'reg___subsample': 0.6951990887029403, 'reg__colsample_bytree': 0.8544708846407317, 'reg__min_child_samples': 11, 'reg__reg_alpha': 9.6183386906202, 'reg__reg_lambda': 39.987332820566536, 'reg__min_split_gain': 4.978333169584532, 'reg__path_smooth': 4.960158350696347}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.07298910801916361, 'reg__num_leaves': 49, 'reg__subsample': 0.6934515176517067, 'reg__colsample_bytree': 0.8990898371347066, 'reg__min_child_samples': 12, 'reg__reg_alpha': 9.659709261583318, 'reg__reg_lambda': 41.656233463955026, 'reg__min_split_gain': 4.9266832928031805, 'reg__path_smooth': 1.0707082860314536}

2.4.2 Scatter



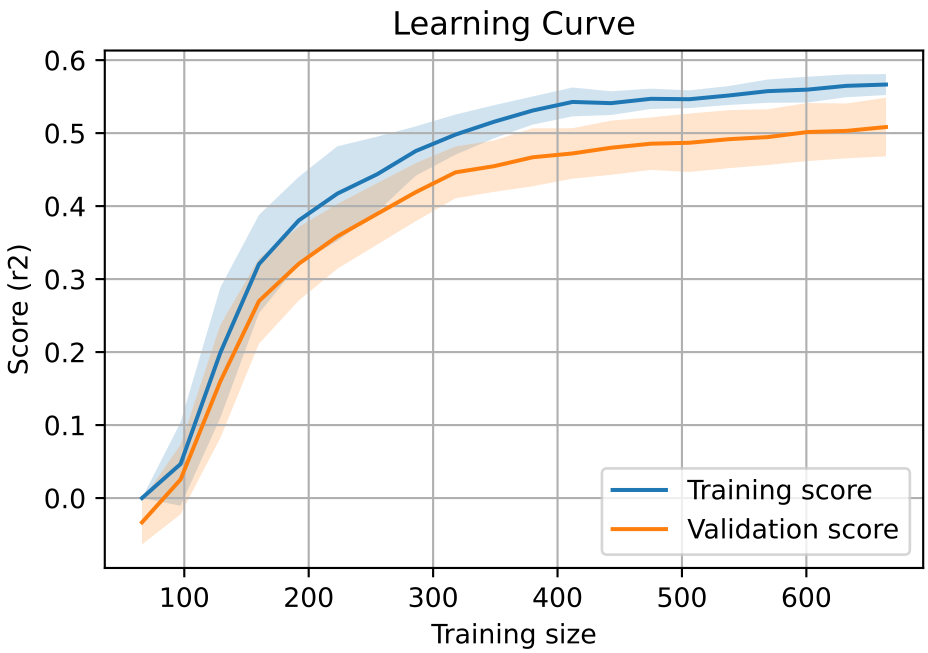
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 3

Len(group) : 1356, **Len(training)** : 950,

Len(training_real) : 665, **Len(test_of_training)** : 285, **Len(test)** : 406

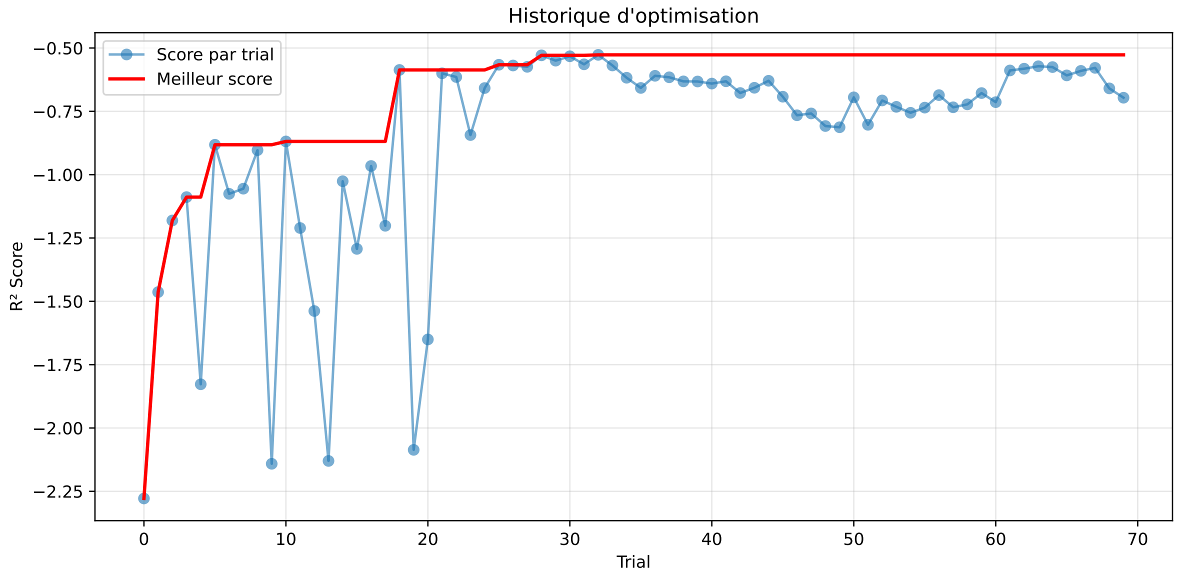


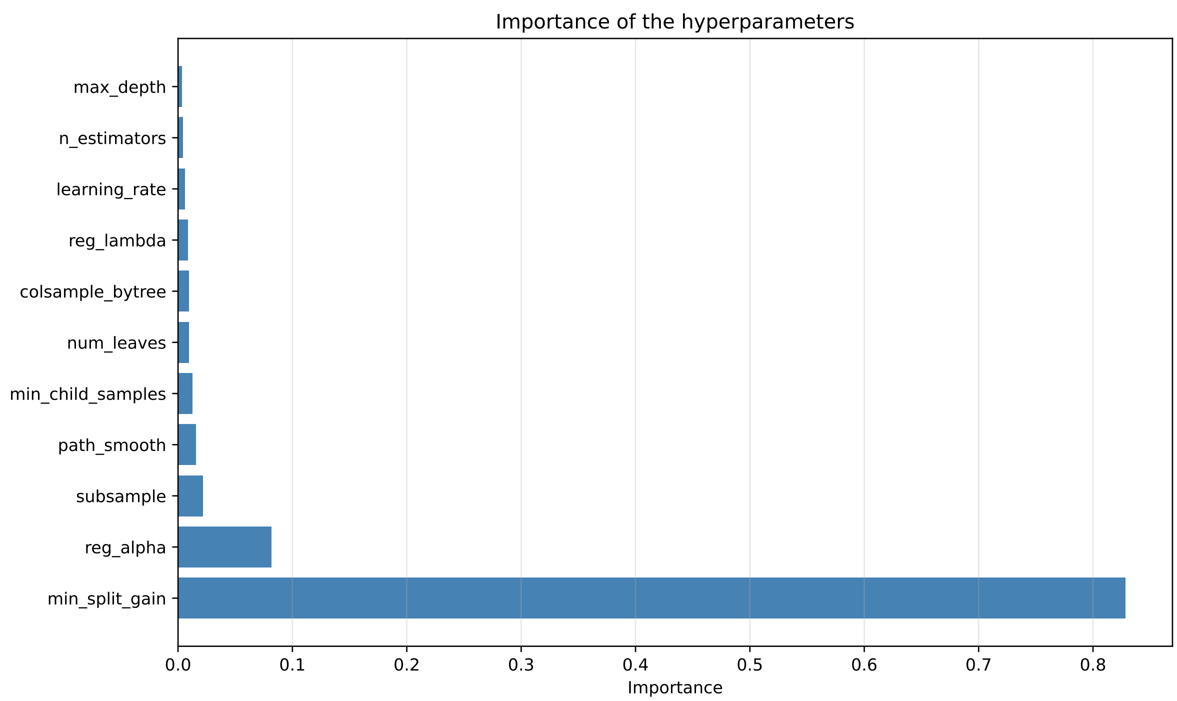
2.4.5 Gridsearch — Group: 4

Len(group) : 1360, **Len(training)** : 952, **Len(test)** : 408

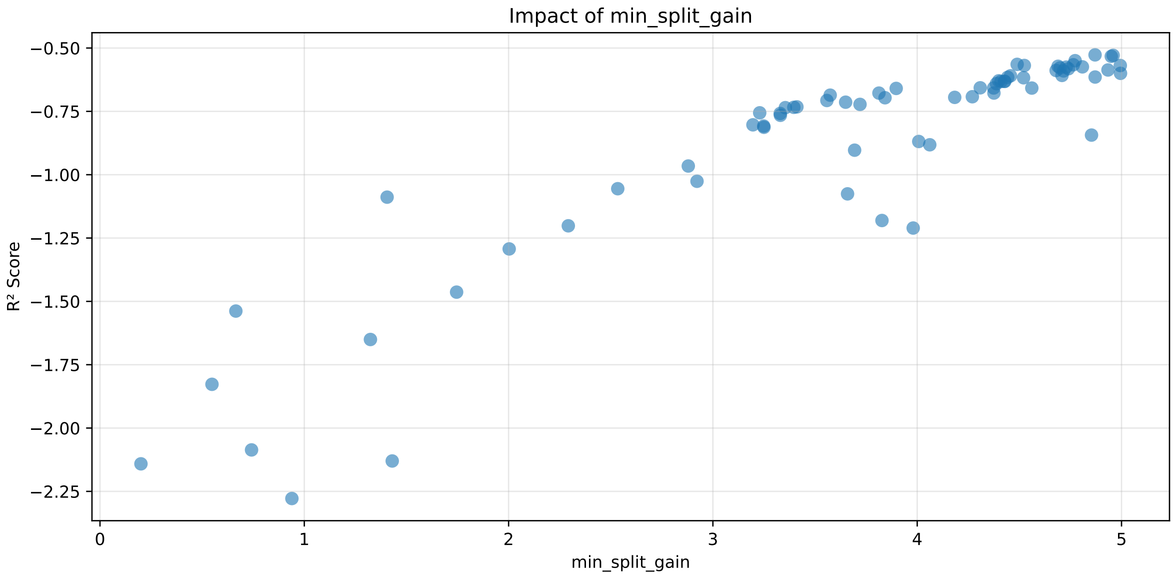
Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383 y_test: mean=2.202, std=0.791, min=1.062, max=7.822

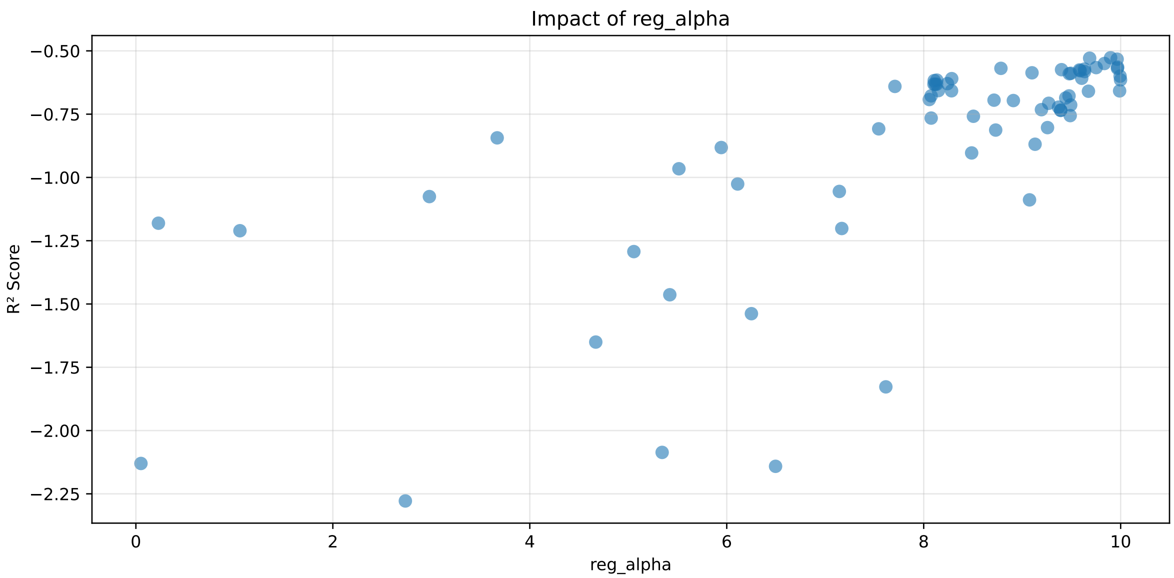
==================== TOP Importance FEATURES ==================== feature importance r_umidity 111 r_THI 73 r_CO2 54 r_NH3 46 r_temperature 34





Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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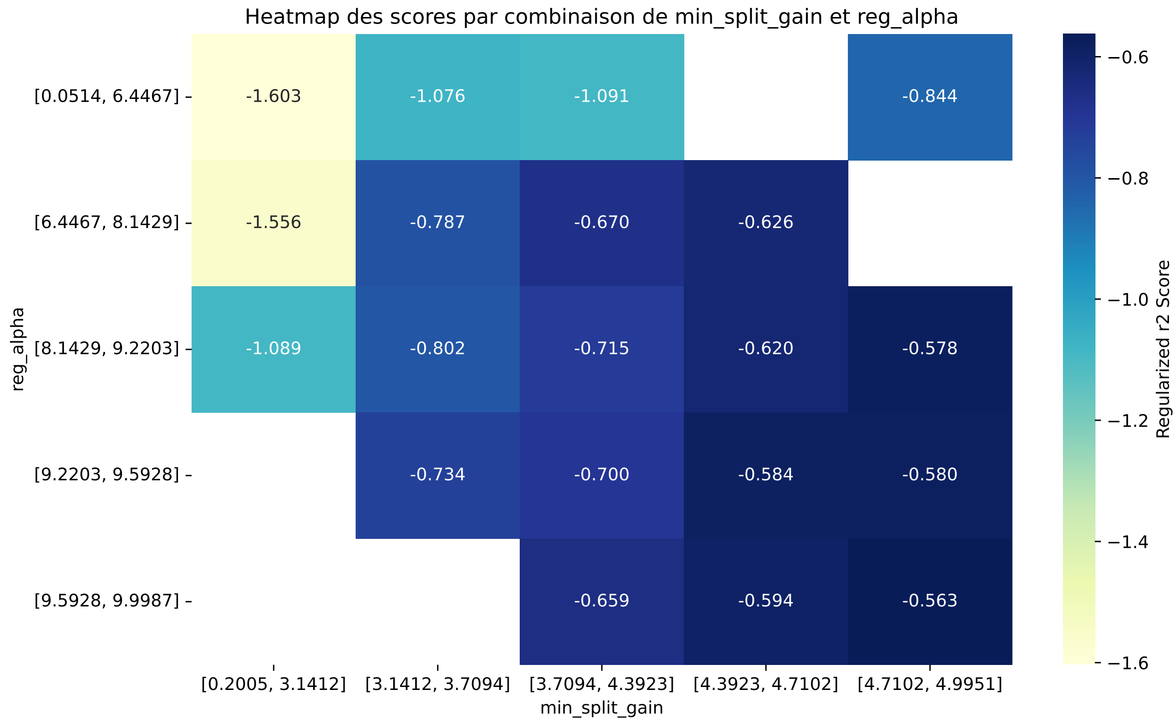
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3013	0.2618	0.2664	0.0391	-0.527	0.3229	0.061	1.233	1.1506
2	0.2943	0.2551	0.2616	0.0386	-0.529	0.3112	0.0561	1.22	1.1537
3	0.2981	0.2585	0.2645	0.04	-0.5327	0.3214	0.0629	1.2431	1.153
4	0.3039	0.2632	0.2695	0.0384	-0.5499	0.3252	0.062	1.2355	1.1545
5	0.3116	0.2698	0.2755	0.0385	-0.5644	0.3327	0.0628	1.2329	1.1546
6	0.3083	0.2666	0.2737	0.0409	-0.5661	0.3284	0.0618	1.2318	1.1562
7	0.3117	0.2698	0.2761	0.0391	-0.5689	0.3317	0.0619	1.2294	1.1554
8	0.3077	0.2659	0.2733	0.0397	-0.569	0.3282	0.0623	1.2343	1.157
9	0.3093	0.2673	0.274	0.0406	-0.572	0.3316	0.0643	1.2406	1.157
10	0.3074	0.2654	0.2727	0.04	-0.5743	0.3283	0.0629	1.2369	1.1582
---	---	---	---	---	---	---	---	---	---
70	0.557	0.4219	0.4558	0.039	-2.2782	0.4958	0.0738	1.1749	1.32

Hyperparameters:

Rank 1: {’reg___n_estimators’: 500, ’reg___max_depth’: 4, ’reg___learning_rate’: 0.018008525468224987, ’reg___num_leaves’: 62, ’reg___subsample’: 0.5757527432020257, ’reg___colsample_bytree’: 0.834577800125603, ’reg___min_child_samples’: 10, ’reg___reg_alpha’: 9.898133616864534, ’reg___reg_lambda’: 49.76498335303305, ’reg___min_split_gain’: 4.870443529350169, ’reg___path_smooth’: 2.2069536066895976}

Rank 2: {’reg__n_estimators’: 200, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.0114436592758462, ’reg__num_leaves’: 60, ’reg__subsample’: 0.5069088416882387, ’reg__colsample_bytree’: 0.5212461239945378, ’reg__min_child_samples’: 33, ’reg__reg_alpha’: 9.685127138277162, ’reg__reg_lambda’: 49.710796881007894, ’reg__min_split_gain’: 4.959116598125295, ’reg__path_smooth’: 0.6589042978188866}

Rank 3: {’reg__n_estimators’: 500, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.01742573186248644, ’reg__num_leaves’: 63, ’reg__subsample’: 0.548388216210724, ’reg__colsample_bytree’: 0.5018796469139297, ’reg__min_child_samples’: 33, ’reg__reg_alpha’: 9.964364895876567, ’reg__reg_lambda’: 49.8611412589511, ’reg__min_split_gain’: 4.9498034151898445, ’reg__path_smooth’: 0.5383496568247028}

Rank 4: {’reg__n_estimators’: 200, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.016518160527078288, ’reg__num_leaves’: 60, ’reg__subsample’: 0.5233283233653875, ’reg__colsample_bytree’: 0.8219655541199025, ’reg__min_child_samples’: 34, ’reg__reg_alpha’: 9.835169213012566, ’reg__reg_lambda’: 47.80890824880316, ’reg__min_split_gain’: 4.772921473677156, ’reg__path_smooth’: 4.608875488841845}

Rank 5: {’reg__n_estimators’: 500, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.02319102438684082, ’reg__num_leaves’: 28, ’reg__subsample’: 0.5807302775619981, ’reg__colsample_bytree’: 0.8197647891996801, ’reg__min_child_samples’: 32, ’reg__reg_alpha’: 9.969206713395474, ’reg__reg_lambda’: 48.824317659792456, ’reg__min_split_gain’: 4.489193976855505, ’reg__path_smooth’: 0.6743917886235006}

Rank 6: {’reg__n_estimators’: 500, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.021819839721461225, ’reg__num_leaves’: 24, ’reg__subsample’: 0.5163053495200212, ’reg__colsample_bytree’: 0.6369548318514626, ’reg__min_child_samples’: 28, ’reg__reg_alpha’: 9.751324402483709, ’reg__reg_lambda’: 44.390686808513735, ’reg__min_split_gain’: 4.7641868272505725, ’reg__path_smooth’: 1.8338466387829586}

Rank 7: {’reg__n_estimators’: 500, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.020615590912604988, ’reg__num_leaves’: 23, ’reg__subsample’: 0.5808971334991775, ’reg__colsample_bytree’: 0.8257104793019621, ’reg__min_child_samples’: 29, ’reg__reg_alpha’: 9.967209366078913, ’reg__reg_lambda’: 47.20344219458358, ’reg__min_split_gain’: 4.524391986061009, ’reg__path_smooth’: 0.7241093284727393}

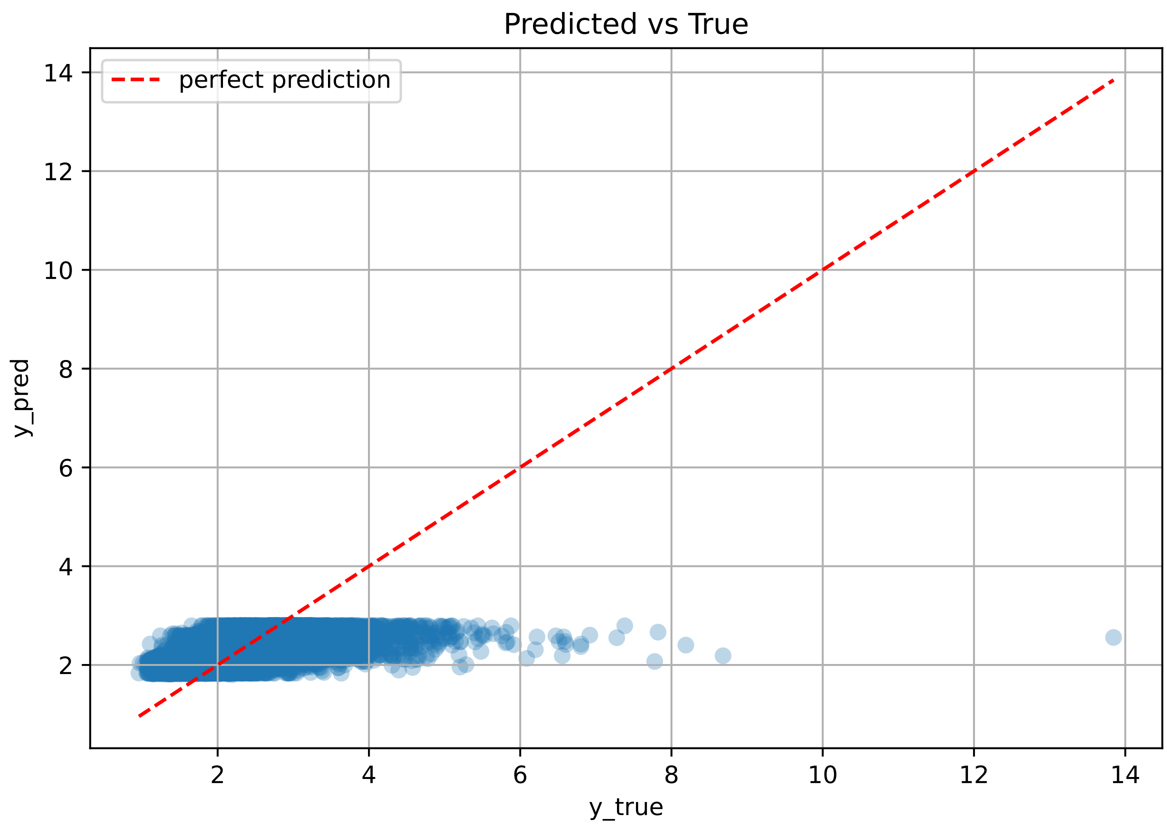
Rank 8: {’reg__n_estimators’: 600, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.02254890893695839, ’reg__num_leaves’: 63, ’reg__subsample’: 0.5507208611449308, ’reg__colsample_bytree’: 0.50875275789051, ’reg__min_child_samples’: 29, ’reg__reg_alpha’: 8.784215534588157, ’reg__reg_lambda’: 48.87001407704411, ’reg__min_split_gain’: 4.994041145225405, ’reg__path_smooth’: 1.8218194432819401}

Rank 9: {’reg__n_estimators’: 500, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.019229605207620983, ’reg__num_leaves’: 17, ’reg__subsample’: 0.5028127037262863,

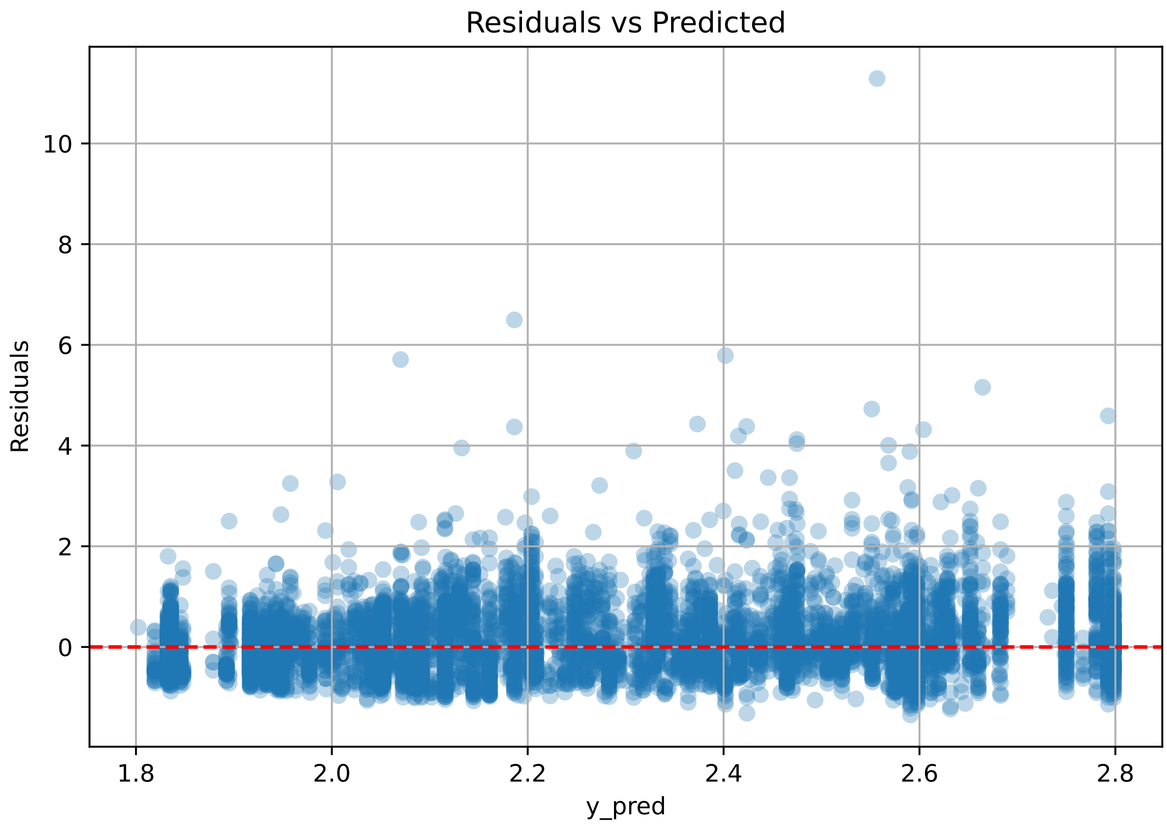
'reg___colsample_bytree': 0.5783590761671281, 'reg___min_child_samples': 32, 'reg___reg_alpha': 9.633047967298335, 'reg___reg_lambda': 47.280106421794585, 'reg___min_split_gain': 4.689472739606158, 'reg___path_smooth': 0.02247678165013367}

Rank 10: {'reg___n_estimators': 500, 'reg___max_depth': 4, 'reg___learning_rate': 0.023161040689147993, 'reg___num_leaves': 30, 'reg___subsample': 0.5572163871556381, 'reg___colsample_bytree': 0.5061767834268908, 'reg___min_child_samples': 36, 'reg___reg_alpha': 9.399697948182228, 'reg___reg_lambda': 47.997422556457906, 'reg___min_split_gain': 4.808540356928596, 'reg___path_smooth': 1.7664617250226033}

2.4.6 Scatter



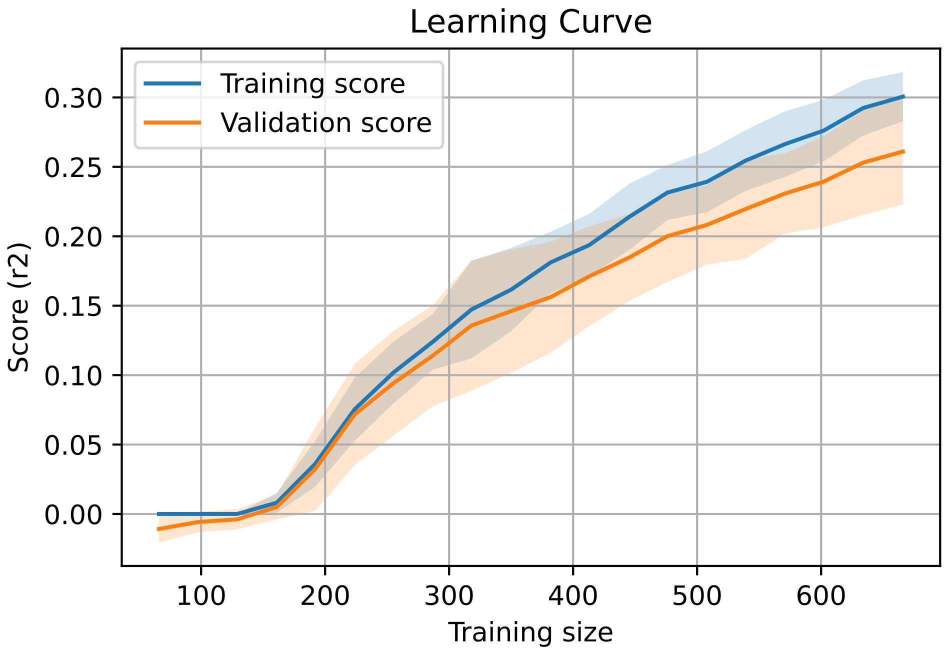
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 4

Len(group) : 1360, **Len(training)** : 952,

Len(training_real) : 667, **Len(test_of_training)** : 285, **Len(test)** : 408

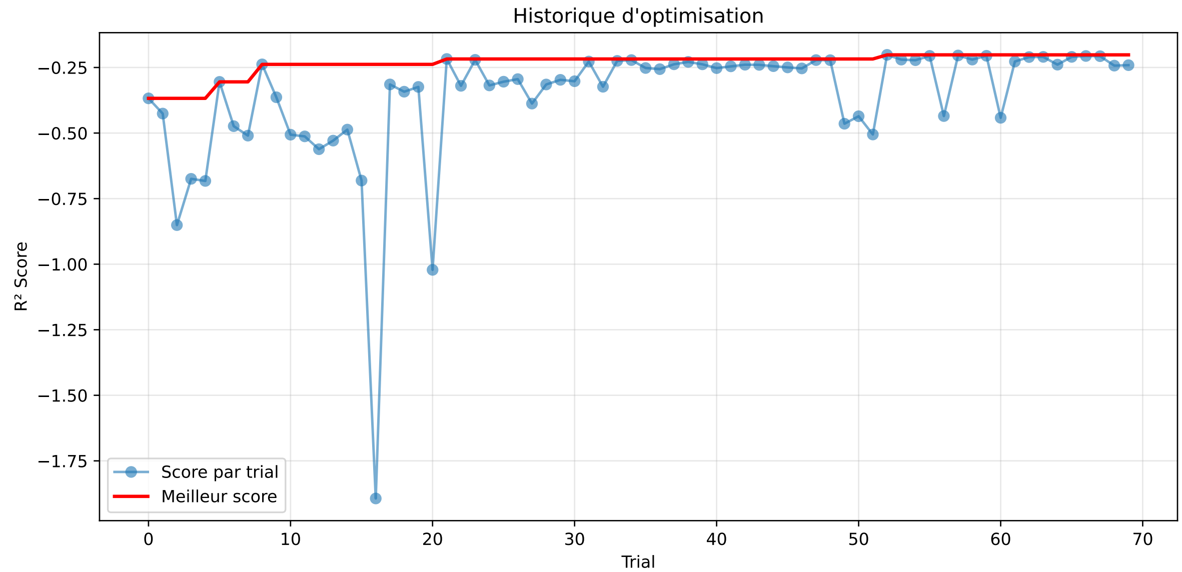


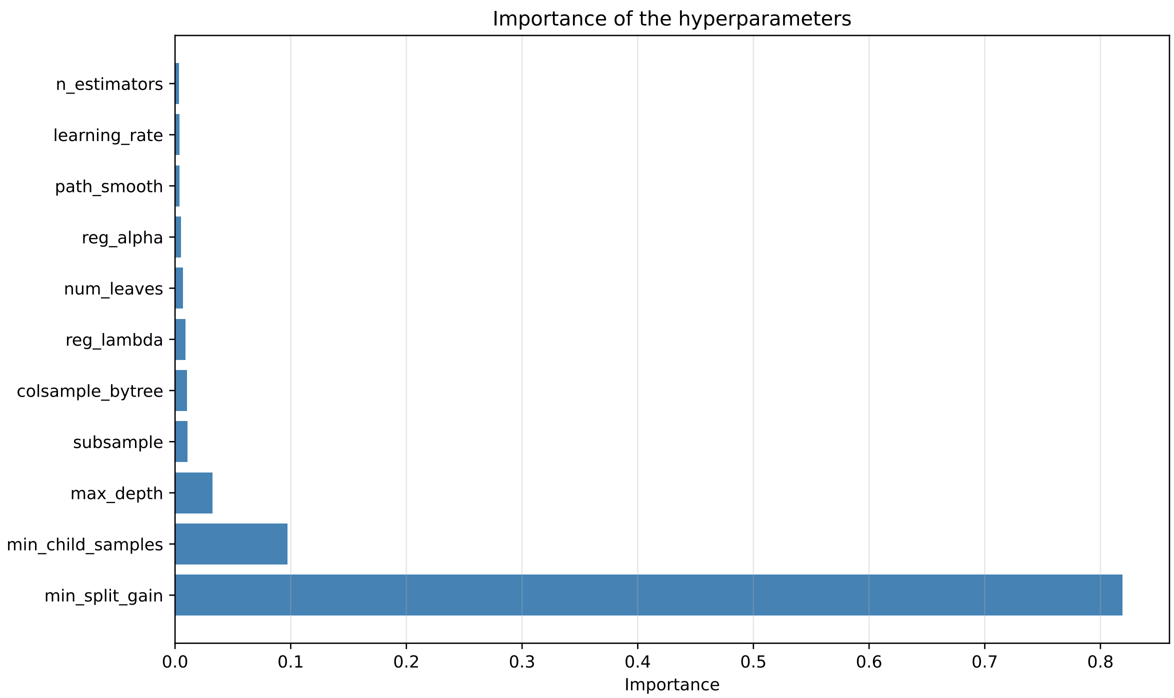
2.4.9 Gridsearch — Group: 5

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

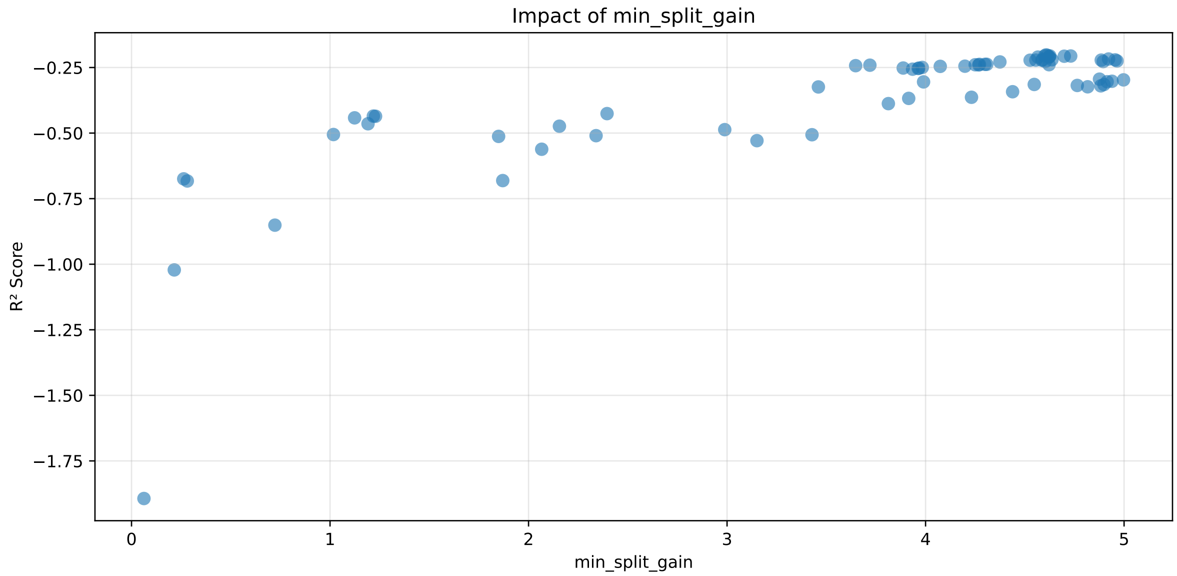
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813 y_test: mean=2.212, std=0.659, min=1.071, max=4.858

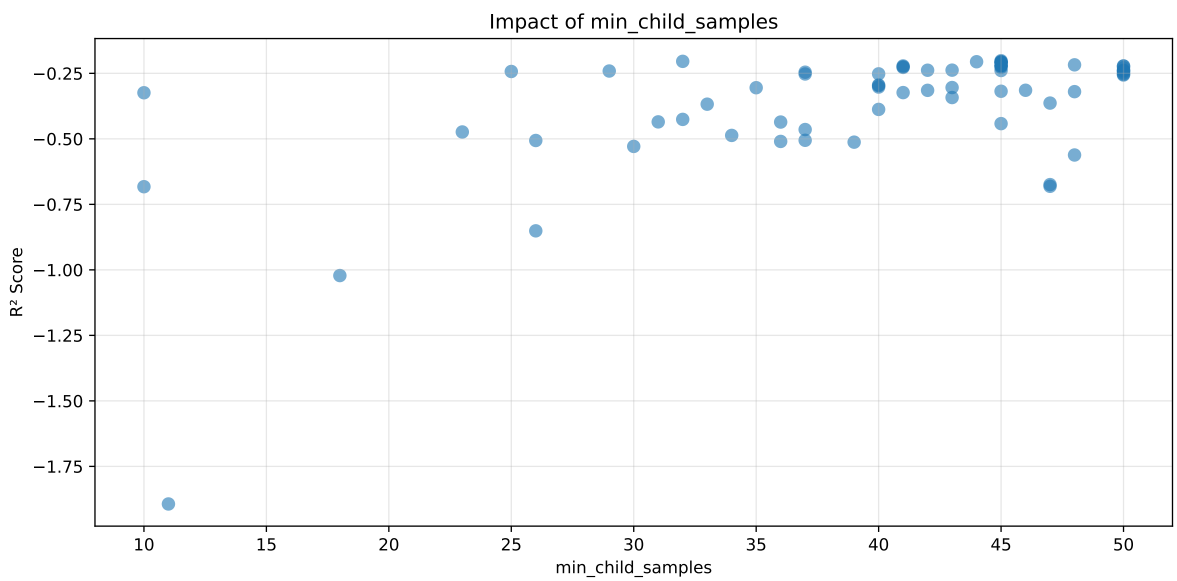
==================== TOP Importance FEATURES ==================== feature importance r_umidity 287 r_temperature 52 r_NH3 50 r_CO2 2 r_THI 0





Paramètres avec >5% d'importance : ['min_split_gain', 'min_child_samples']



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```

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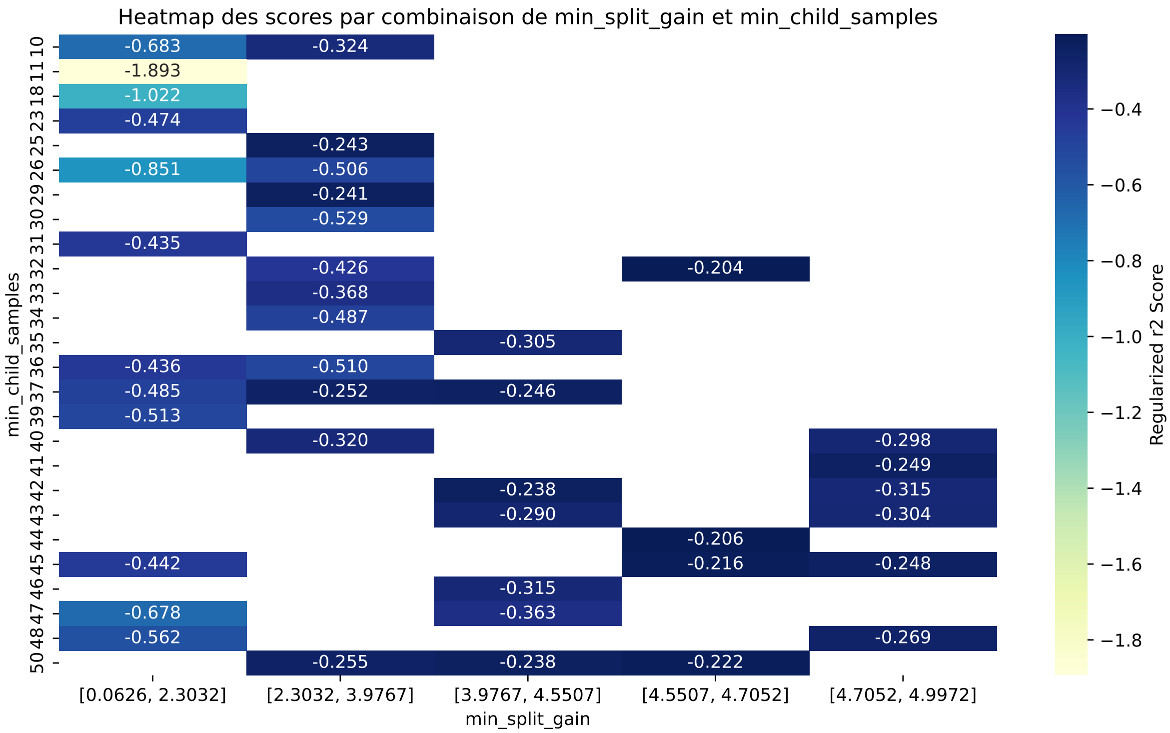
The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4075	0.3785	0.434	0.0221	-0.2018	0.451	0.0725	1.1916	1.0767
2	0.4081	0.3789	0.4346	0.0224	-0.2042	0.453	0.0741	1.1957	1.0769
3	0.4085	0.3792	0.4349	0.0221	-0.2054	0.4544	0.0752	1.1982	1.0771
4	0.4079	0.3787	0.4344	0.0219	-0.2057	0.4539	0.0752	1.1986	1.0772
5	0.4087	0.3795	0.4353	0.0221	-0.2061	0.4532	0.0738	1.1944	1.0772
6	0.4082	0.3789	0.4349	0.0221	-0.2069	0.4539	0.075	1.1979	1.0773
7	0.4092	0.3797	0.4357	0.0221	-0.2096	0.4553	0.0755	1.1989	1.0776
8	0.4095	0.38	0.436	0.022	-0.2096	0.4548	0.0748	1.1969	1.0776
9	0.4094	0.3799	0.4359	0.0221	-0.2102	0.4551	0.0752	1.198	1.0777
10	0.4042	0.3746	0.4326	0.0222	-0.2175	0.4481	0.0735	1.1962	1.079
---	---	---	---	---	---	---	---	---	---
70	0.6001	0.4814	0.5482	0.0271	-1.8934	0.5668	0.0855	1.1776	1.2467

Hyperparameters:

Rank 1: {'reg___n_estimators': 800, 'reg___max_depth': 7, 'reg___learning_rate': 0.012660869551840075, 'reg___num_leaves': 59, 'reg___subsample': 0.8653310306838501, 'reg___colsample_bytree': 0.7267061395932104, 'reg___min_child_samples': 45, 'reg___reg_alpha': 9.9852818239974, 'reg___reg_lambda': 42.95691149098625, 'reg___min_split_gain': 4.607302394126417, 'reg___path_smooth': 4.607067900147257}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.014740766974117268, 'reg__num_leaves': 54, 'reg__subsample': 0.7074862758549642, 'reg__colsample_bytree': 0.7084366485753134, 'reg__min_child_samples': 32, 'reg__reg_alpha': 9.414645279419865, 'reg__reg_lambda': 43.467152903381276, 'reg__min_split_gain': 4.61409677243826, 'reg__path_smooth': 4.5650811118027566}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.015794661812418984, 'reg__num_leaves': 37, 'reg__subsample': 0.7343286057070955, 'reg__colsample_bytree': 0.7146670340212009, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.866760639554913, 'reg__reg_lambda': 35.39989794184777, 'reg__min_split_gain': 4.600222809181097, 'reg__path_smooth': 4.642033069327331}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.015640561312168552, 'reg__num_leaves': 59, 'reg__subsample': 0.8272006405847782, 'reg__colsample_bytree': 0.7234695811053358, 'reg__min_child_samples': 44, 'reg__reg_alpha': 9.964632182626598, 'reg__reg_lambda': 34.6030205824464, 'reg__min_split_gain': 4.625950183278426, 'reg__path_smooth': 4.633400892527133}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.01610605552479242, 'reg__num_leaves': 54, 'reg__subsample': 0.8266298714976275, 'reg__colsample_bytree': 0.7260376952513656, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.264203065349083, 'reg__reg_lambda': 36.6767534166411, 'reg__min_split_gain': 4.731153356577217, 'reg__path_smooth': 4.272059985018182}

Rank 6: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.015915087380992402, 'reg__num_leaves': 53, 'reg__subsample': 0.825802178509464, 'reg__colsample_bytree': 0.7210479799401829, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.398656123781095, 'reg__reg_lambda': 36.48870343439519, 'reg__min_split_gain': 4.69868652425149, 'reg__path_smooth': 4.116698517900062}

Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.01572982788501325, 'reg__num_leaves': 53, 'reg__subsample': 0.8144550574941551, 'reg__colsample_bytree': 0.7087355215399799, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.34749185451063, 'reg__reg_lambda': 36.023703768271716, 'reg__min_split_gain': 4.622424383886923, 'reg__path_smooth': 4.24036390892723}

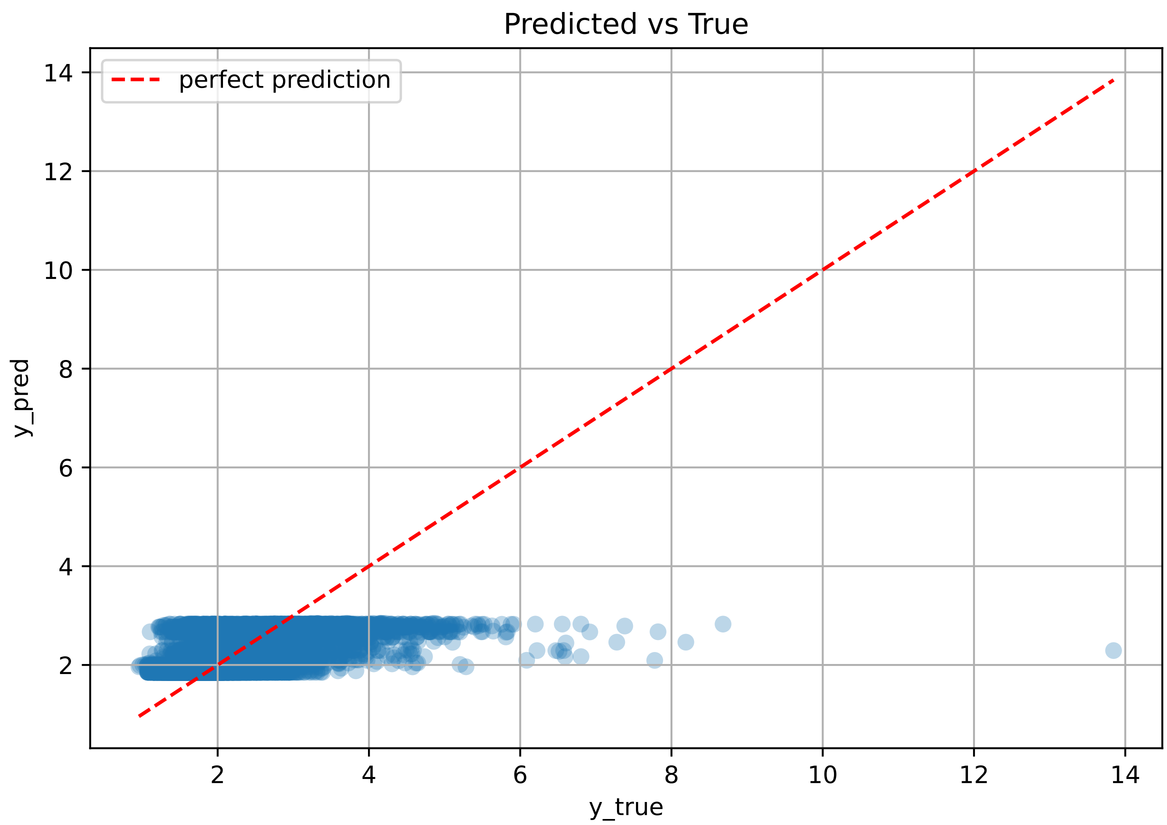
Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.01596727736710415, 'reg__num_leaves': 35, 'reg__subsample': 0.8234204743574272, 'reg__colsample_bytree': 0.7140470569340032, 'reg__min_child_samples': 45, 'reg__reg_alpha': 9.30346317191591, 'reg__reg_lambda': 35.0353598838857, 'reg__min_split_gain': 4.622720762484688, 'reg__path_smooth': 3.175322823962456}

Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.016176627475346653, 'reg__num_leaves': 37, 'reg__subsample': 0.8206369241065748,

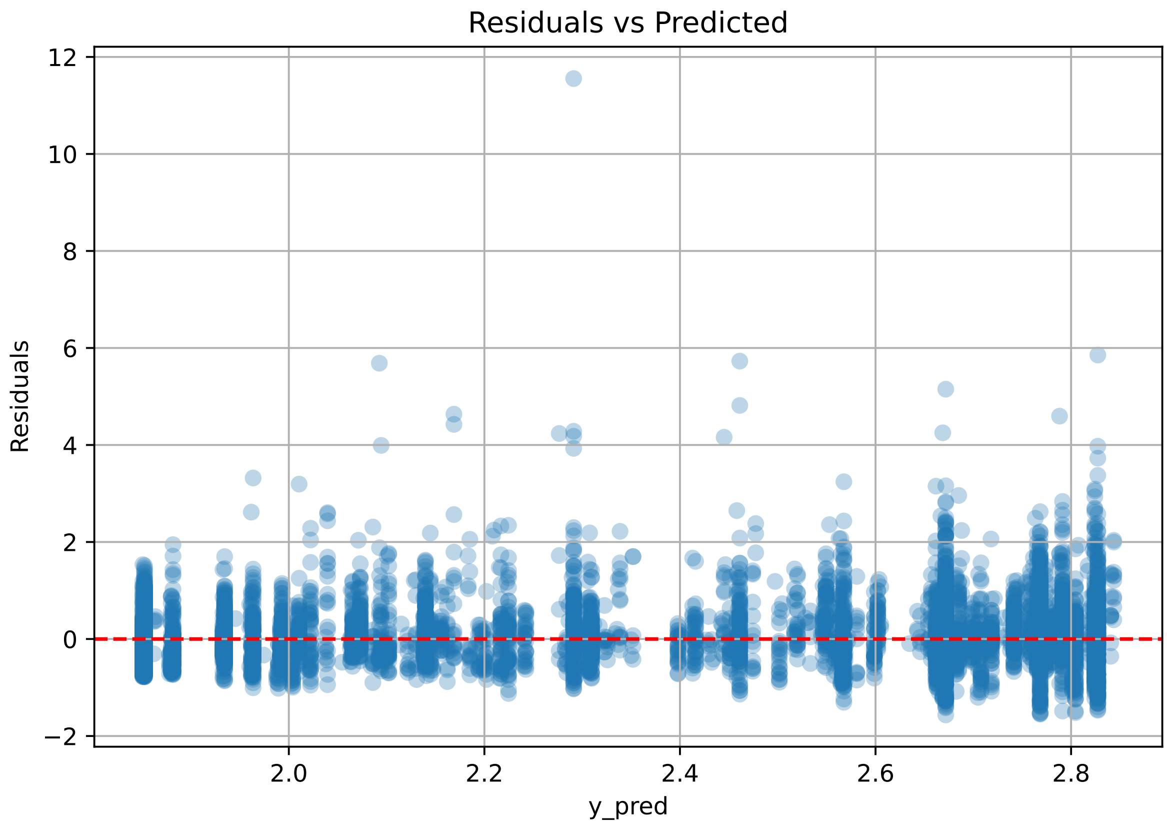
'reg___colsample_bytree': 0.7228560022958278, 'reg___min_child_samples': 45, 'reg___reg_alpha': 9.446365189343114, 'reg___reg_lambda': 34.24222198873332, 'reg___min_split_gain': 4.565770641940738, 'reg___path_smooth': 3.1850255020391116}

Rank 10: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.011977969834985313, 'reg___num_leaves': 62, 'reg___subsample': 0.8969389851095873, 'reg___colsample_bytree': 0.6510806065187895, 'reg___min_child_samples': 48, 'reg___reg_alpha': 9.39777747823219, 'reg___reg_lambda': 47.252027498502684, 'reg___min_split_gain': 4.921873517258551, 'reg___path_smooth': 4.752361009583063}

2.4.10 Scatter



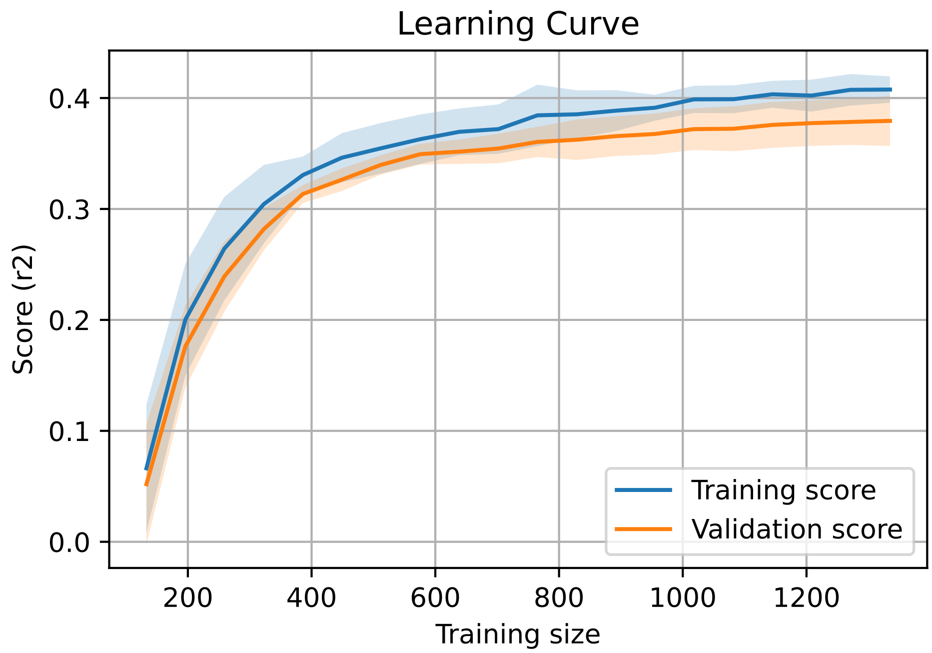
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 5

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817

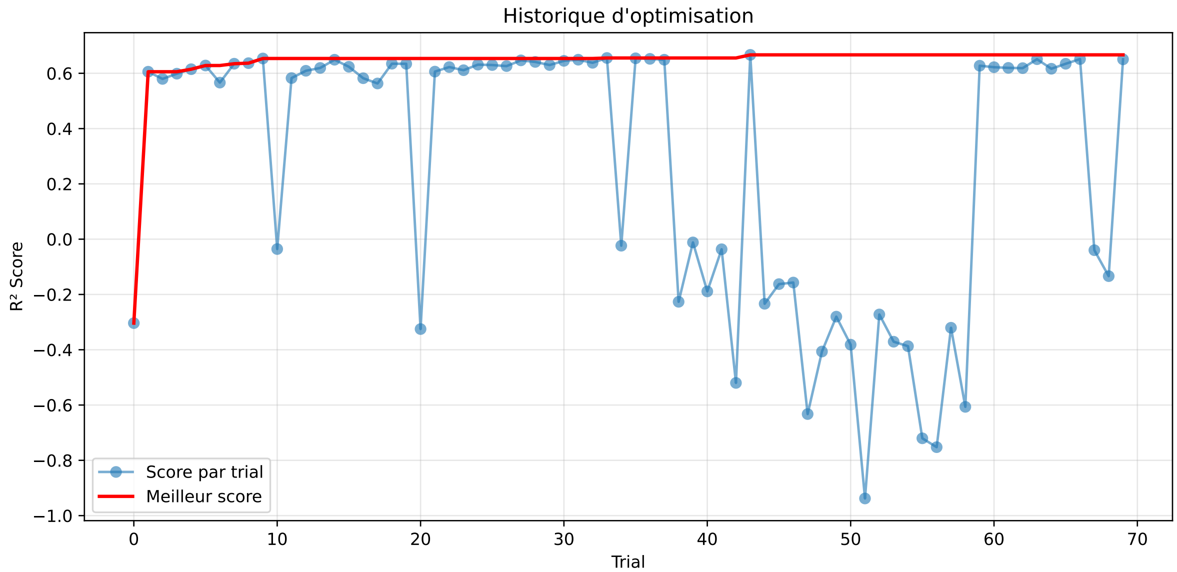


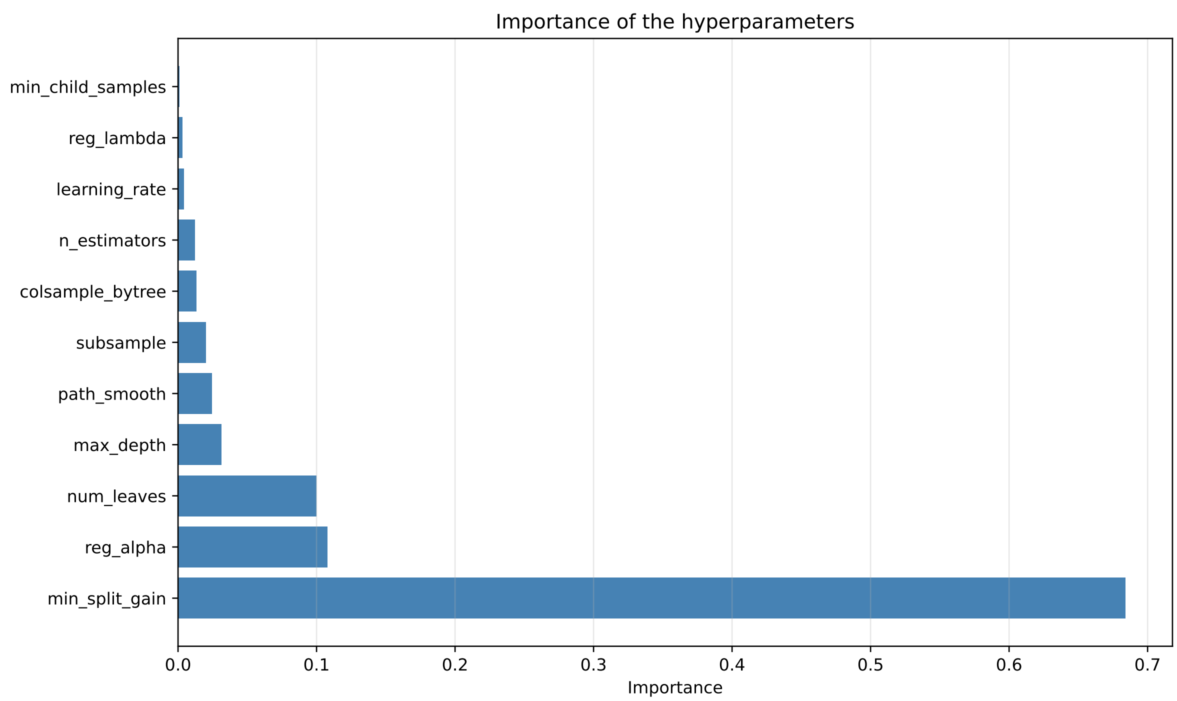
2.4.13 Gridsearch — Group: 11

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

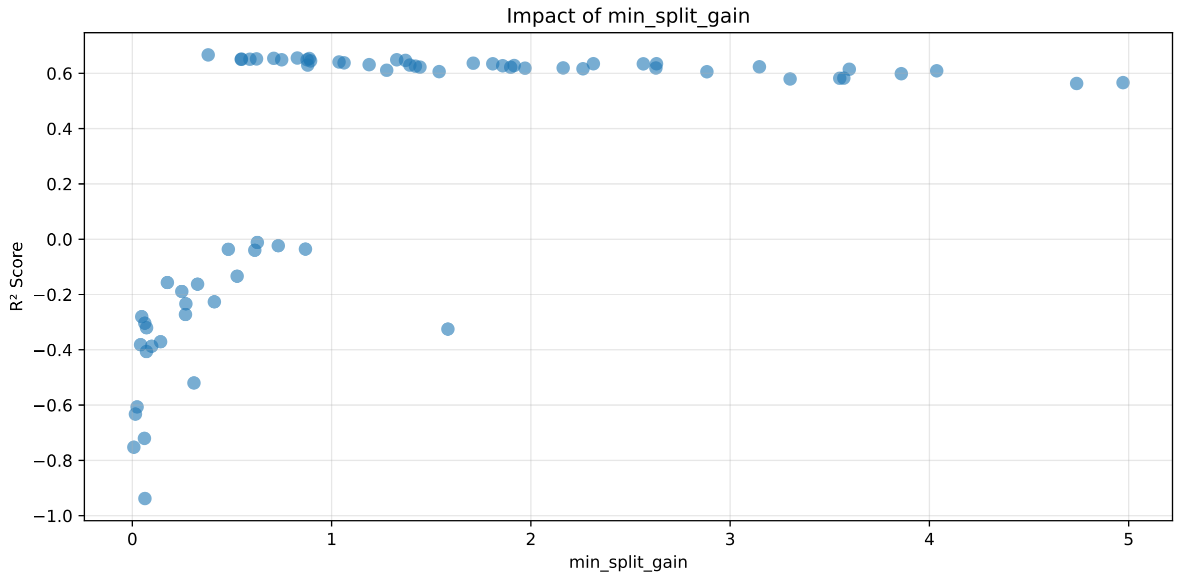
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682 y_test: mean=2.561, std=0.947, min=1.120, max=13.845

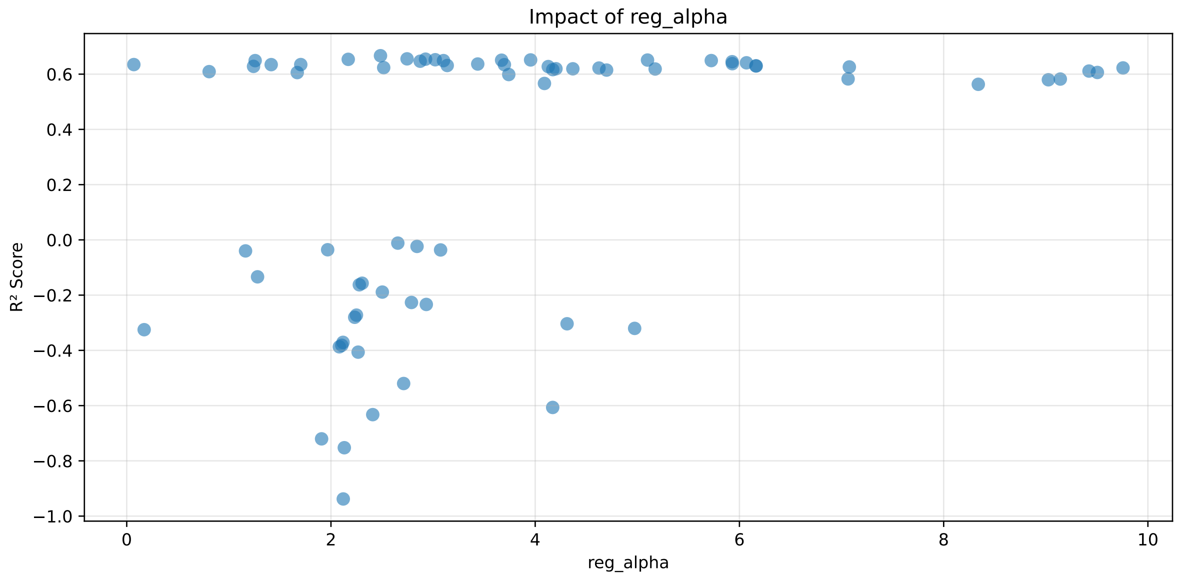
==================== TOP Importance FEATURES ==================== feature importance r_NH3 512 r_umidity 496 r_CO2 402 r_THI 248 r_temperature 154

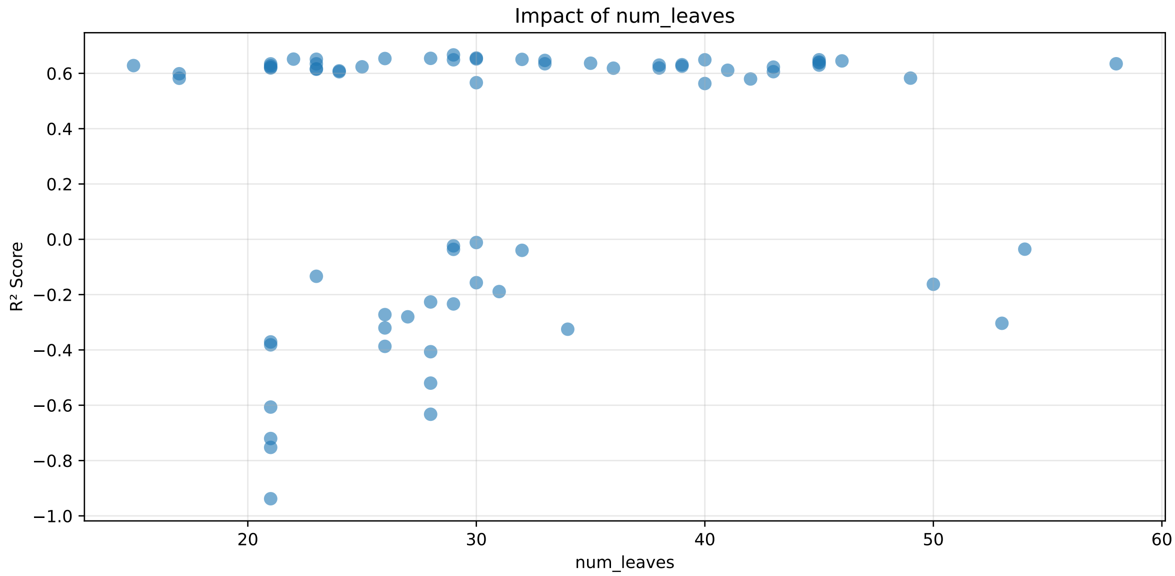




Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'num_leaves']

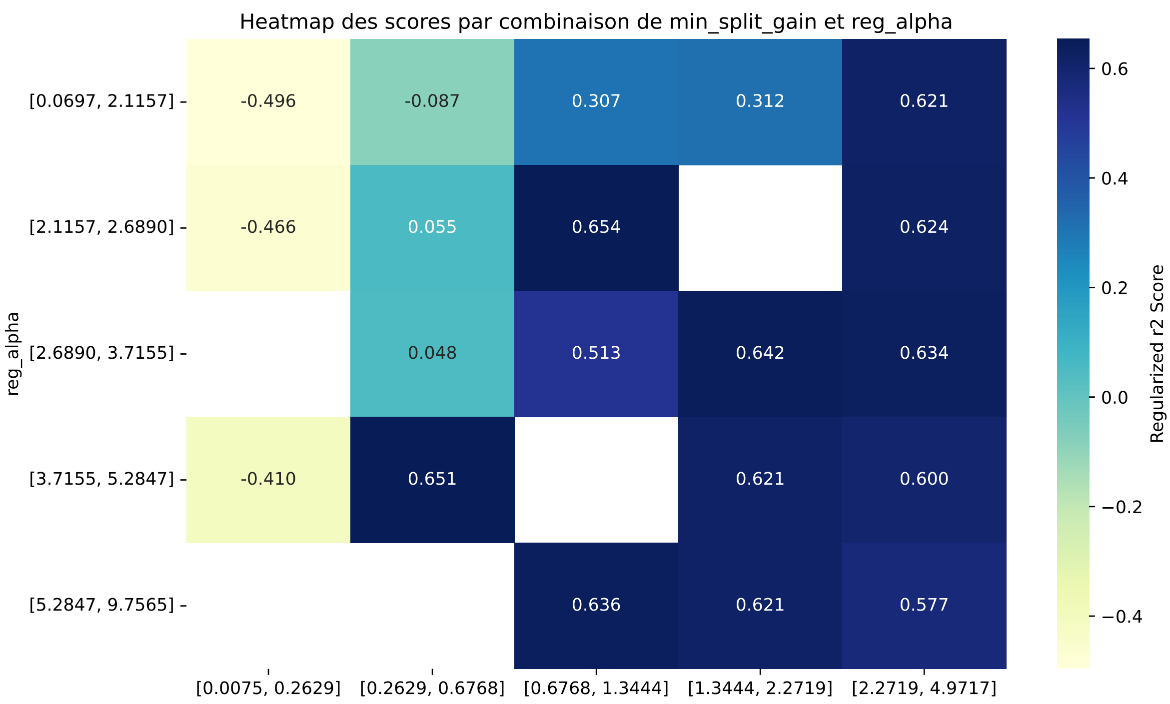

Impact of min_split_gain
R² Score
min_split_gain
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
1
2
3
4
5


Impact of reg_alpha
R² Score
reg_alpha
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
2
4
6
8
10




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6974	0.6667	0.5657	0.0144	0.6667	0.5802	-0.0865	0.8703	1.046
2	0.6799	0.6554	0.5549	0.0151	0.6554	0.5679	-0.0875	0.8665	1.0374
3	0.6869	0.6543	0.5552	0.0121	0.6543	0.5719	-0.0823	0.8742	1.0499
4	0.6825	0.6536	0.5597	0.0125	0.6536	0.5703	-0.0833	0.8726	1.0442
5	0.6848	0.6522	0.5562	0.0124	0.6522	0.5739	-0.0783	0.8799	1.0499
6	0.6833	0.6515	0.5582	0.0125	0.6515	0.5695	-0.082	0.8742	1.0488
7	0.6806	0.651	0.5548	0.0125	0.651	0.5668	-0.0842	0.8706	1.0454
8	0.6806	0.6505	0.555	0.0122	0.6505	0.571	-0.0795	0.8778	1.0462
9	0.6791	0.6491	0.5499	0.0133	0.6491	0.5681	-0.081	0.8752	1.0461
10	0.6795	0.649	0.5536	0.0121	0.649	0.5707	-0.0784	0.8793	1.047
---	---	---	---	---	---	---	---	---	---
70	0.7771	0.6954	0.5767	0.0122	-0.938	0.585	-0.1104	0.8413	1.1174

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.022920944249042455, 'reg__num_leaves': 29, 'reg__subsample': 0.8436790854764074, 'reg__colsample_bytree': 0.7695530579096084, 'reg__min_child_samples': 37, 'reg__reg_alpha': 2.4849932021573826, 'reg__reg_lambda': 26.648343197365175, 'reg__min_split_gain': 0.3806997330166576, 'reg__path_smooth': 0.08539630416999155}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.02188368484191861, 'reg__num_leaves': 30, 'reg__subsample': 0.7463373563120452, 'reg__colsample_bytree': 0.8210684157921552, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.7453675981854526, 'reg__reg_lambda': 13.761560651107018, 'reg__min_split_gain': 0.8280611935919634, 'reg__path_smooth': 1.3871996226301064}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.022031896038947105, 'reg__num_leaves': 28, 'reg__subsample': 0.7443689916235611, 'reg__colsample_bytree': 0.637979367209574, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.9262703529502994, 'reg__reg_lambda': 14.838398780755607, 'reg__min_split_gain': 0.7097561906782106, 'reg__path_smooth': 2.5203820632009}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.022380493827217154, 'reg__num_leaves': 26, 'reg__subsample': 0.8746773950550037, 'reg__colsample_bytree': 0.6183059289661397, 'reg__min_child_samples': 25, 'reg__reg_alpha': 2.169233600703534, 'reg__reg_lambda': 18.857451326587487, 'reg__min_split_gain': 0.8880173474607267, 'reg__path_smooth': 1.1553571311959536}

Rank 5: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.021379541156224636, 'reg__num_leaves': 30, 'reg__subsample': 0.7457837121020735,

'reg___colsample_bytree': 0.6298288517074248, 'reg___min_child_samples': 37, 'reg___reg_alpha': 3.021193123380837, 'reg___reg_lambda': 27.698809040868788, 'reg___min_split_gain': 0.6231957202785672, 'reg___path_smooth': 2.452039145075024}

Rank 6: {'reg___n_estimators': 600, 'reg___max_depth': 4, 'reg___learning_rate': 0.01711113591652954, 'reg___num_leaves': 22, 'reg___subsample': 0.7748858386634546, 'reg___colsample_bytree': 0.6705216186009867, 'reg___min_child_samples': 32, 'reg___reg_alpha': 3.955105031211788, 'reg___reg_lambda': 31.727087600068437, 'reg___min_split_gain': 0.5473913227592839, 'reg___path_smooth': 2.024465174746362}

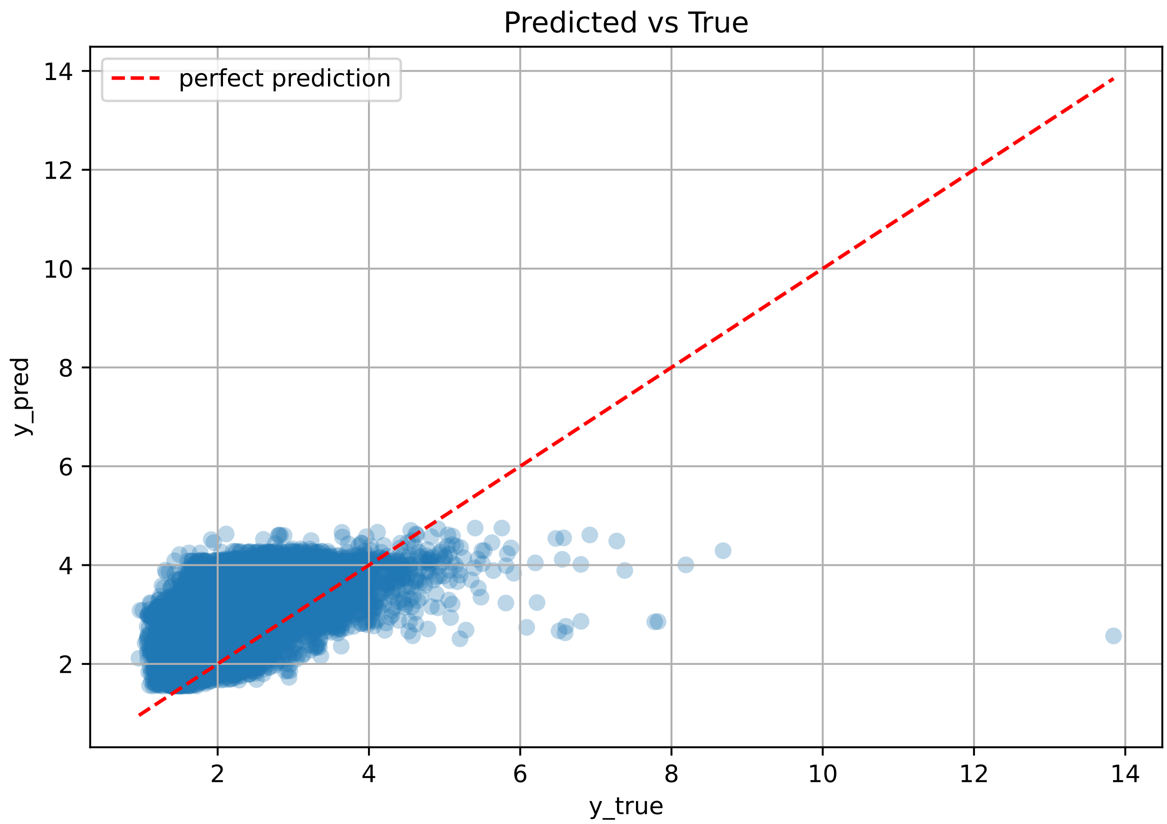
Rank 7: {'reg___n_estimators': 600, 'reg___max_depth': 4, 'reg___learning_rate': 0.026661340628375062, 'reg___num_leaves': 23, 'reg___subsample': 0.6991852544871138, 'reg___colsample_bytree': 0.6843633141603819, 'reg___min_child_samples': 32, 'reg___reg_alpha': 5.0999338473690985, 'reg___reg_lambda': 16.18671428584608, 'reg___min_split_gain': 0.5899457975468221, 'reg___path_smooth': 2.218724363131614}

Rank 8: {'reg___n_estimators': 600, 'reg___max_depth': 4, 'reg___learning_rate': 0.02005976014181701, 'reg___num_leaves': 32, 'reg___subsample': 0.867642290126332, 'reg___colsample_bytree': 0.5727670596374406, 'reg___min_child_samples': 39, 'reg___reg_alpha': 3.670950443953198, 'reg___reg_lambda': 31.771019593738508, 'reg___min_split_gain': 0.5465292520344678, 'reg___path_smooth': 1.3315791867367748}

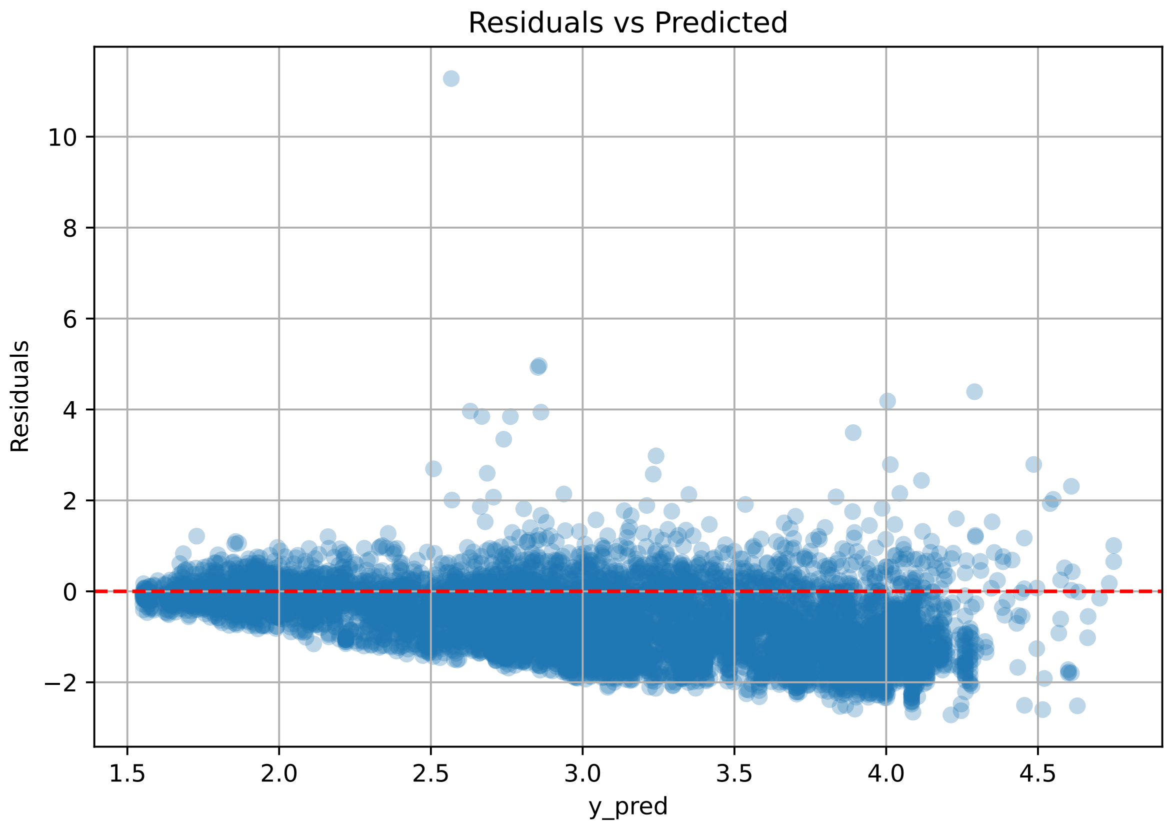
Rank 9: {'reg___n_estimators': 700, 'reg___max_depth': 5, 'reg___learning_rate': 0.0175835564832564, 'reg___num_leaves': 45, 'reg___subsample': 0.8952106510660502, 'reg___colsample_bytree': 0.7217566911677635, 'reg___min_child_samples': 37, 'reg___reg_alpha': 5.724326390855027, 'reg___reg_lambda': 13.213999702977551, 'reg___min_split_gain': 0.8770809788338908, 'reg___path_smooth': 0.8682247758647259}

Rank 10: {'reg___n_estimators': 700, 'reg___max_depth': 5, 'reg___learning_rate': 0.022273177904964456, 'reg___num_leaves': 29, 'reg___subsample': 0.8989819017884143, 'reg___colsample_bytree': 0.6548539013334529, 'reg___min_child_samples': 37, 'reg___reg_alpha': 3.102209311267279, 'reg___reg_lambda': 26.564641131308175, 'reg___min_split_gain': 0.7493697486149814, 'reg___path_smooth': 1.5466166944769928}

2.4.14 Scatter



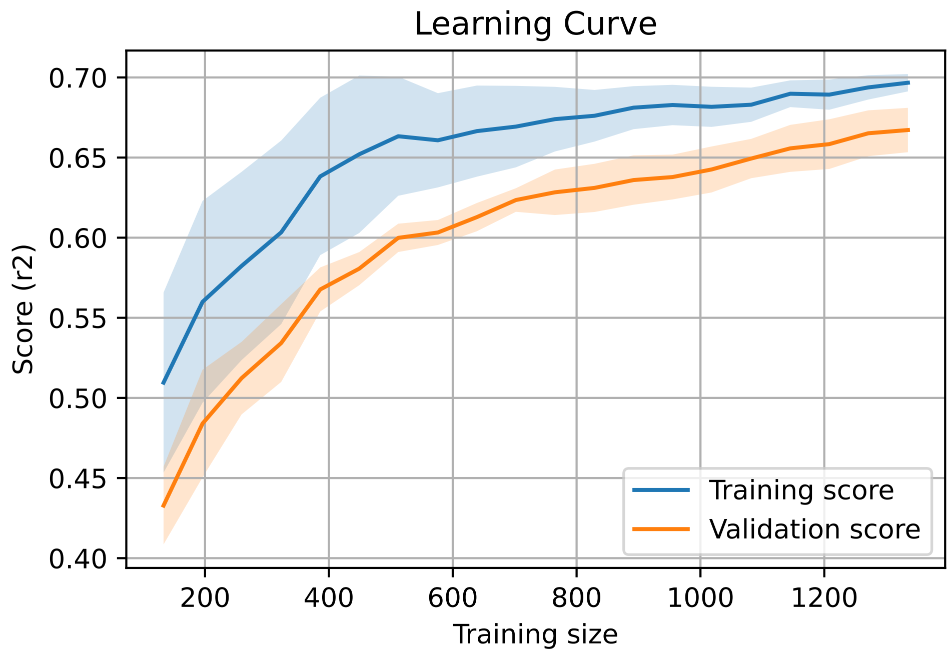
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 11

$\textbf{Len(group)} : 2726$, $\textbf{Len(training)} : 1909$,

$\textbf{Len(training\_real)} : 1337$, $\textbf{Len(test\_of\_training)} : 572$, $\textbf{Len(test)} : 817$



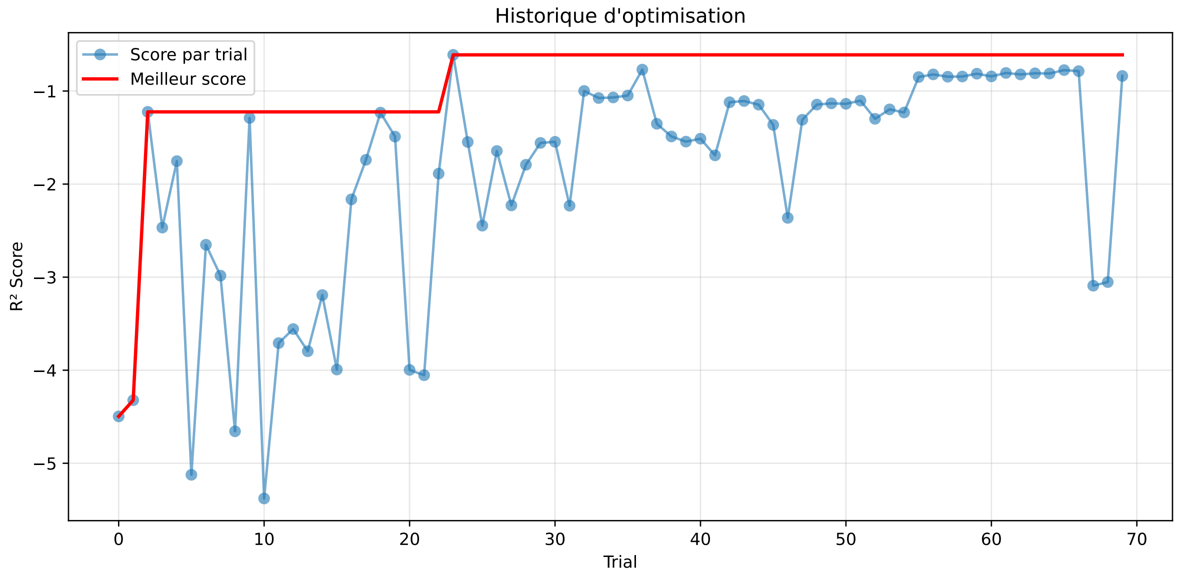
2.5 Model : CatBoost

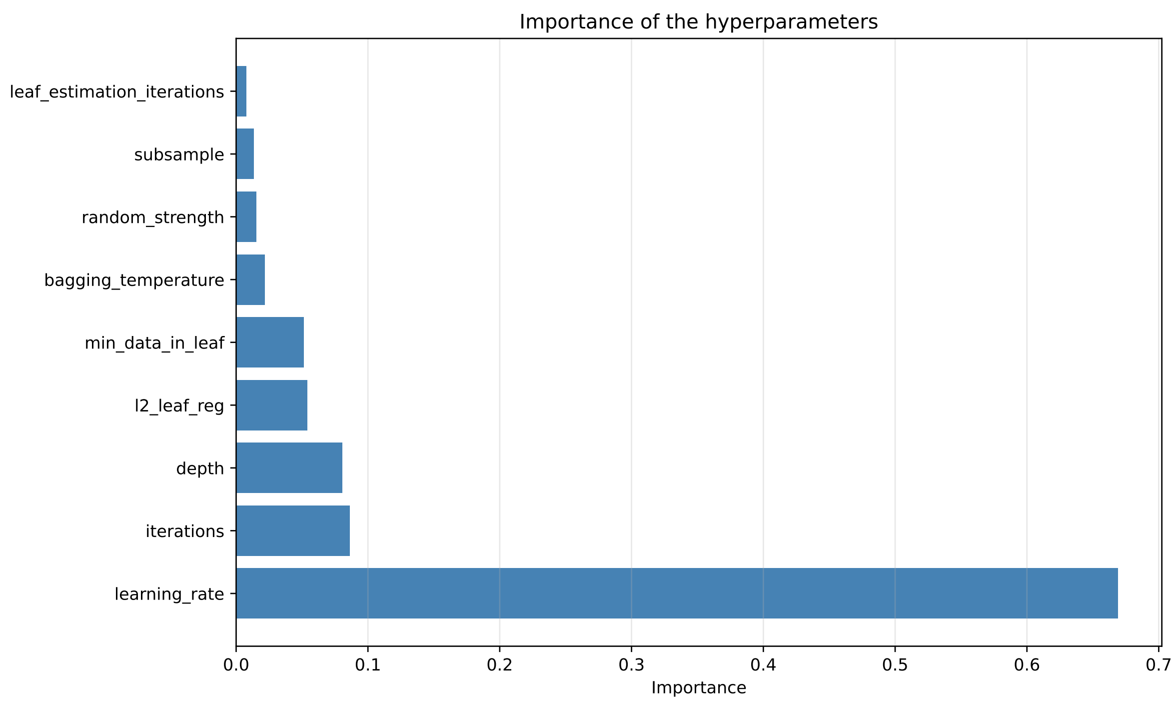
2.5.1 Gridsearch — Group: 3

Len(group) : 1356, **Len(training)** : 950, **Len(test)** : 406

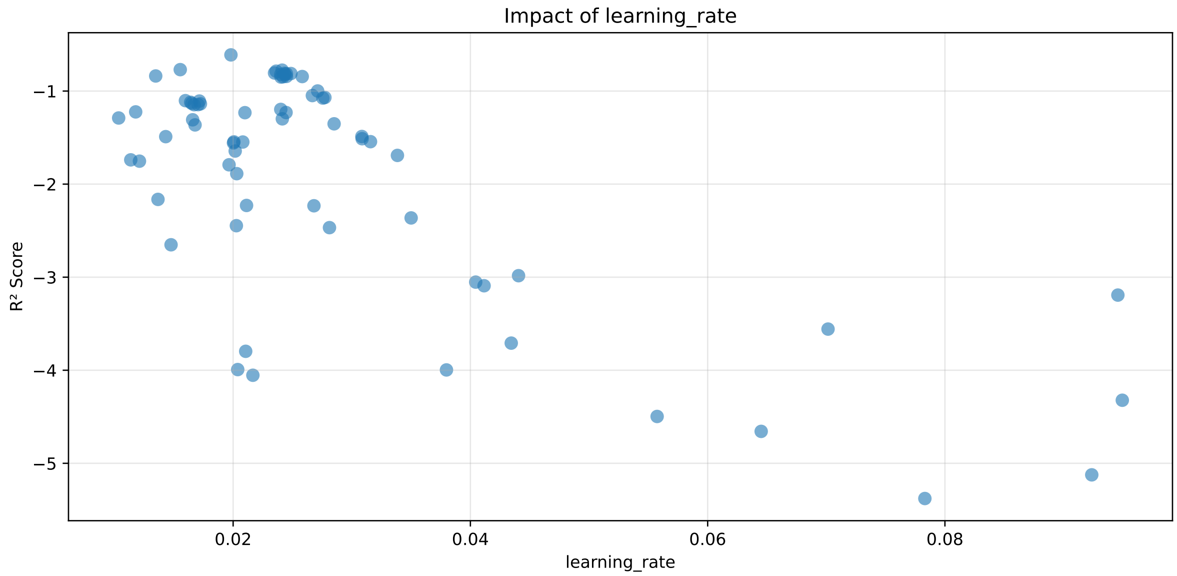
Distribution y_train vs y_test: y_train: mean=2.317, std=0.708, min=1.041, max=6.086 y_test: mean=2.318, std=0.740, min=1.068, max=5.447

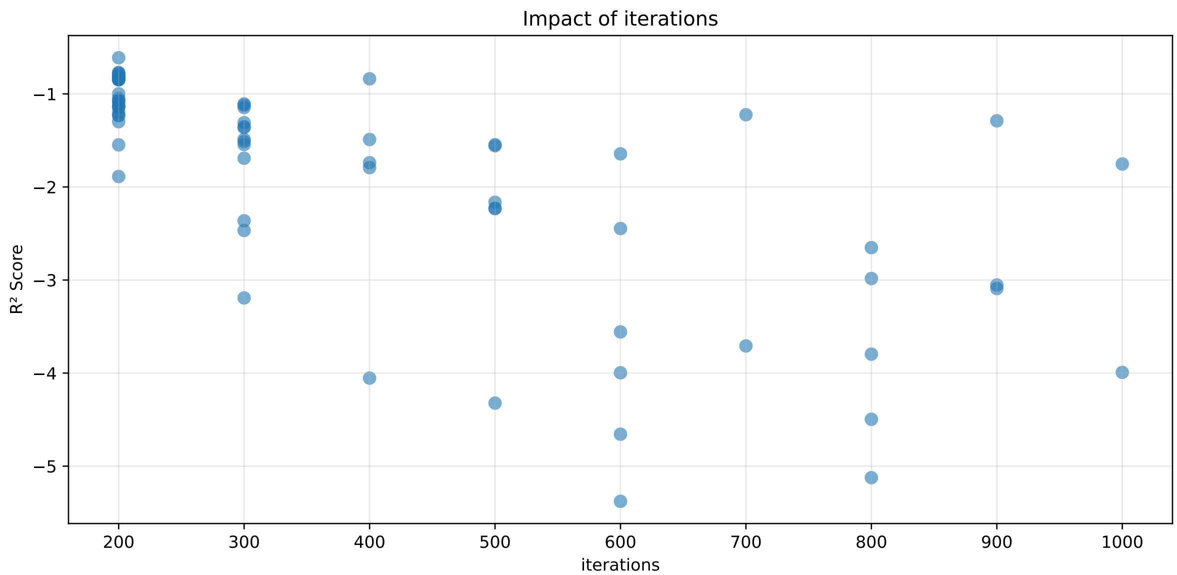
==================== TOP Importance FEATURES ==================== feature importance r_umidity 60.840798 r_NH3 12.993238 r_temperature 10.599043 r_THI 10.073419 r_CO2 5.493502

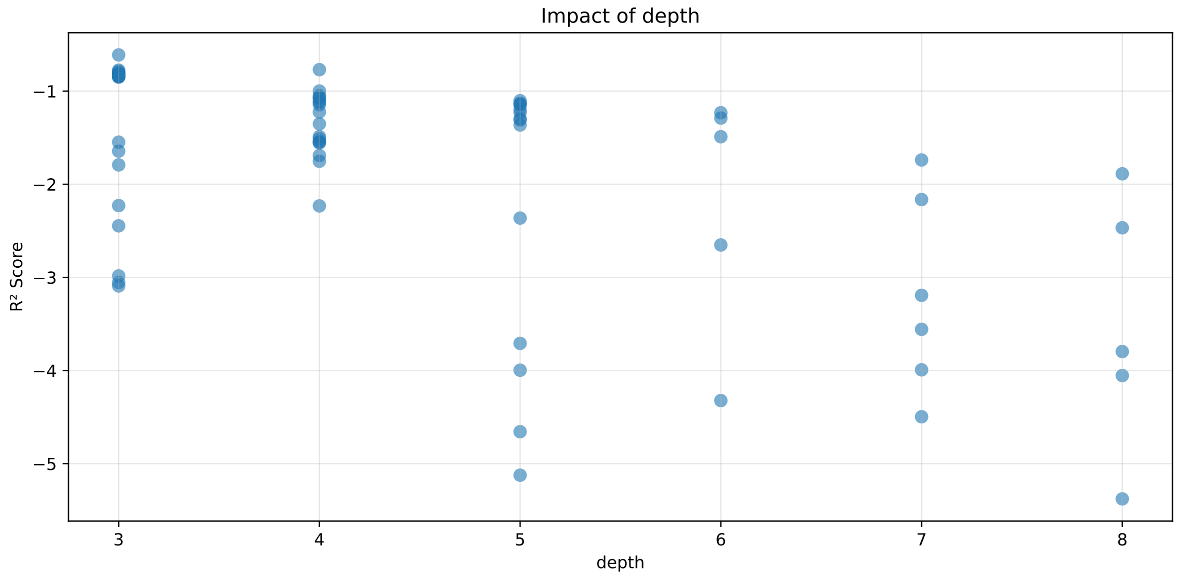


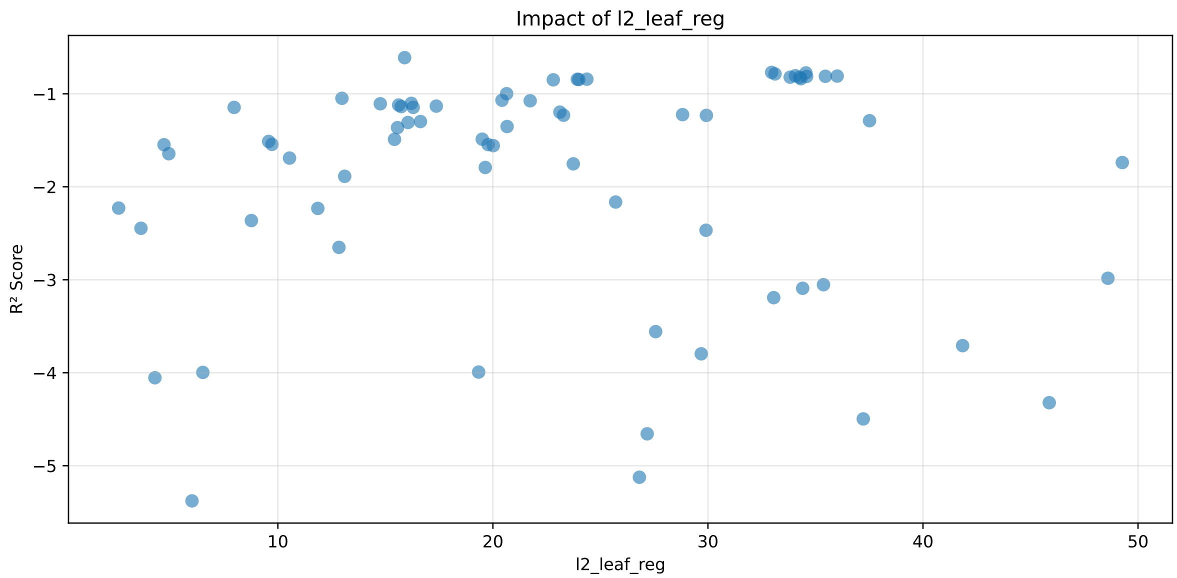


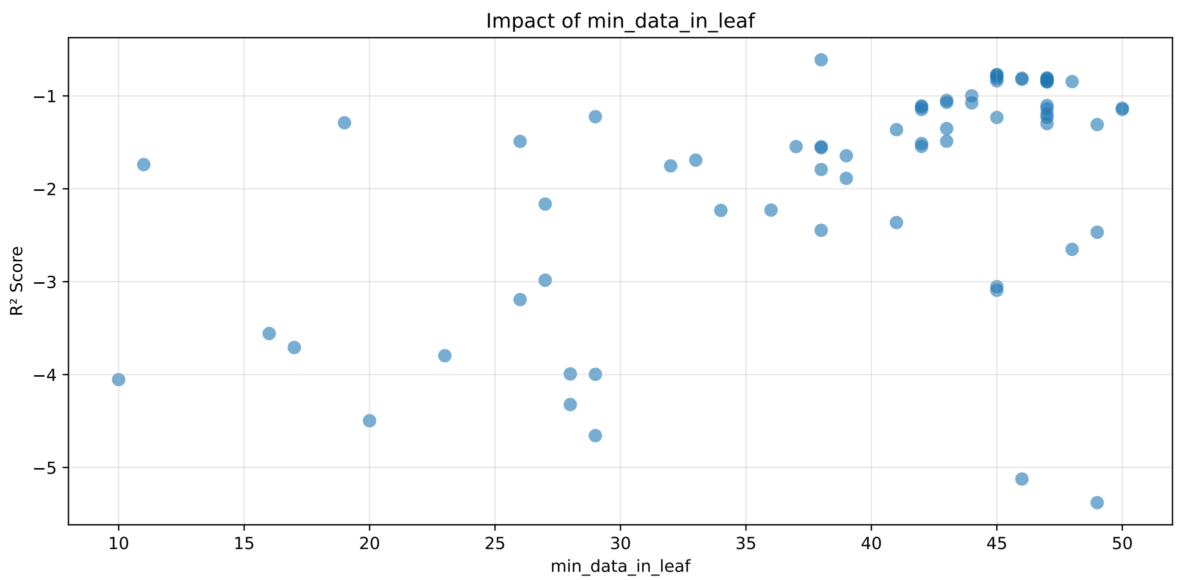
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'depth', 'l2_leaf_reg', 'min_data_in_leaf']


Impact of learning_rate
R² Score
−1
−2
−3
−4
−5
0.02
0.04
0.06
0.08
learning_rate


Impact of iterations
R² Score
−1
−2
−3
−4
−5
200
300
400
500
600
700
800
900
1000
iterations

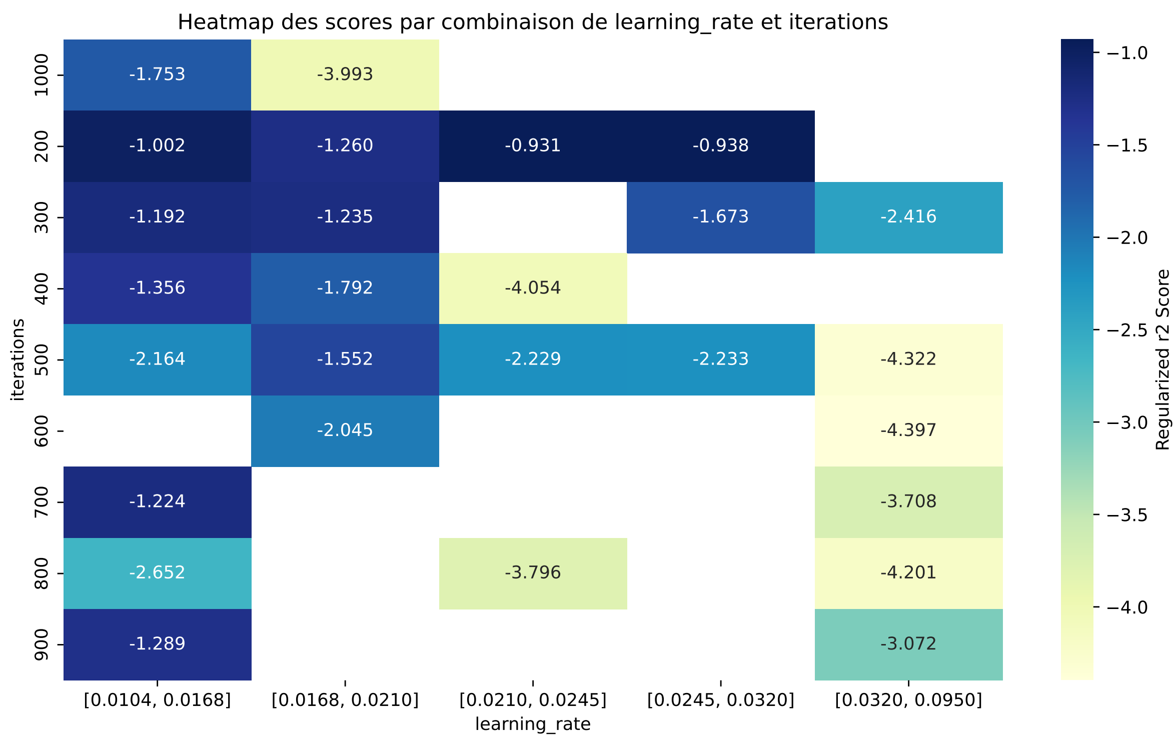

Impact of depth
R² Score
−1
−2
−3
−4
−5
3
4
5
6
7
8
depth


Impact of l2_leaf_reg
R² Score
−1
−2
−3
−4
−5
10
20
30
40
50
l2_leaf_reg




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6587	0.5983	0.6763	0.037	-0.6112	0.6823	0.084	1.1404	1.1011
2	0.6681	0.5997	0.6794	0.0368	-0.7698	0.6827	0.083	1.1385	1.1142
3	0.6712	0.6023	0.6851	0.0368	-0.7752	0.6914	0.0892	1.148	1.1144
4	0.6715	0.6021	0.6854	0.0379	-0.7863	0.6901	0.088	1.1462	1.1153
5	0.675	0.6045	0.6885	0.0373	-0.8057	0.6951	0.0906	1.1499	1.1166
6	0.6788	0.608	0.6912	0.0366	-0.8095	0.6962	0.0883	1.1452	1.1166
7	0.6731	0.6024	0.686	0.037	-0.8118	0.6916	0.0892	1.1481	1.1174
8	0.6771	0.6061	0.6899	0.0369	-0.8137	0.6962	0.0901	1.1487	1.1171
9	0.6775	0.6061	0.6902	0.0368	-0.8212	0.6946	0.0884	1.1459	1.1177
10	0.678	0.6066	0.6906	0.0372	-0.8219	0.6966	0.09	1.1483	1.1178
---	---	---	---	---	---	---	---	---	---
70	0.9971	0.6935	0.743	0.0139	-5.3783	0.7589	0.0653	1.0942	1.4377

Hyperparameters:

Rank 1: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.019815471876070318, 'reg___l2_leaf_reg': 15.895330267054945, 'reg___bagging_temperature': 3.718398311209085, 'reg___random_strength': 4.8909035481841805, 'reg___subsample': 0.6183617702687056, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 9}

Rank 2: {'reg___iterations': 200, 'reg___depth': 4, 'reg___learning_rate': 0.015542810673678895, 'reg___l2_leaf_reg': 32.962667425239545, 'reg___bagging_temperature': 2.8926361285465014, 'reg___random_strength': 3.3553783922807385, 'reg___subsample': 0.5769248468028103, 'reg___min_data_in_leaf': 45, 'reg___leaf_estimation_iterations': 7}

Rank 3: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.024132199560398672, 'reg___l2_leaf_reg': 34.55702108875851, 'reg___bagging_temperature': 4.364448990454567, 'reg___random_strength': 3.969598811491819, 'reg___subsample': 0.5515282912225756, 'reg___min_data_in_leaf': 45, 'reg___leaf_estimation_iterations': 7}

Rank 4: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.023626927448698067, 'reg___l2_leaf_reg': 33.11953148213204, 'reg___bagging_temperature': 2.6152130123591997, 'reg___random_strength': 3.887320064636559, 'reg___subsample': 0.5463591034331259, 'reg___min_data_in_leaf': 45, 'reg___leaf_estimation_iterations': 7}

Rank 5: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.023490062658463547, 'reg___l2_leaf_reg': 34.067287654613075, 'reg___bagging_temperature': 4.776894185370167, 'reg___random_strength': 2.9504335265784998, 'reg___subsample': 0.520736148128282, 'reg___min_data_in_leaf': 47, 'reg___leaf_estimation_iterations': 7}

Rank 6: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.024327211003863588, 'reg___l2_leaf_reg': 36.01166952199049, 'reg___bagging_temperature': 2.514010909858946, 'reg___random_strength': 2.827954536984605, 'reg___subsample': 0.523826806925335, 'reg___min_data_in_leaf': 46, 'reg___leaf_estimation_iterations': 7}

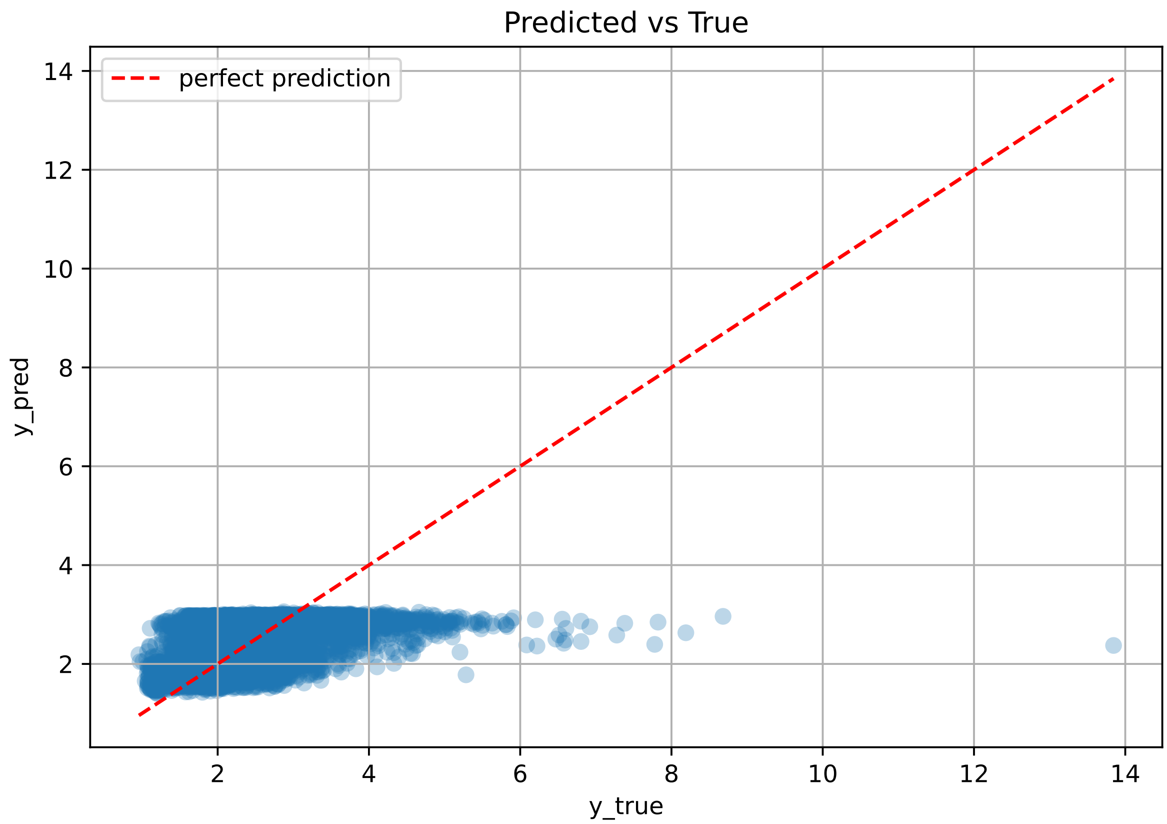
Rank 7: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.02486966994903841, 'reg___l2_leaf_reg': 35.46183587193506, 'reg___bagging_temperature': 2.6101544180948837, 'reg___random_strength': 3.9781849231843367, 'reg___subsample': 0.5239524438060024, 'reg___min_data_in_leaf': 45, 'reg___leaf_estimation_iterations': 7}

Rank 8: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.024491985870727963, 'reg___l2_leaf_reg': 34.589264481284246, 'reg___bagging_temperature': 4.5708393786301915, 'reg___random_strength': 3.0693566123292273, 'reg___subsample': 0.5208075176573284, 'reg___min_data_in_leaf': 47, 'reg___leaf_estimation_iterations': 7}

Rank 9: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.024011708756433287, 'reg___l2_leaf_reg': 33.824953743043494, 'reg___bagging_temperature': 4.675519411058846, 'reg___random_strength': 3.0246297601035628, 'reg___subsample': 0.5556933034015988, 'reg___min_data_in_leaf': 47, 'reg___leaf_estimation_iterations': 7}

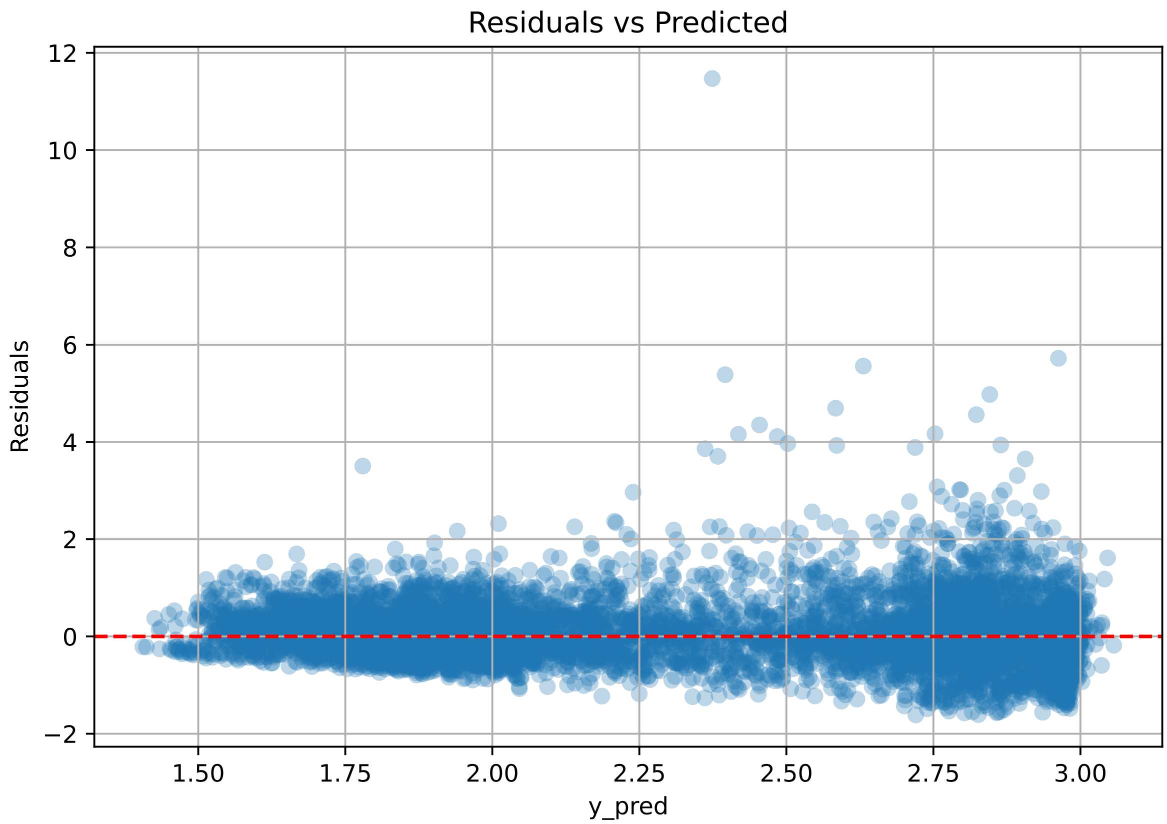
Rank 10: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.024360217361028825, 'reg___l2_leaf_reg': 34.26372505398085, 'reg___bagging_temperature': 4.777702563621042, 'reg___random_strength': 2.9411224956533935, 'reg___subsample': 0.5251166059450129, 'reg___min_data_in_leaf': 46, 'reg___leaf_estimation_iterations': 7}

2.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14
2
4
6
8
10
12
14

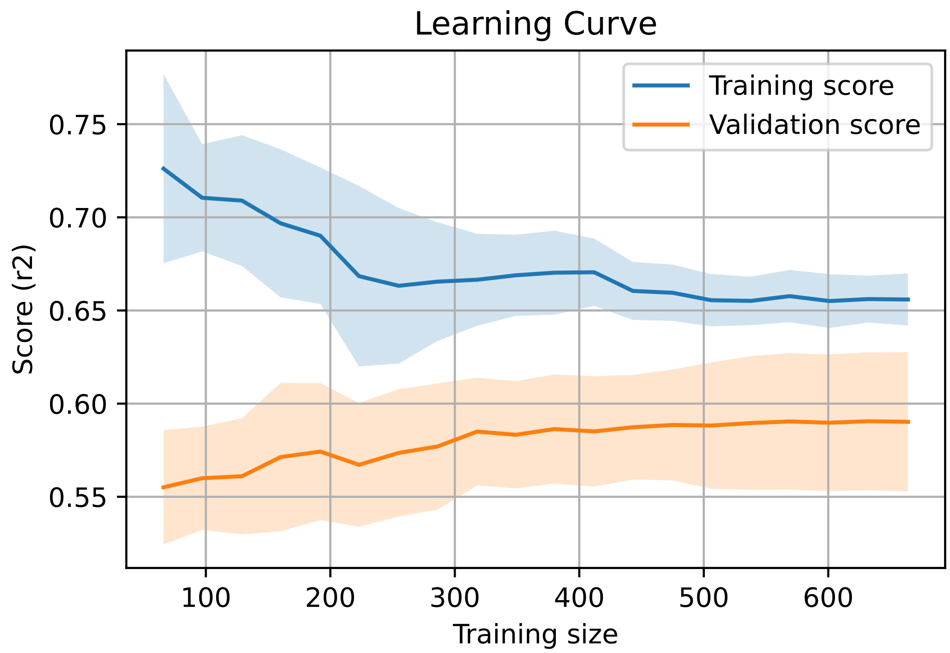
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 3

Len(group) : 1356, **Len(training)** : 950,

Len(training_real) : 665, **Len(test_of_training)** : 285, **Len(test)** : 406

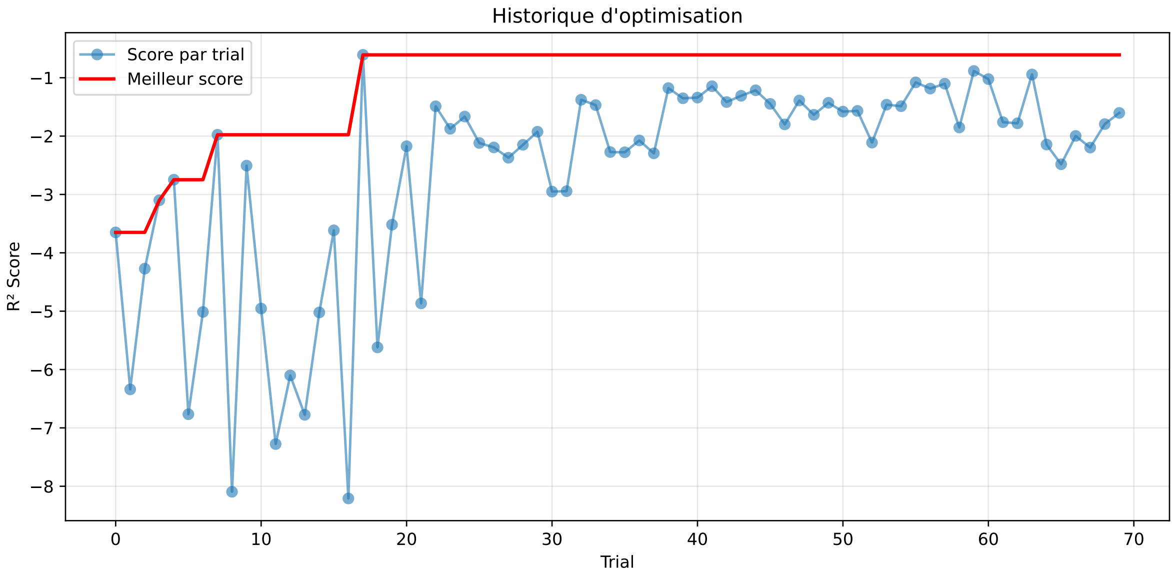


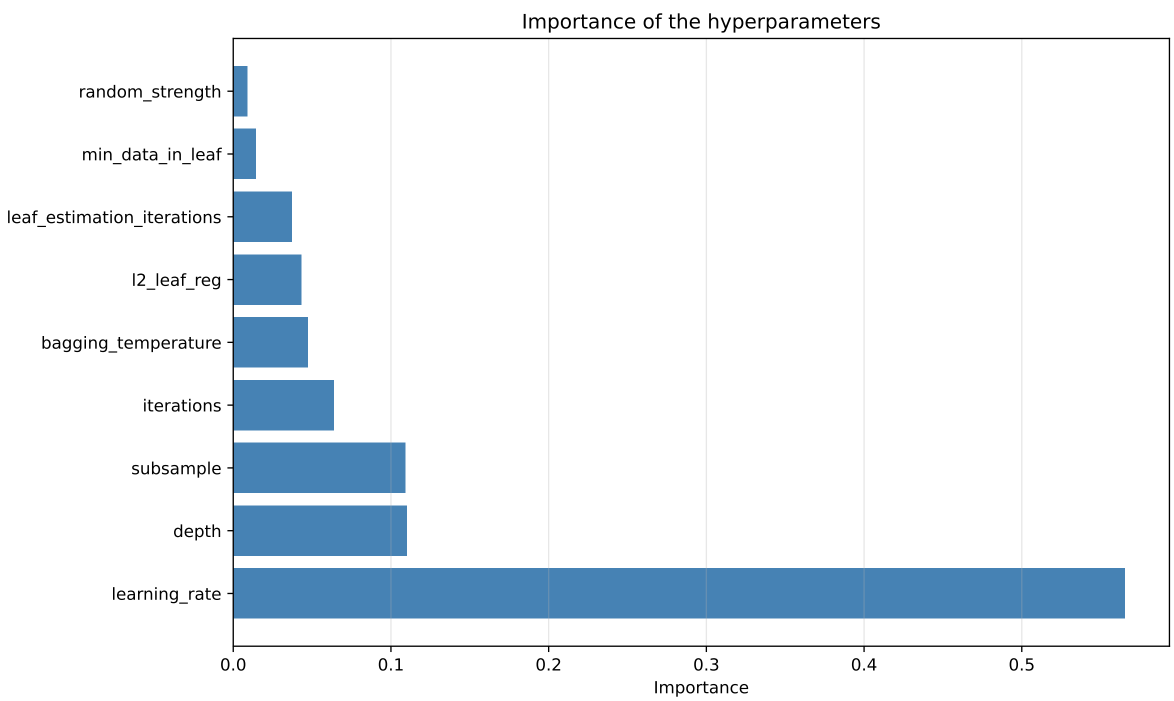
2.5.5 Gridsearch — Group: 4

Len(group) : 1360, **Len(training)** : 952, **Len(test)** : 408

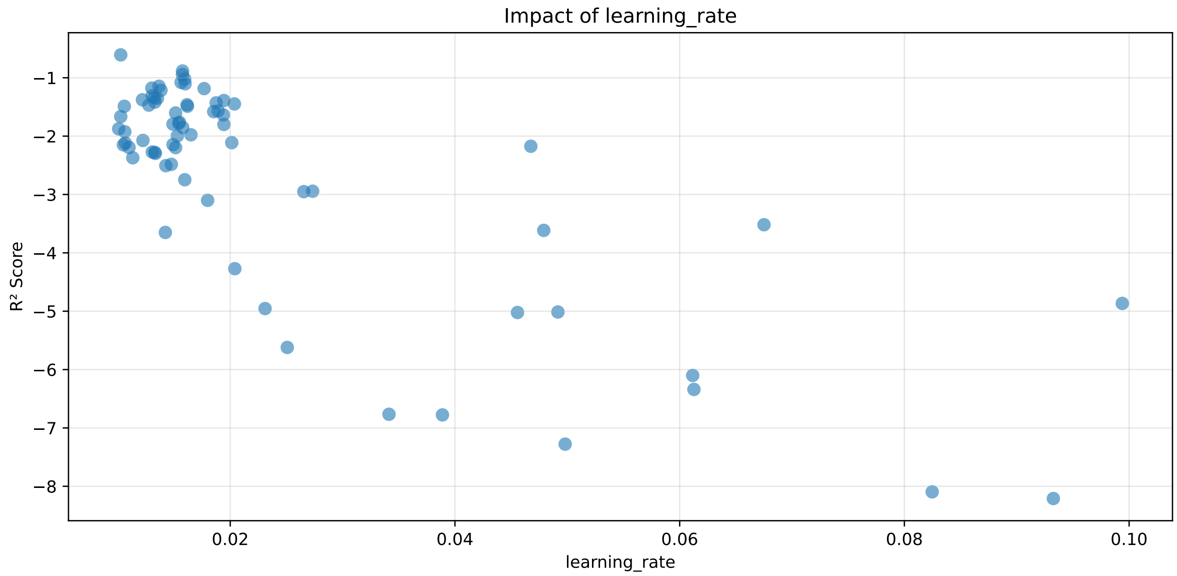
Distribution y_train vs y_test: y_train: mean=2.239, std=0.831, min=0.959, max=7.383 y_test: mean=2.202, std=0.791, min=1.062, max=7.822

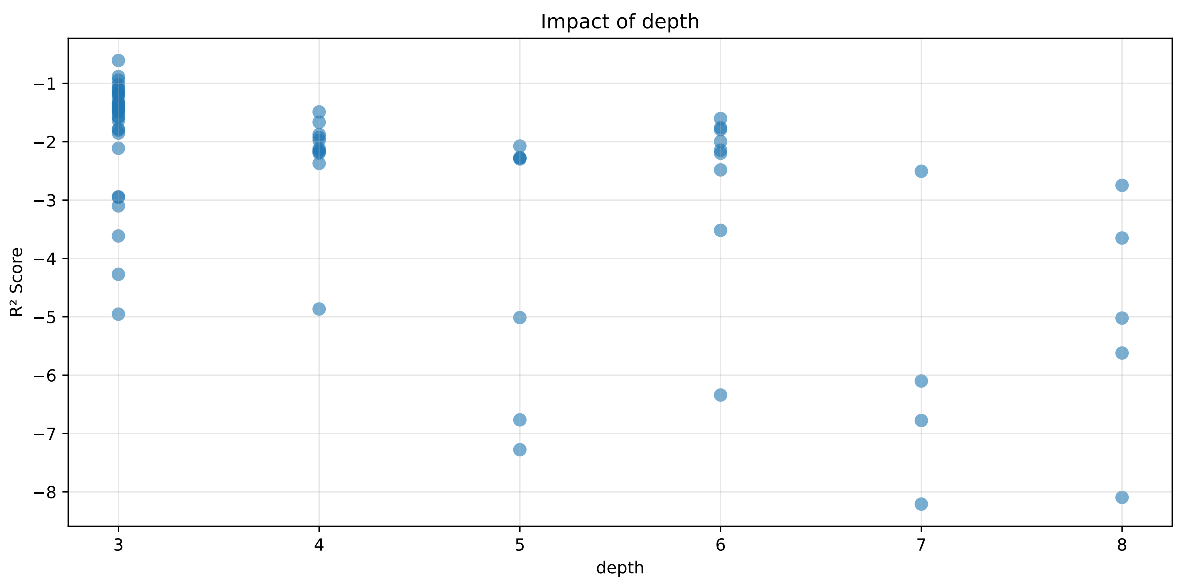
==================== TOP Importance FEATURES ==================== feature importance r_umidity 48.142149 r_THI 23.705168 r_CO2 13.089969 r_NH3 9.746130 r_temperature 5.316584

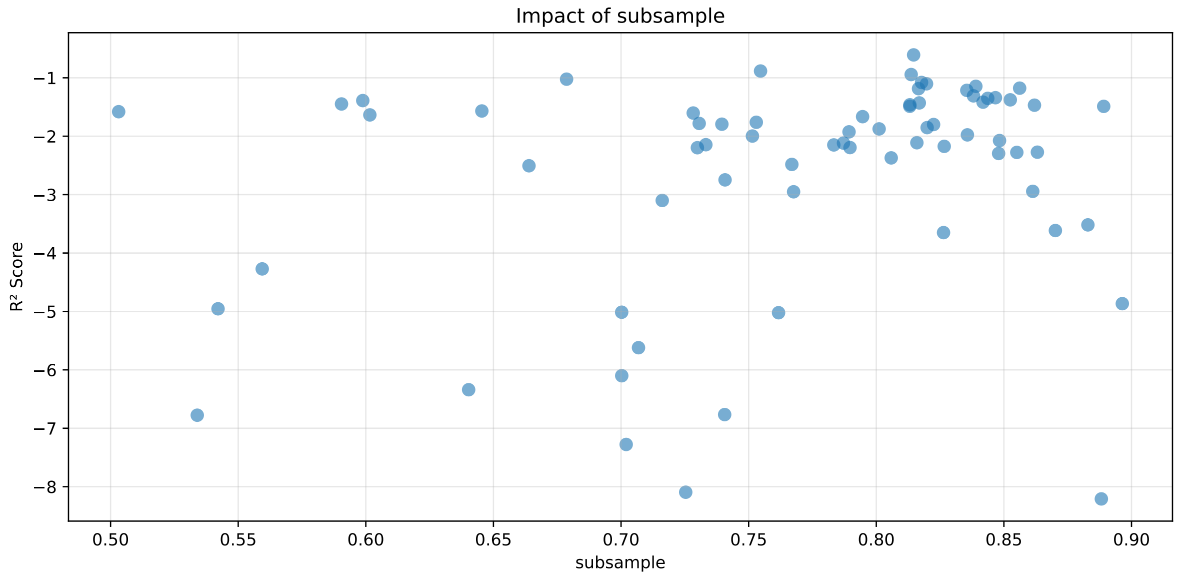


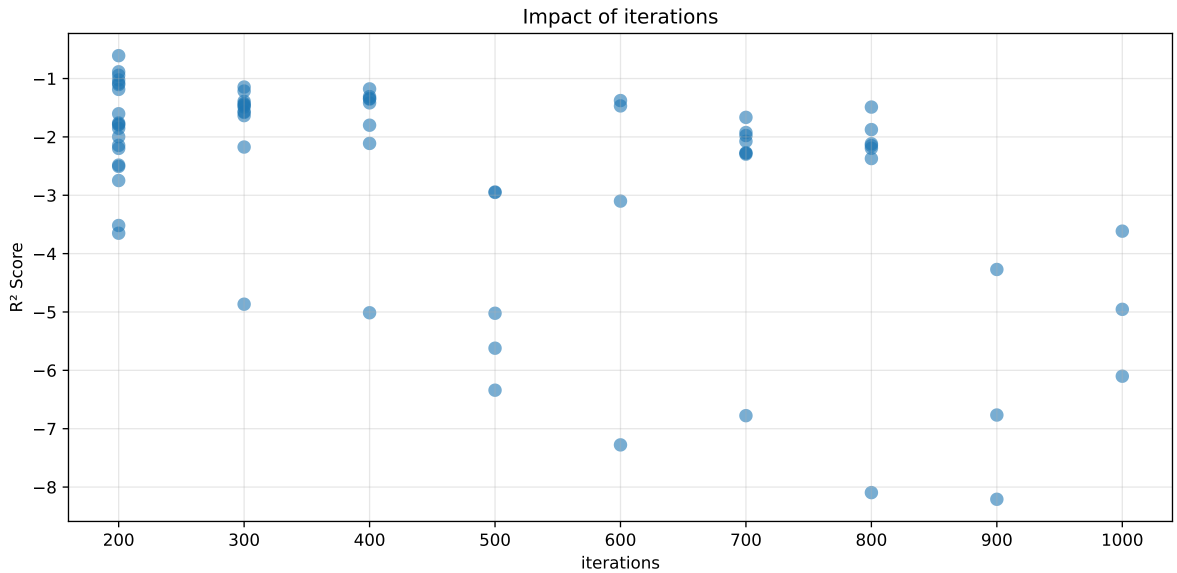


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'subsample', 'iterations']


Impact of learning_rate
R² Score
−1
−2
−3
−4
−5
−6
−7
−8
0.02
0.04
0.06
0.08
0.10
learning_rate

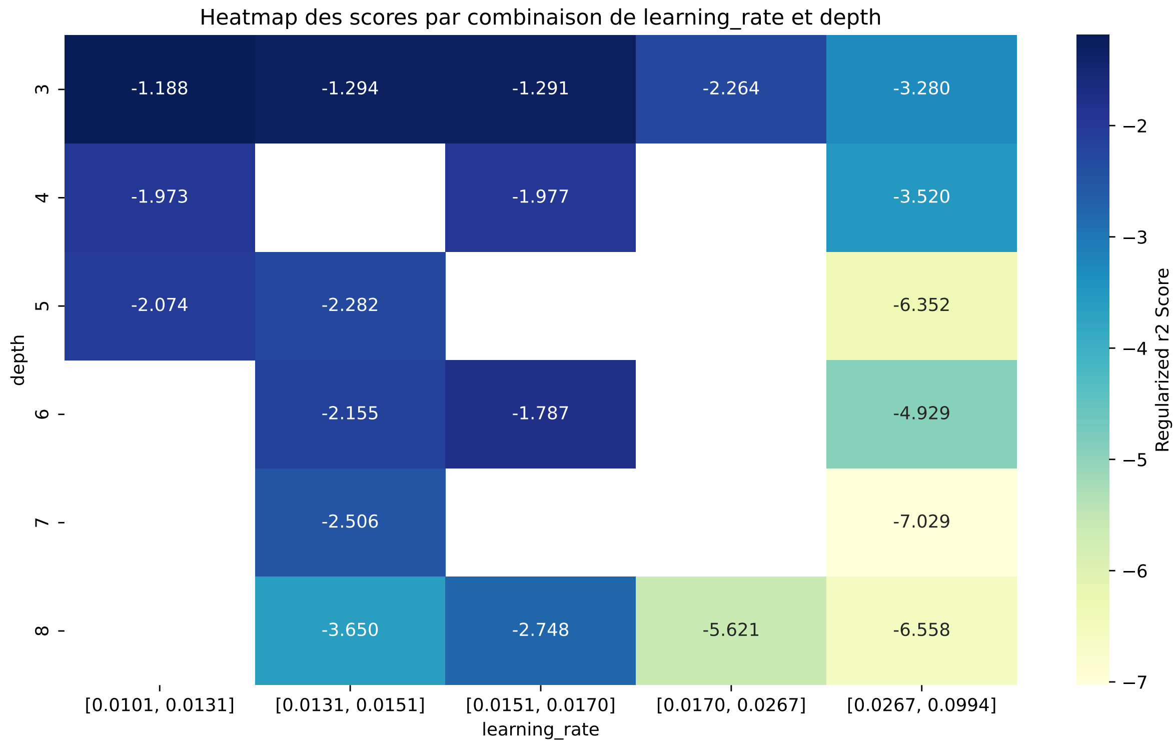

Impact of depth
R² Score
−1
−2
−3
−4
−5
−6
−7
−8
3
4
5
6
7
8
depth






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3751	0.3283	0.3619	0.0339	-0.6081	0.3645	0.0362	1.1102	1.1426
2	0.4405	0.3774	0.4235	0.0409	-0.8842	0.4224	0.045	1.1193	1.1671
3	0.4585	0.3917	0.4326	0.0408	-0.9444	0.4364	0.0447	1.1141	1.1705
4	0.4662	0.3953	0.4387	0.0359	-1.0235	0.4354	0.0401	1.1015	1.1795
5	0.477	0.4028	0.4461	0.0387	-1.0801	0.4489	0.0461	1.1144	1.1841
6	0.4807	0.4053	0.45	0.0391	-1.1028	0.4524	0.047	1.1161	1.186
7	0.4949	0.4168	0.4658	0.0406	-1.1455	0.4707	0.054	1.1295	1.1874
8	0.5115	0.4311	0.4805	0.0391	-1.1774	0.486	0.0549	1.1273	1.1866
9	0.4944	0.4143	0.4586	0.0391	-1.186	0.4655	0.0512	1.1235	1.1931
10	0.5041	0.4223	0.4726	0.0386	-1.215	0.4736	0.0514	1.1217	1.1939
---	---	---	---	---	---	---	---	---	---
70	0.9961	0.5577	0.5949	0.0284	-8.2093	0.6396	0.0818	1.1468	1.786

Hyperparameters:

Rank 1: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.010254252927187894, ’reg__l2_leaf_reg’: 18.968089872452854, ’reg__bagging_temperature’: 2.596657107913388, ’reg__random_strength’: 4.124660473571545, ’reg__subsample’: 0.8146274276951542, ’reg__min_data_in_leaf’: 44, ’reg__leaf_estimation_iterations’: 4}

Rank 2: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.015751213577315425, ’reg__l2_leaf_reg’: 18.375251422964215, ’reg__bagging_temperature’: 1.6226558453401052, ’reg__random_strength’: 4.6563310280528585, ’reg__subsample’: 0.7546559062769557, ’reg__min_data_in_leaf’: 10, ’reg__leaf_estimation_iterations’: 8}

Rank 3: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.015746664830213537, ’reg__l2_leaf_reg’: 18.747965418813283, ’reg__bagging_temperature’: 1.7200656087418071, ’reg__random_strength’: 3.6029409579094804, ’reg__subsample’: 0.8136738453763722, ’reg__min_data_in_leaf’: 37, ’reg__leaf_estimation_iterations’: 8}

Rank 4: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.0159479571632368, ’reg__l2_leaf_reg’: 18.45408506950239, ’reg__bagging_temperature’: 1.6379840808002535, ’reg__random_strength’: 3.5876463591184633, ’reg__subsample’: 0.678661672178789, ’reg__min_data_in_leaf’: 33, ’reg__leaf_estimation_iterations’: 7}

Rank 5: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.015624797416680256, ’reg__l2_leaf_reg’: 17.939633424628326, ’reg__bagging_temperature’: 1.712560926179686, ’reg__random_strength’: 2.877949570615288, ’reg__subsample’: 0.8177404525790879, ’reg__min_data_in_leaf’: 47, ’reg__leaf_estimation_iterations’: 7}

Rank 6: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.01598277395986005, ’reg__l2_leaf_reg’: 18.91549083034821, ’reg__bagging_temperature’: 1.6644867471053937, ’reg__random_strength’: 2.8742094766872244, ’reg__subsample’: 0.8196942727010271, ’reg__min_data_in_leaf’: 48, ’reg__leaf_estimation_iterations’: 7}

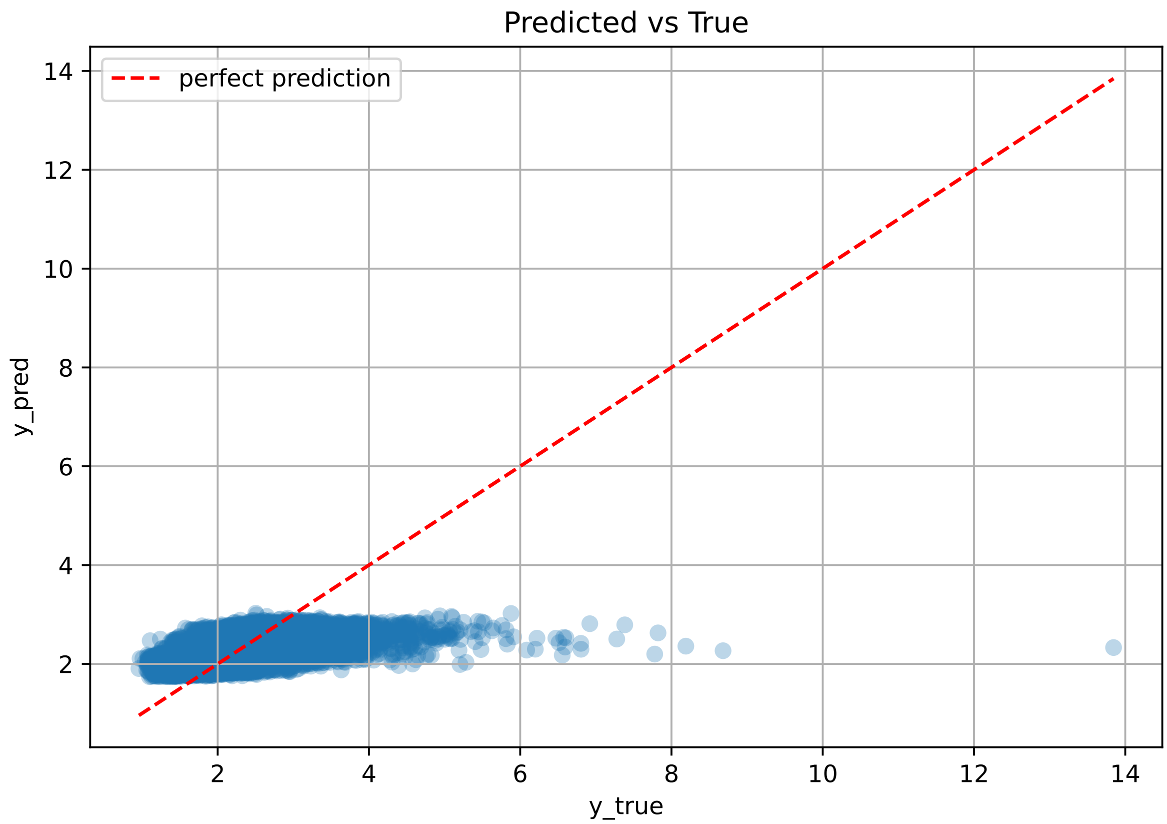
Rank 7: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.0136590021110927, ’reg__l2_leaf_reg’: 25.03696112811747, ’reg__bagging_temperature’: 2.3339398066061983, ’reg__random_strength’: 3.8706808712937333, ’reg__subsample’: 0.8390115082801198, ’reg__min_data_in_leaf’: 46, ’reg__leaf_estimation_iterations’: 6}

Rank 8: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.01301834469883348, ’reg__l2_leaf_reg’: 41.15096069272615, ’reg__bagging_temperature’: 2.360248759002039, ’reg__random_strength’: 4.471714775870632, ’reg__subsample’: 0.856176325914334, ’reg__min_data_in_leaf’: 37, ’reg__leaf_estimation_iterations’: 6}

Rank 9: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.01767634506924535, ’reg__l2_leaf_reg’: 33.85008963783647, ’reg__bagging_temperature’: 3.472645949728096, ’reg__random_strength’: 2.8517416382013234, ’reg__subsample’: 0.8165472039899373, ’reg__min_data_in_leaf’: 48, ’reg__leaf_estimation_iterations’: 7}

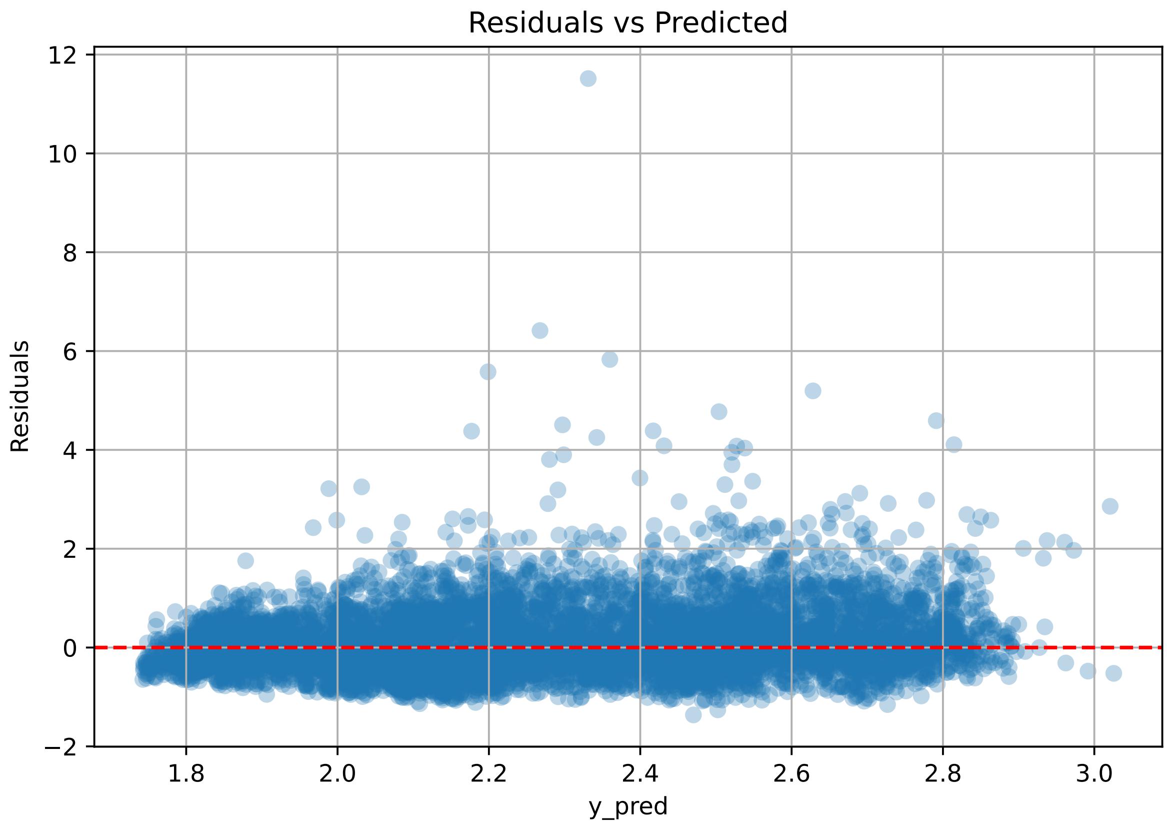
Rank 10: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.013834513035774272, ’reg__l2_leaf_reg’: 1.0853196835760528, ’reg__bagging_temperature’: 2.3358977445857416, ’reg__random_strength’: 3.7664439862915966, ’reg__subsample’: 0.8354714774021443, ’reg__min_data_in_leaf’: 45, ’reg__leaf_estimation_iterations’: 6}

2.5.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

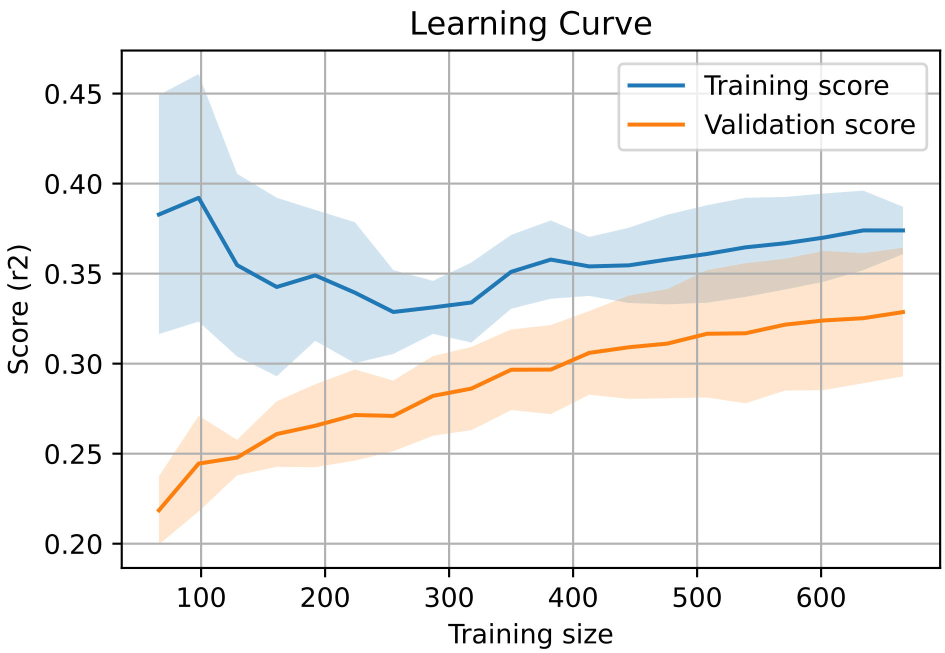
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 4

Len(group) : 1360, **Len(training)** : 952,

Len(training_real) : 667, **Len(test_of_training)** : 285, **Len(test)** : 408

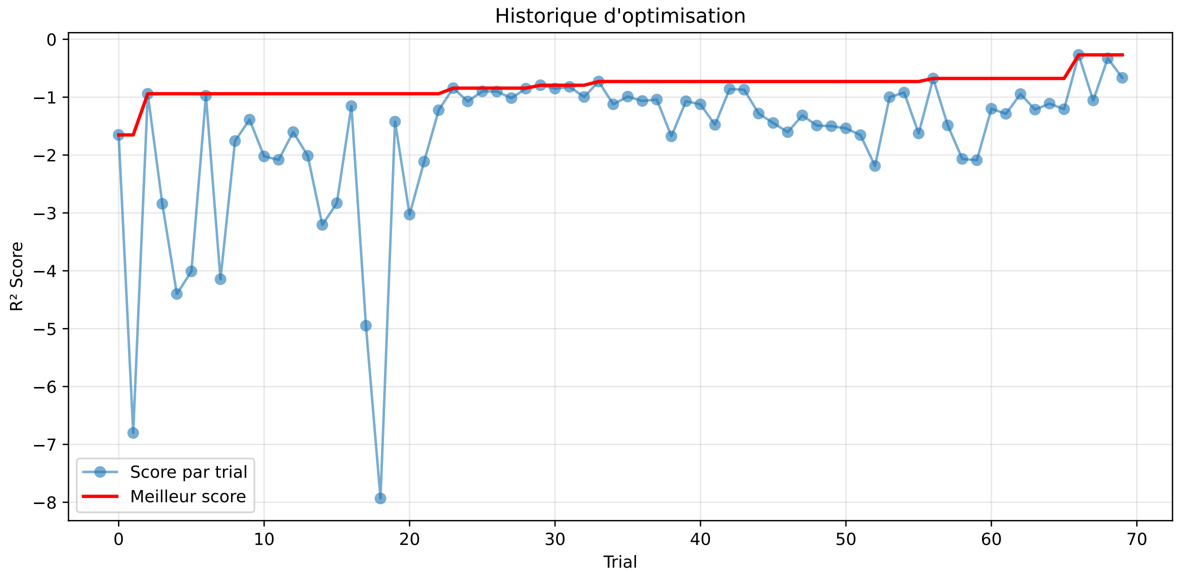


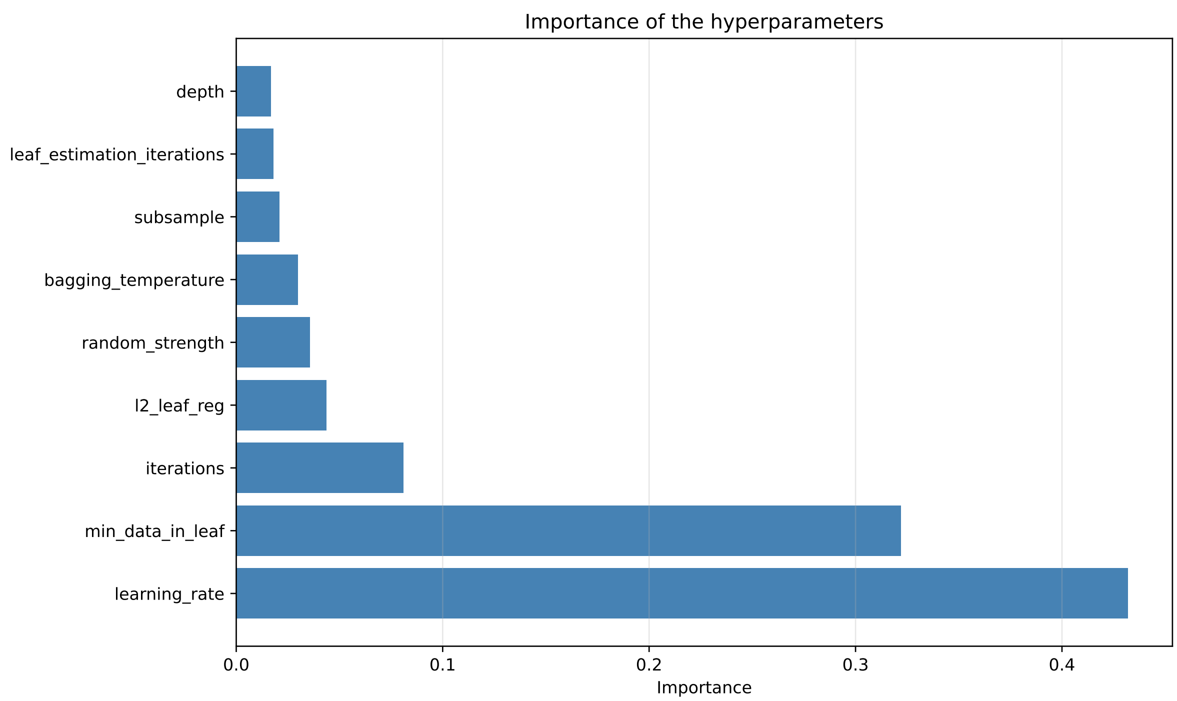
2.5.9 Gridsearch — Group: 5

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

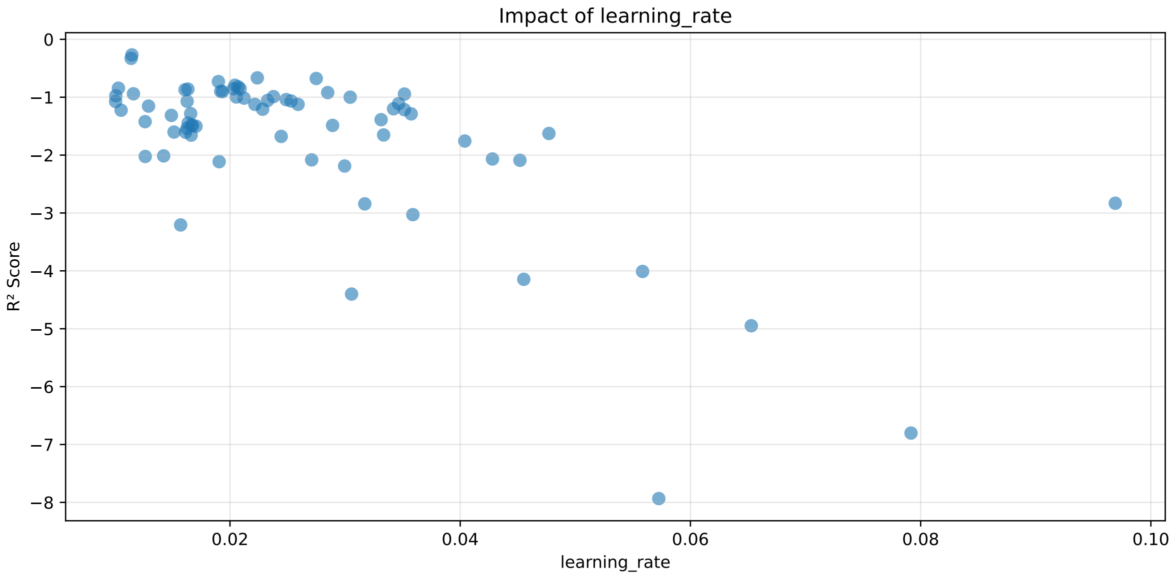
Distribution y_train vs y_test: y_train: mean=2.238, std=0.681, min=0.973, max=5.813 y_test: mean=2.212, std=0.659, min=1.071, max=4.858

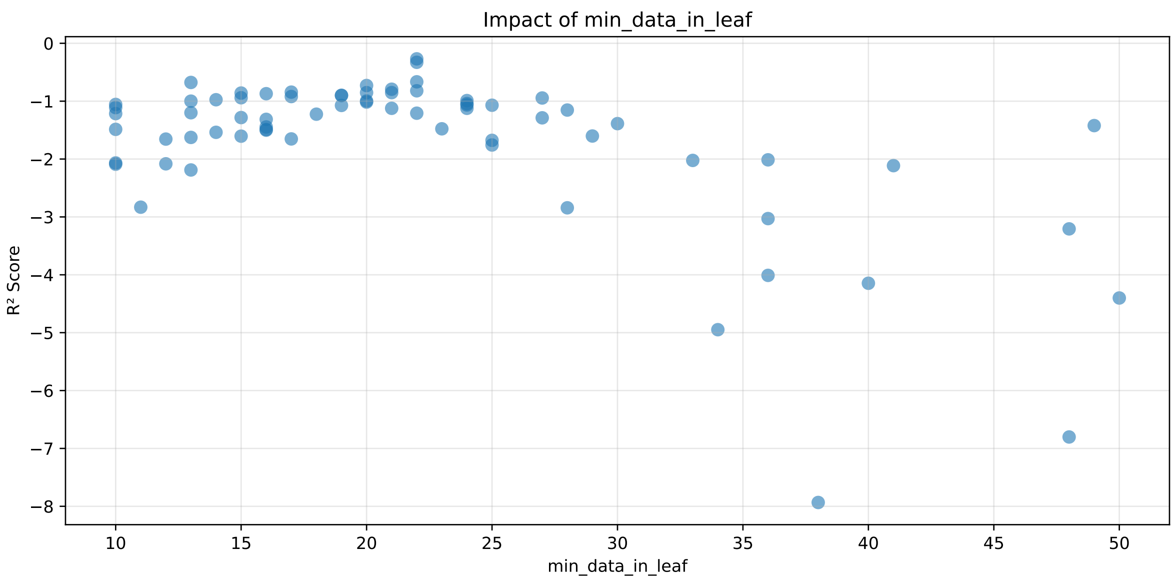
==================== TOP Importance FEATURES ==================== feature importance r_umidity 59.407744 r_temperature 13.200717 r_NH3 11.123909 r_CO2 10.632902 r_THI 5.634728

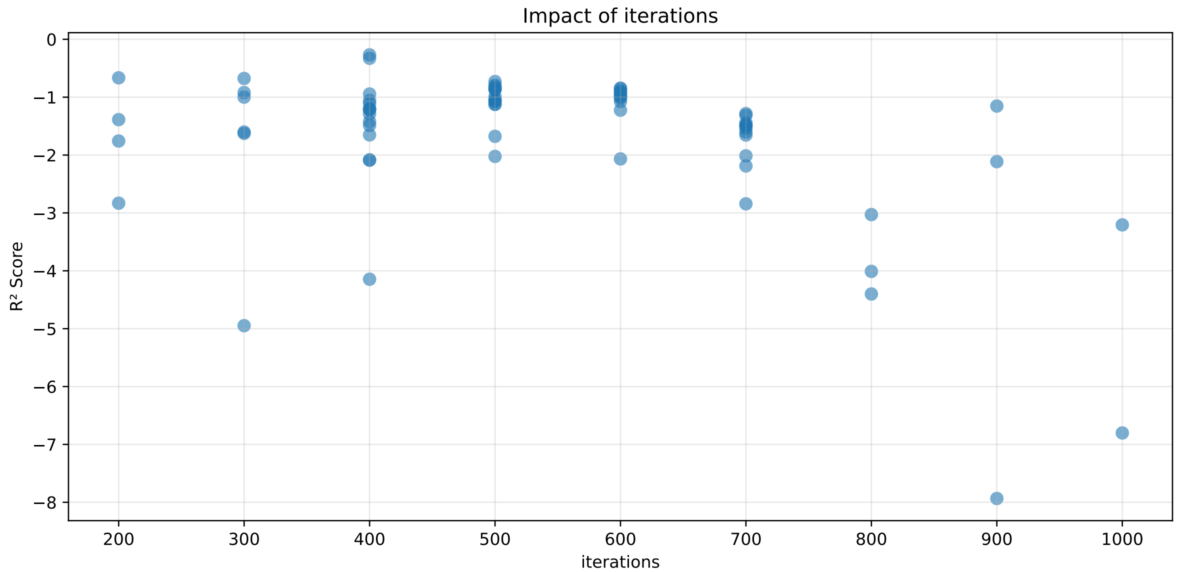




Paramètres avec >5% d'importance : ['learning_rate', 'min_data_in_leaf', 'iterations']

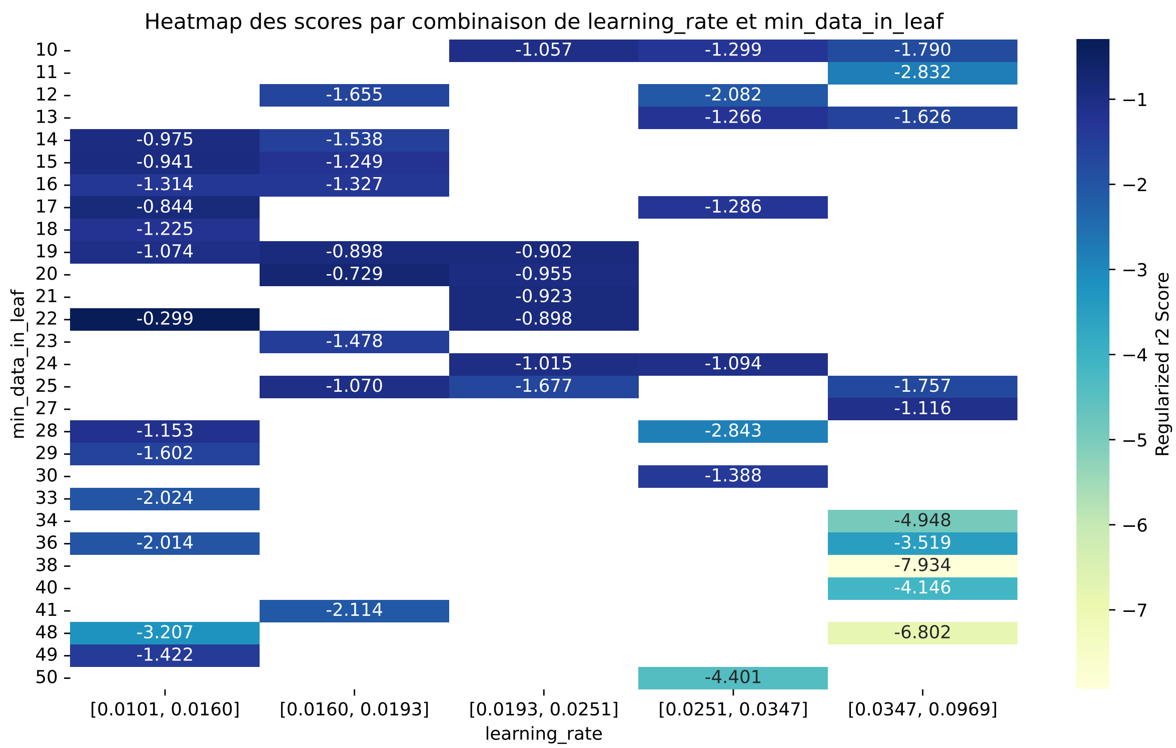

Impact of learning_rate
R² Score
learning_rate
0.02
0.04
0.06
0.08
0.10
0
−1
−2
−3
−4
−5
−6
−7
−8


Impact of min_data_in_leaf
R² Score
min_data_in_leaf
10
15
20
25
30
35
40
45
50
0
−1
−2
−3
−4
−5
−6
−7
−8



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.478	0.4425	0.5045	0.0218	-0.2687	0.5096	0.0671	1.1518	1.0804
2	0.488	0.4491	0.512	0.0225	-0.3285	0.5173	0.0682	1.1519	1.0866
3	0.5368	0.4795	0.5384	0.0202	-0.6655	0.5473	0.0677	1.1413	1.1194
4	0.5357	0.478	0.5406	0.024	-0.6765	0.544	0.066	1.138	1.1208
5	0.5431	0.4825	0.5454	0.0245	-0.7294	0.551	0.0685	1.1419	1.1256
6	0.5485	0.4846	0.5478	0.0237	-0.7935	0.5547	0.0701	1.1446	1.1319
7	0.5507	0.4854	0.5489	0.0239	-0.8208	0.5558	0.0704	1.145	1.1346
8	0.5602	0.4933	0.5555	0.0212	-0.8437	0.5618	0.0685	1.1389	1.1355
9	0.5524	0.4856	0.5497	0.0237	-0.8521	0.5569	0.0714	1.147	1.1377
10	0.5498	0.483	0.5464	0.0231	-0.8524	0.5544	0.0714	1.1478	1.1382
---	---	---	---	---	---	---	---	---	---
70	0.9844	0.5597	0.6021	0.0303	-7.9339	0.6256	0.0659	1.1177	1.7587

Hyperparameters:

Rank 1: {'reg___iterations': 400, 'reg___depth': 4, 'reg___learning_rate': 0.011478151537954952, 'reg___l2_leaf_reg': 35.562434761279405, 'reg___bagging_temperature': 1.488983572158661, 'reg___random_strength': 4.997758907785784, 'reg___subsample': 0.8863131906076801, 'reg___min_data_in_leaf': 22, 'reg___leaf_estimation_iterations': 1}

Rank 2: {'reg___iterations': 400, 'reg___depth': 4, 'reg___learning_rate': 0.01141421439284256, 'reg___l2_leaf_reg': 23.470703039622023, 'reg___bagging_temperature': 1.638121524685228, 'reg___random_strength': 4.567116874043977, 'reg___subsample': 0.876024228259785, 'reg___min_data_in_leaf': 22, 'reg___leaf_estimation_iterations': 1}

Rank 3: {'reg___iterations': 200, 'reg___depth': 7, 'reg___learning_rate': 0.022374250770912745, 'reg___l2_leaf_reg': 23.3776947724456, 'reg___bagging_temperature': 1.5728453599365513, 'reg___random_strength': 3.523802042315689, 'reg___subsample': 0.5838318081675629, 'reg___min_data_in_leaf': 22, 'reg___leaf_estimation_iterations': 1}

Rank 4: {'reg___iterations': 300, 'reg___depth': 4, 'reg___learning_rate': 0.027493349345591472, 'reg___l2_leaf_reg': 25.512707738508187, 'reg___bagging_temperature': 2.1686918436548233, 'reg___random_strength': 4.67255762925713, 'reg___subsample': 0.8994305455419213, 'reg___min_data_in_leaf': 13, 'reg___leaf_estimation_iterations': 1}

Rank 5: {'reg___iterations': 500, 'reg___depth': 4, 'reg___learning_rate': 0.018993467952229077, 'reg___l2_leaf_reg': 30.229381335195303, 'reg___bagging_temperature': 2.1113252407589314, 'reg___random_strength': 4.67460267454618, 'reg___subsample': 0.8986163572430247, 'reg___min_data_in_leaf': 20, 'reg___leaf_estimation_iterations': 1}

Rank 6: {'reg___iterations': 500, 'reg___depth': 4, 'reg___learning_rate': 0.02042895803146675, 'reg___l2_leaf_reg': 30.183121977026378, 'reg___bagging_temperature': 1.886858995055542, 'reg___random_strength': 4.933264770019834, 'reg___subsample': 0.8979689556318682, 'reg___min_data_in_leaf': 21, 'reg___leaf_estimation_iterations': 1}

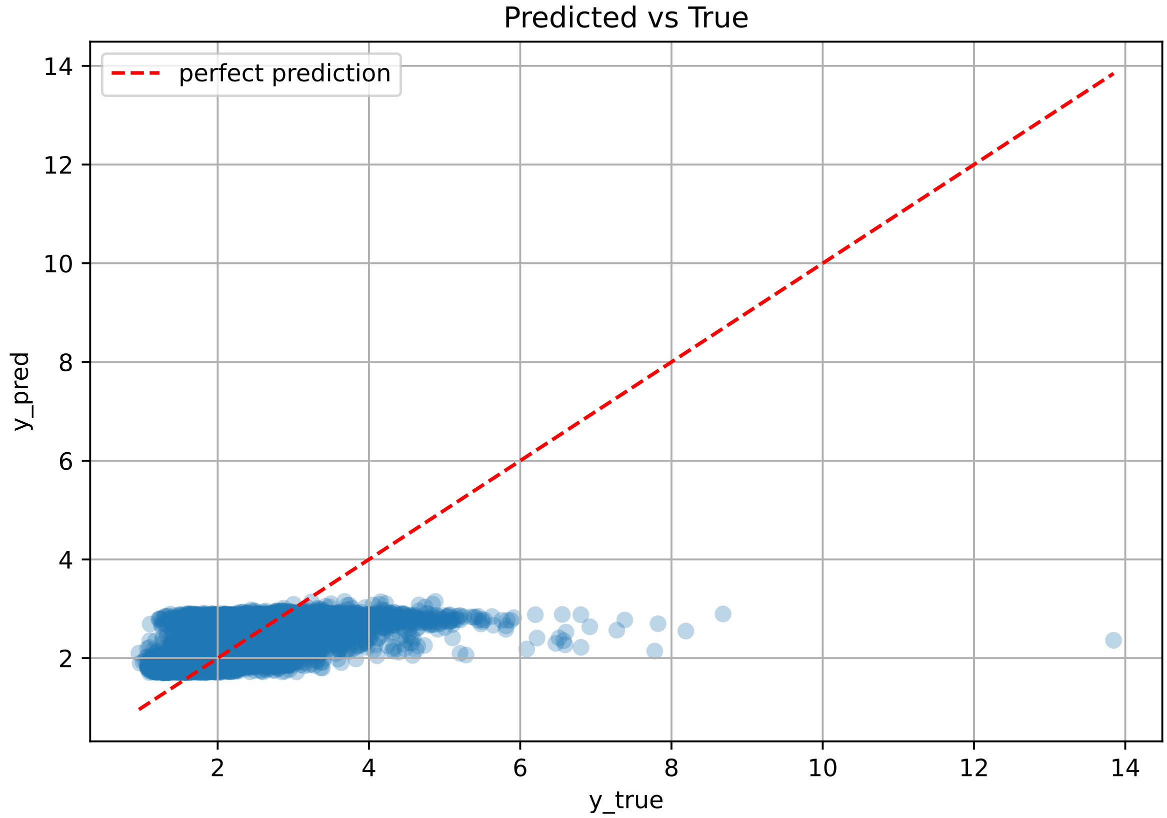
Rank 7: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.020710742366584226, 'reg__l2_leaf_reg': 29.61494927329012, 'reg__bagging_temperature': 1.9254050024917881, 'reg__random_strength': 4.9229880793521925, 'reg__subsample': 0.8920586931353741, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 1}

Rank 8: {'reg__iterations': 600, 'reg__depth': 6, 'reg__learning_rate': 0.010307621342193117, 'reg__l2_leaf_reg': 35.39293758595118, 'reg__bagging_temperature': 2.543440088733146, 'reg__random_strength': 4.637456201917818, 'reg__subsample': 0.8836553124034459, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 2}

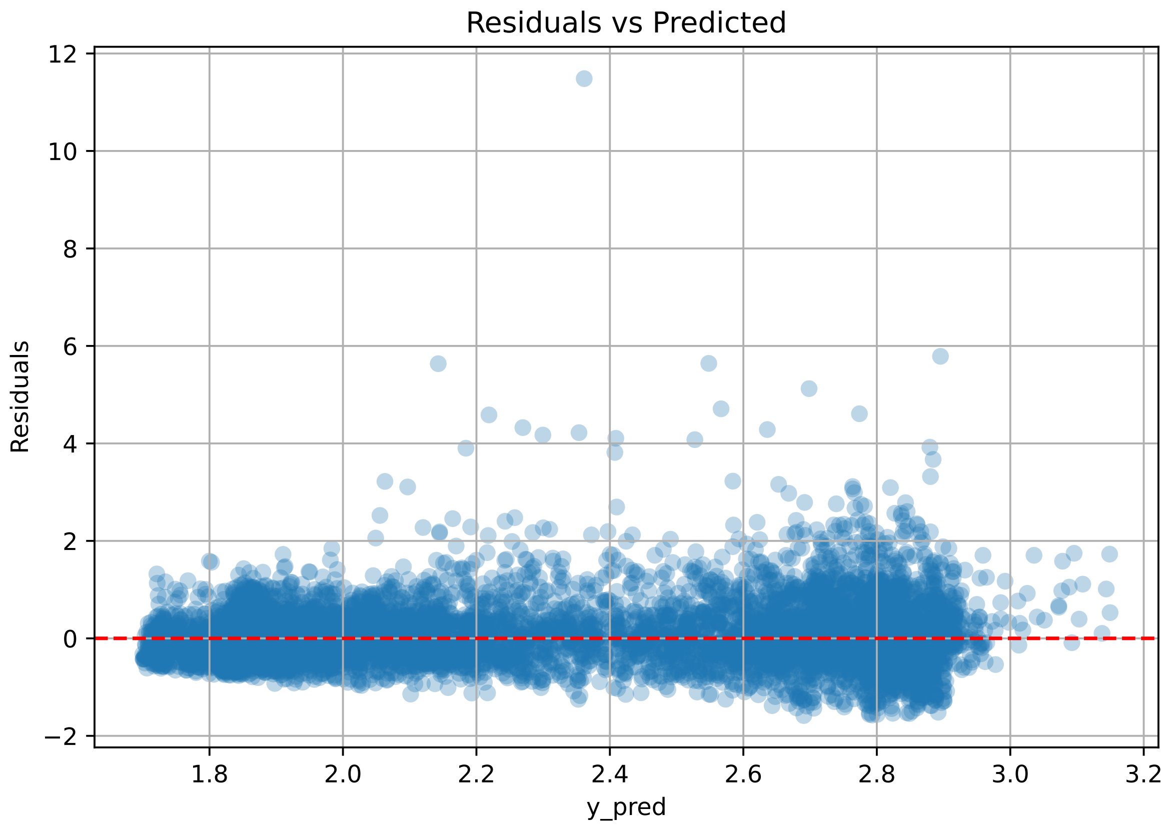
Rank 9: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.020877695399644833, 'reg__l2_leaf_reg': 28.50302925613448, 'reg__bagging_temperature': 1.8948597951792658, 'reg__random_strength': 4.655030798881805, 'reg__subsample': 0.8983763644035503, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 1}

Rank 10: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.020308383328901045, 'reg__l2_leaf_reg': 49.81568897120886, 'reg__bagging_temperature': 1.9582507526181416, 'reg__random_strength': 4.8019986810987945, 'reg__subsample': 0.8984623009467342, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1}

2.5.10 Scatter



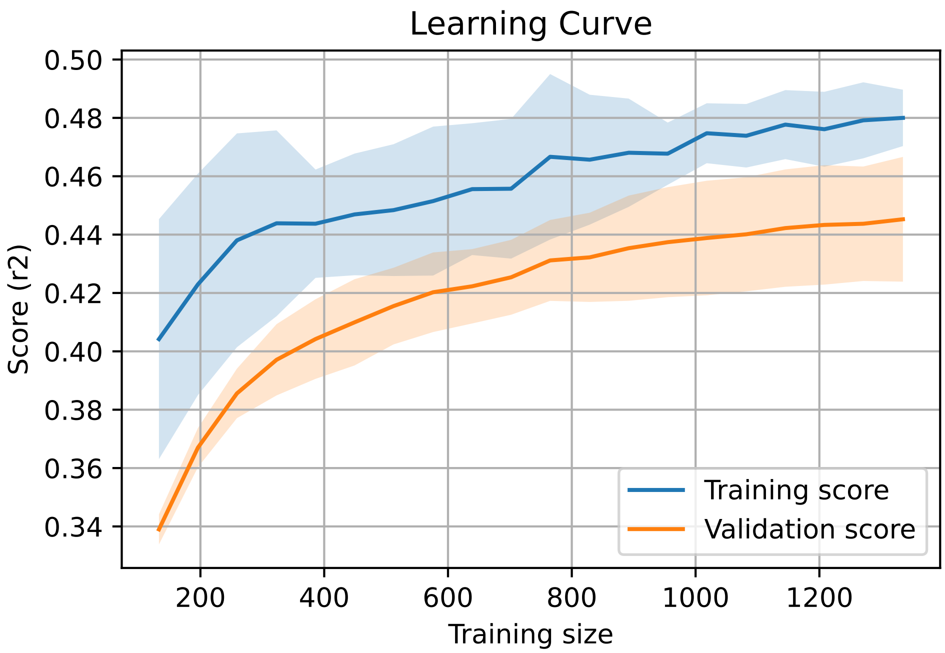
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 5

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817

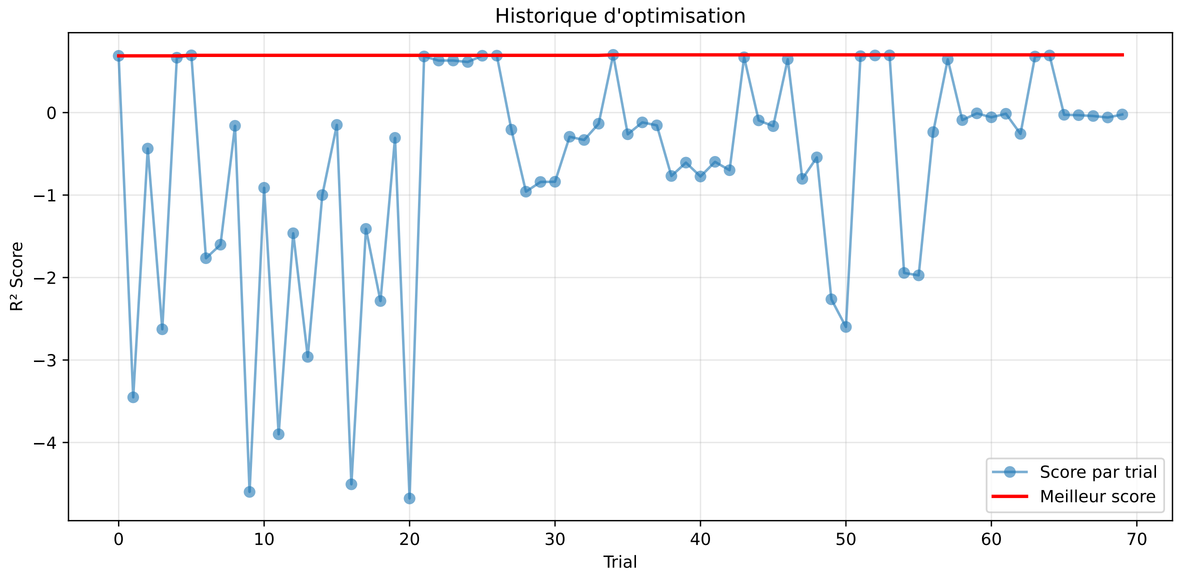


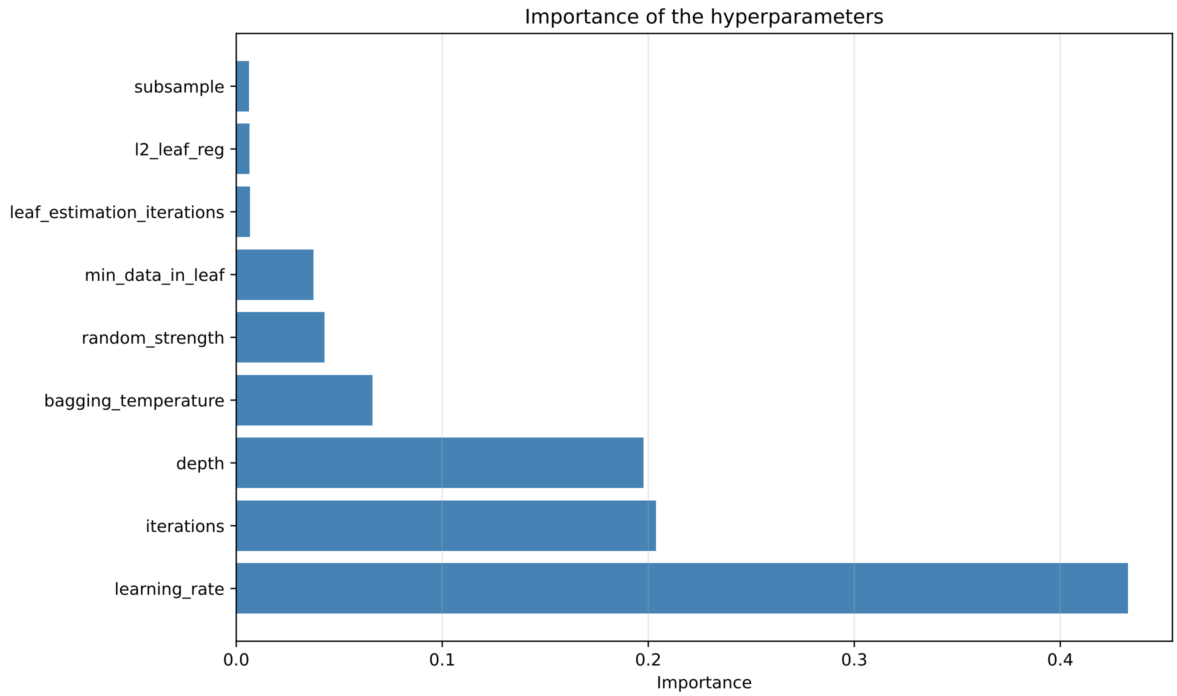
2.5.13 Gridsearch — Group: 11

Len(group) : 2726, **Len(training)** : 1909, **Len(test)** : 817

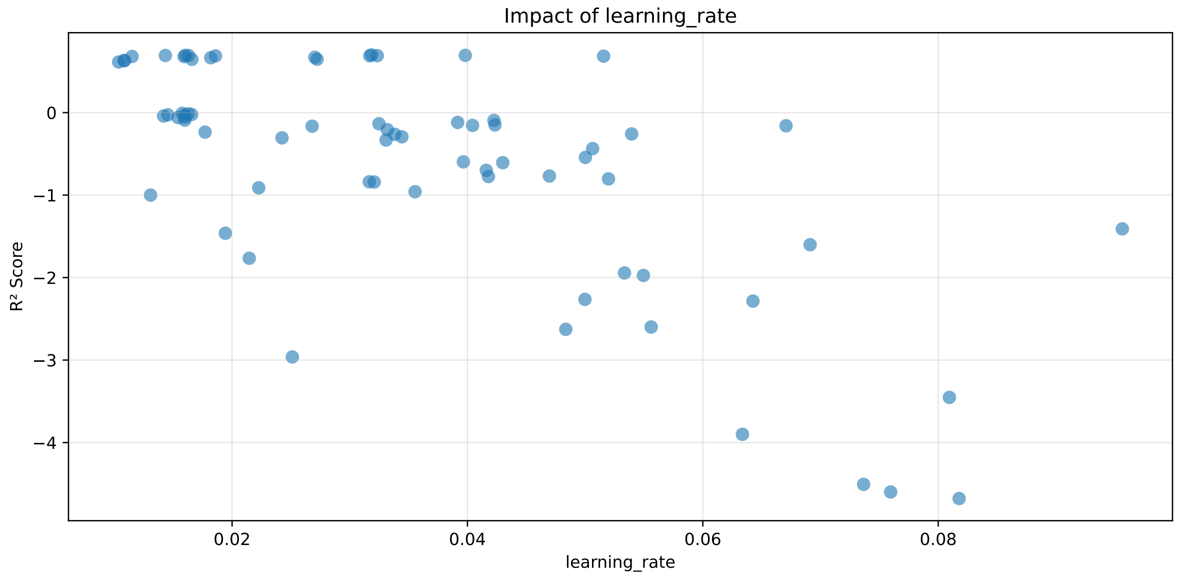
Distribution y_train vs y_test: y_train: mean=2.540, std=0.859, min=1.150, max=8.682 y_test: mean=2.561, std=0.947, min=1.120, max=13.845

==================== TOP Importance FEATURES ==================== feature importance r_umidity 37.791877 r_NH3 31.901623 r_CO2 14.433355 r_temperature 8.078971 r_THI 7.794174

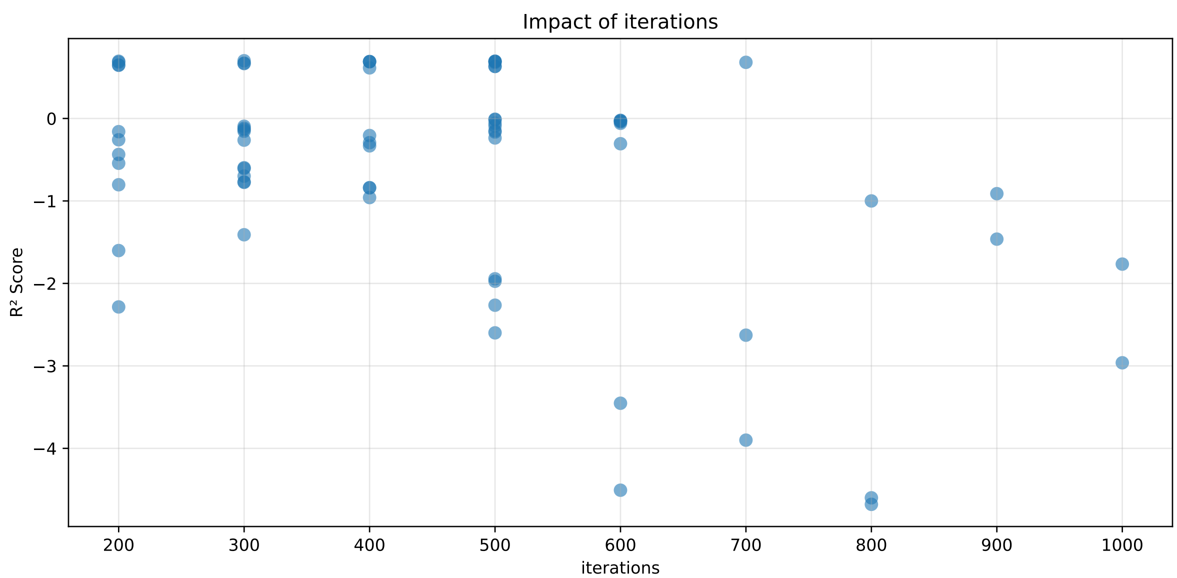




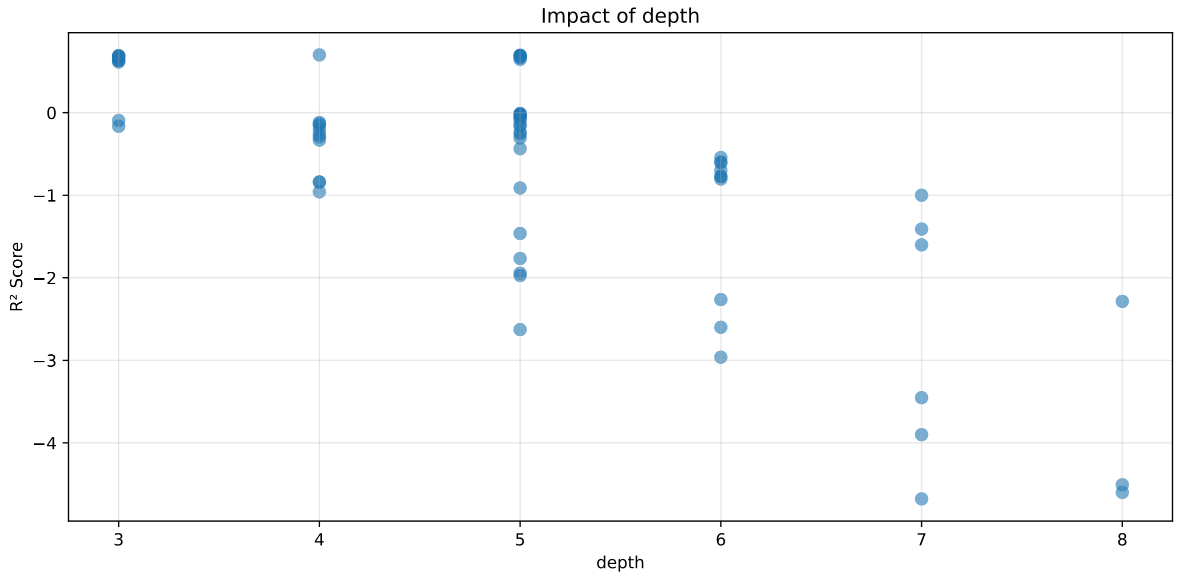
Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'depth', 'bagging_temperature']

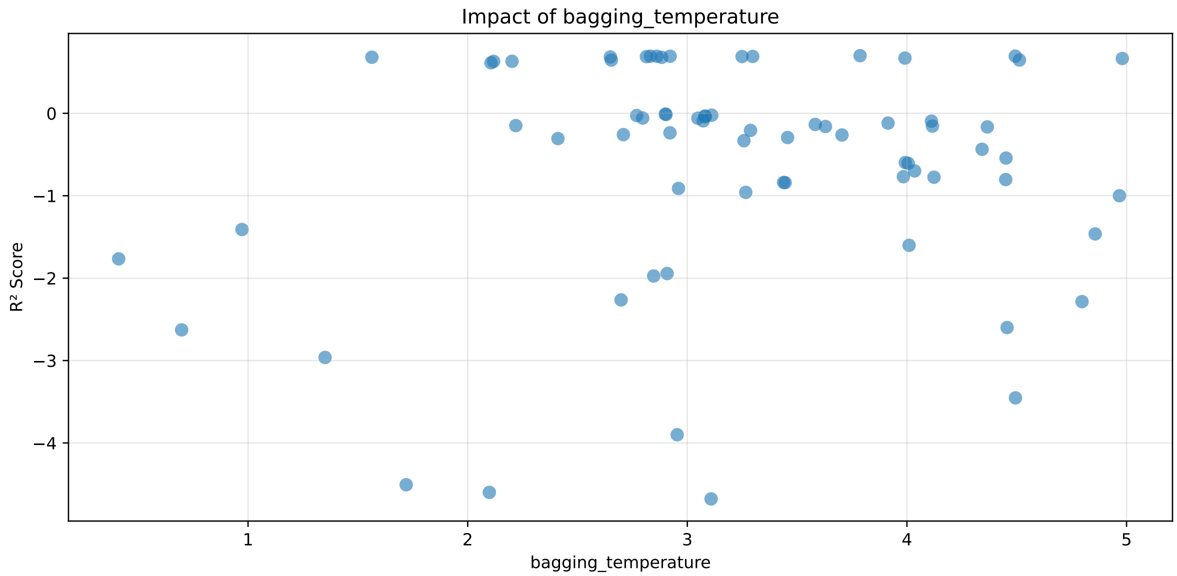


Impact of learning_rate
R² Score
0
−1
−2
−3
−4
0.02
0.04
0.06
0.08
learning_rate



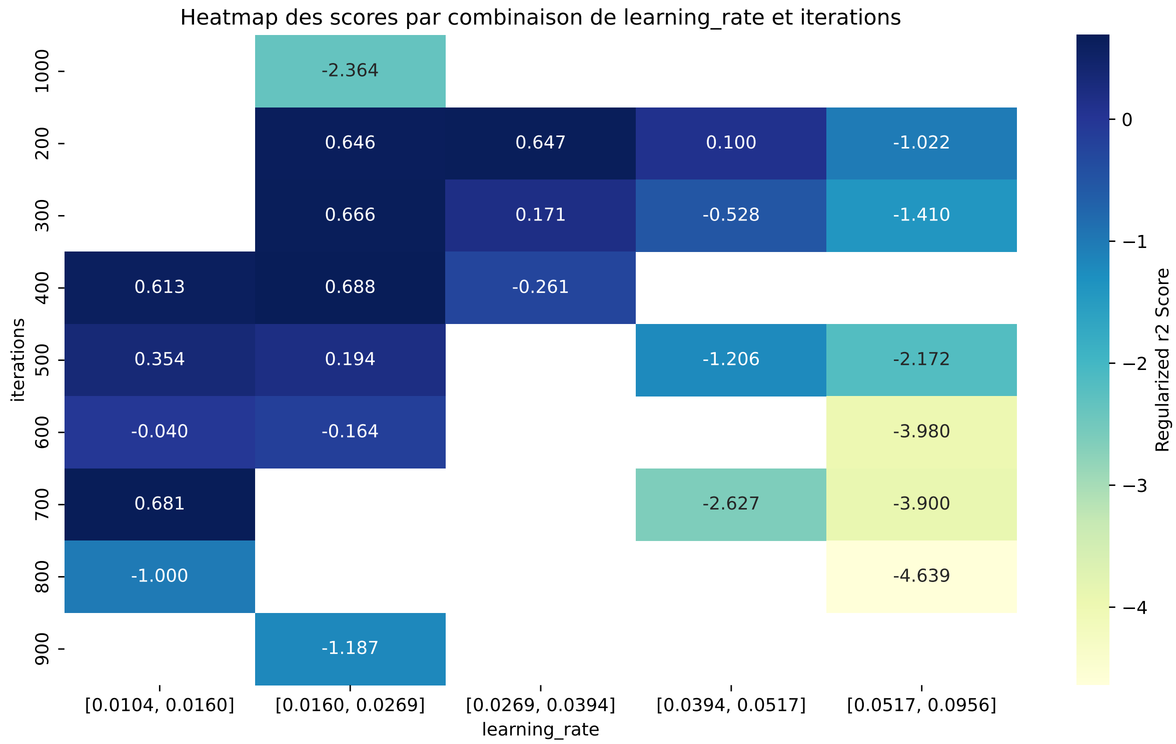
Impact of iterations
R² Score
0
−1
−2
−3
−4
200
300
400
500
600
700
800
900
1000
iterations





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7338	0.7	0.6024	0.0162	0.7	0.6028	-0.0972	0.8612	1.0482
2	0.7278	0.6941	0.6011	0.0143	0.6941	0.6049	-0.0892	0.8715	1.0485
3	0.7267	0.6938	0.5984	0.0144	0.6938	0.6058	-0.088	0.8732	1.0474
4	0.7265	0.6927	0.5991	0.0141	0.6927	0.6043	-0.0884	0.8723	1.0488
5	0.7258	0.6924	0.5983	0.0152	0.6924	0.6064	-0.086	0.8758	1.0483
6	0.7247	0.6904	0.5956	0.0168	0.6904	0.5968	-0.0936	0.8644	1.0496
7	0.7226	0.6893	0.5954	0.0191	0.6893	0.5979	-0.0915	0.8673	1.0482
8	0.7198	0.6881	0.5957	0.0133	0.6881	0.6046	-0.0835	0.8787	1.046
9	0.7093	0.6842	0.591	0.0162	0.6842	0.5882	-0.0959	0.8598	1.0368
10	0.7065	0.6806	0.5873	0.0195	0.6806	0.5878	-0.0928	0.8636	1.038
---	---	---	---	---	---	---	---	---	---
70	0.9759	0.7066	0.5876	0.0153	-4.6786	0.6053	-0.1013	0.8566	1.381

Hyperparameters:

Rank 1: {'reg___iterations': 300, 'reg___depth': 4, 'reg___learning_rate': 0.0318587121442995, 'reg___l2_leaf_reg': 43.485541066034294, 'reg___bagging_temperature': 3.7866590686414328, 'reg___random_strength': 2.509062740989326, 'reg___subsample': 0.8054673337466368, 'reg___min_data_in_leaf': 15, 'reg___leaf_estimation_iterations': 8}

Rank 2: {’reg__iterations’: 200, ’reg__depth’: 5, ’reg__learning_rate’: 0.03981605431567943, ’reg__l2_leaf_reg’: 25.900366680255754, ’reg__bagging_temperature’: 4.493131064574564, ’reg__random_strength’: 4.543364375097674, ’reg__subsample’: 0.8029379401217402, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 8}

Rank 3: {’reg__iterations’: 500, ’reg__depth’: 5, ’reg__learning_rate’: 0.01602312433024485, ’reg__l2_leaf_reg’: 37.26827856596932, ’reg__bagging_temperature’: 2.8322147239730824, ’reg__random_strength’: 2.81558817591498, ’reg__subsample’: 0.7564777909186962, ’reg__min_data_in_leaf’: 18, ’reg__leaf_estimation_iterations’: 4}

Rank 4: {’reg__iterations’: 500, ’reg__depth’: 5, ’reg__learning_rate’: 0.014328874500586465, ’reg__l2_leaf_reg’: 38.51132103568809, ’reg__bagging_temperature’: 2.861625921423669, ’reg__random_strength’: 2.755206483536973, ’reg__subsample’: 0.8223726226571102, ’reg__min_data_in_leaf’: 19, ’reg__leaf_estimation_iterations’: 7}

Rank 5: {’reg__iterations’: 500, ’reg__depth’: 5, ’reg__learning_rate’: 0.016301337099578327, ’reg__l2_leaf_reg’: 34.79715461840558, ’reg__bagging_temperature’: 2.9218336142931944, ’reg__random_strength’: 3.27951978002383, ’reg__subsample’: 0.7593431833598144, ’reg__min_data_in_leaf’: 25, ’reg__leaf_estimation_iterations’: 4}

Rank 6: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.03233727606423259, ’reg__l2_leaf_reg’: 48.18065184729305, ’reg__bagging_temperature’: 3.2978826233473573, ’reg__random_strength’: 3.433113828770553, ’reg__subsample’: 0.7166289213295539, ’reg__min_data_in_leaf’: 10, ’reg__leaf_estimation_iterations’: 6}

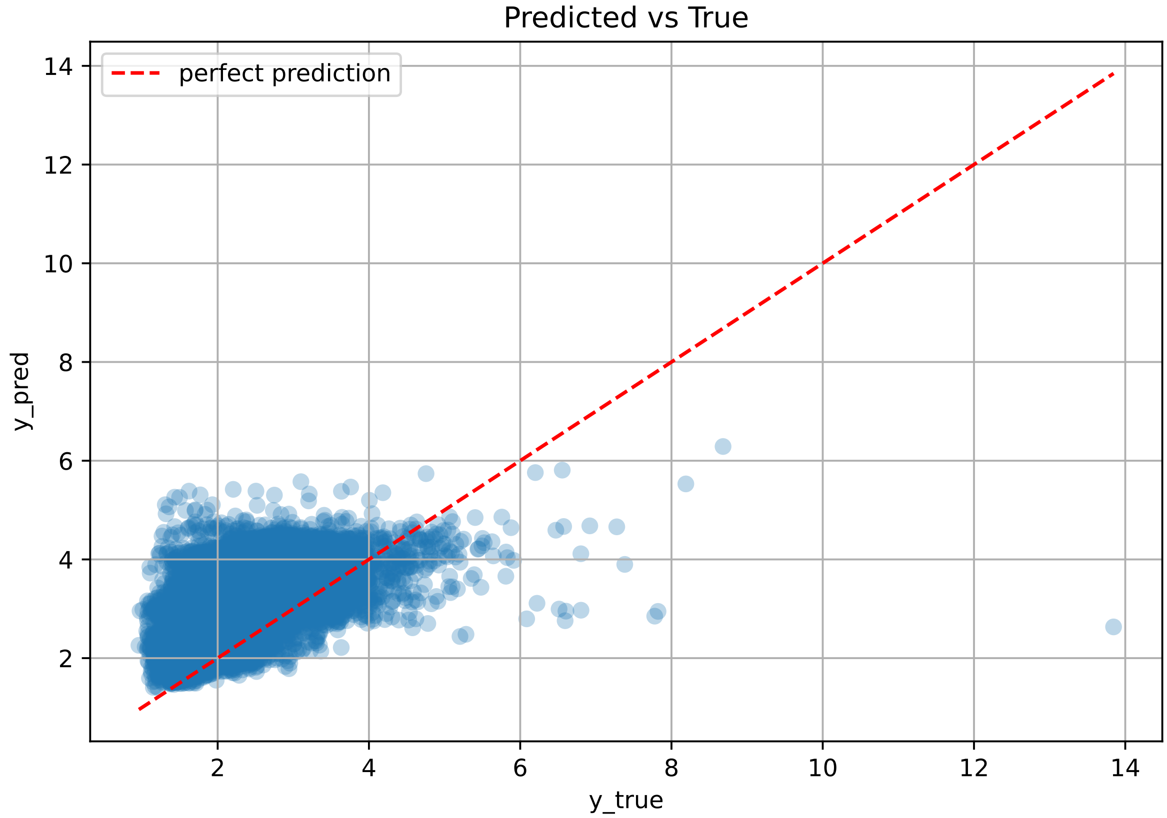
Rank 7: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.03169932413603339, ’reg__l2_leaf_reg’: 49.85932600325498, ’reg__bagging_temperature’: 3.249518573981015, ’reg__random_strength’: 3.3652555139473987, ’reg__subsample’: 0.7355737379162113, ’reg__min_data_in_leaf’: 11, ’reg__leaf_estimation_iterations’: 7}

Rank 8: {’reg__iterations’: 400, ’reg__depth’: 5, ’reg__learning_rate’: 0.018592958483373122, ’reg__l2_leaf_reg’: 44.683806444051264, ’reg__bagging_temperature’: 2.813836963236204, ’reg__random_strength’: 3.180960264553904, ’reg__subsample’: 0.7604909885092038, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 6}

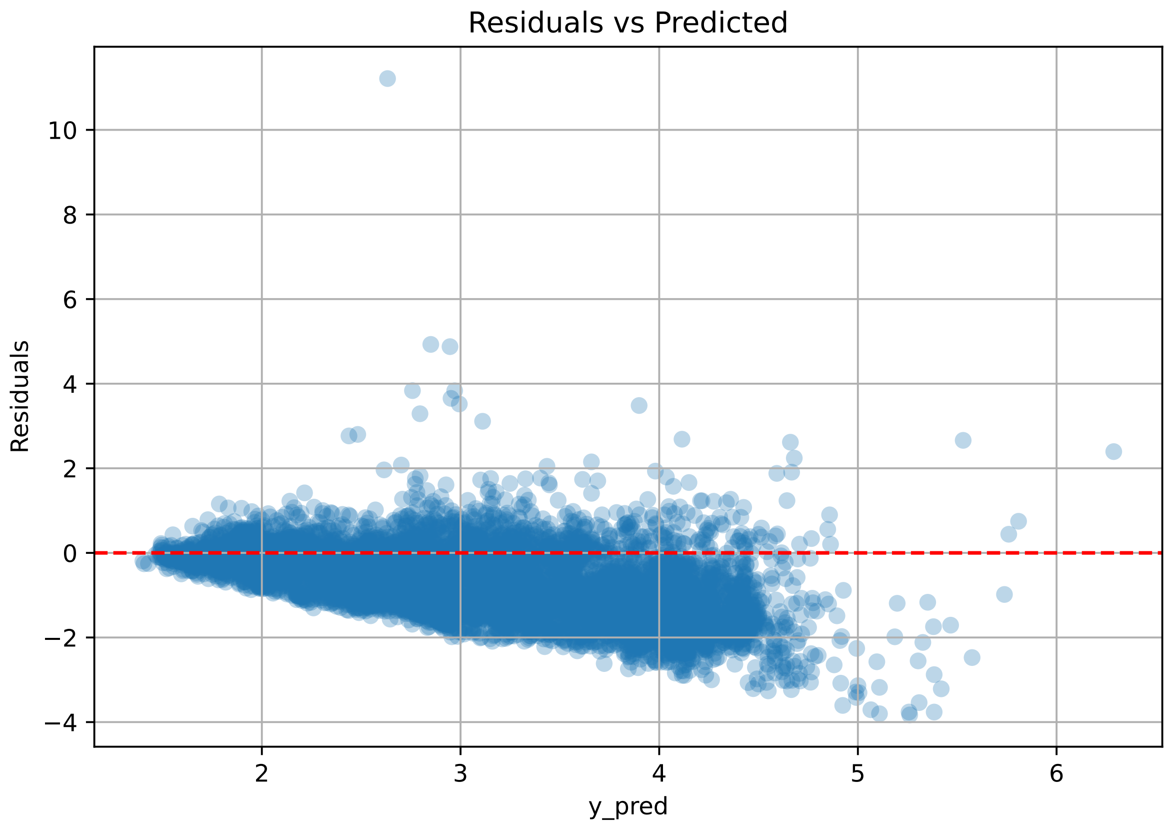
Rank 9: {’reg__iterations’: 200, ’reg__depth’: 3, ’reg__learning_rate’: 0.05156956444585479, ’reg__l2_leaf_reg’: 37.31876504701524, ’reg__bagging_temperature’: 2.6495388950099993, ’reg__random_strength’: 2.872961675445528, ’reg__subsample’: 0.5013383067768755, ’reg__min_data_in_leaf’: 17, ’reg__leaf_estimation_iterations’: 7}

Rank 10: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.011503766205432536, ’reg__l2_leaf_reg’: 17.102223866431217, ’reg__bagging_temperature’: 1.5639315948054993, ’reg__random_strength’: 1.6619284506296963, ’reg__subsample’: 0.6499125411268813, ’reg__min_data_in_leaf’: 25, ’reg__leaf_estimation_iterations’: 9}

2.5.14 Scatter



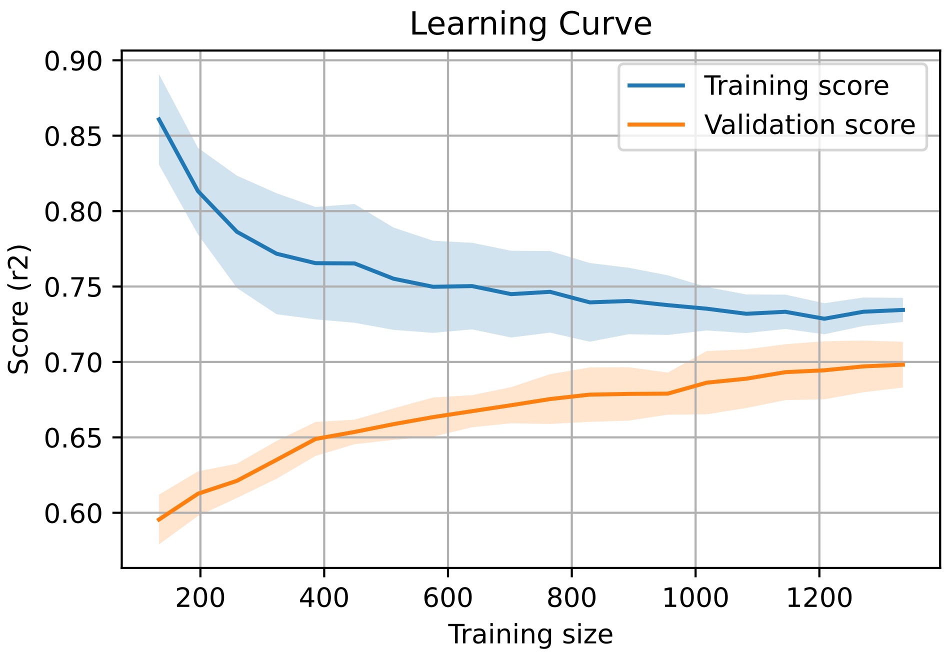
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 11

Len(group) : 2726, **Len(training)** : 1909,

Len(training_real) : 1337, **Len(test_of_training)** : 572, **Len(test)** : 817



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	3	950	406	1356	0.596	0.55	0.04	0.64	0.641	0.091	1.165	1.083	nan
Linear	4	952	408	1360	0.302	0.278	0.049	0.276	0.277	-0.001	0.997	1.087	nan
Linear	5	1909	817	2726	0.431	0.404	0.026	0.457	0.457	0.054	1.133	1.069	nan
Linear	11	1909	817	2726	0.463	0.463	0.017	0.394	0.394	-0.069	0.852	1.0	nan
Linear	MEAN	5720	2448	8168	0.448	0.424	0.033	0.442	0.442	0.019	1.037	1.06	nan
LGBM	3	950	406	1356	0.57	0.51	0.039	0.581	0.61	0.1	1.196	1.116	-0.6781
LGBM	4	952	408	1360	0.301	0.262	0.039	0.266	0.323	0.061	1.233	1.151	-0.527
LGBM	5	1909	817	2726	0.408	0.378	0.022	0.434	0.451	0.072	1.192	1.077	-0.2018
LGBM	11	1909	817	2726	0.697	0.667	0.014	0.566	0.58	-0.086	0.87	1.046	0.6667
LGBM	MEAN	5720	2448	8168	0.494	0.454	0.029	0.462	0.491	0.037	1.123	1.097	-0.185
CatBoost	3	950	406	1356	0.659	0.598	0.037	0.676	0.682	0.084	1.14	1.101	-0.6112
CatBoost	4	952	408	1360	0.375	0.328	0.034	0.362	0.364	0.036	1.11	1.143	-0.6081
CatBoost	5	1909	817	2726	0.478	0.442	0.022	0.504	0.51	0.067	1.152	1.08	-0.2687
CatBoost	11	1909	817	2726	0.734	0.7	0.016	0.602	0.603	-0.097	0.861	1.048	0.7
CatBoost	MEAN	5720	2448	8168	0.561	0.517	0.027	0.536	0.54	0.023	1.066	1.093	-0.197

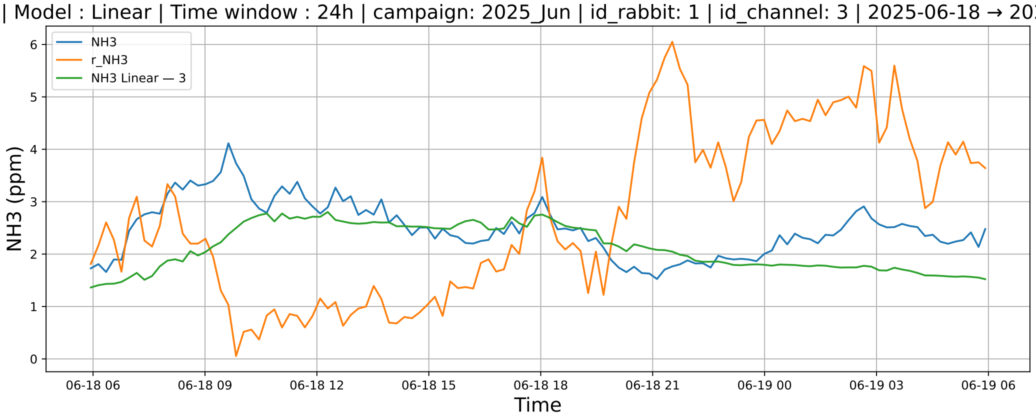
3 Plot

3.1 Standalone plots

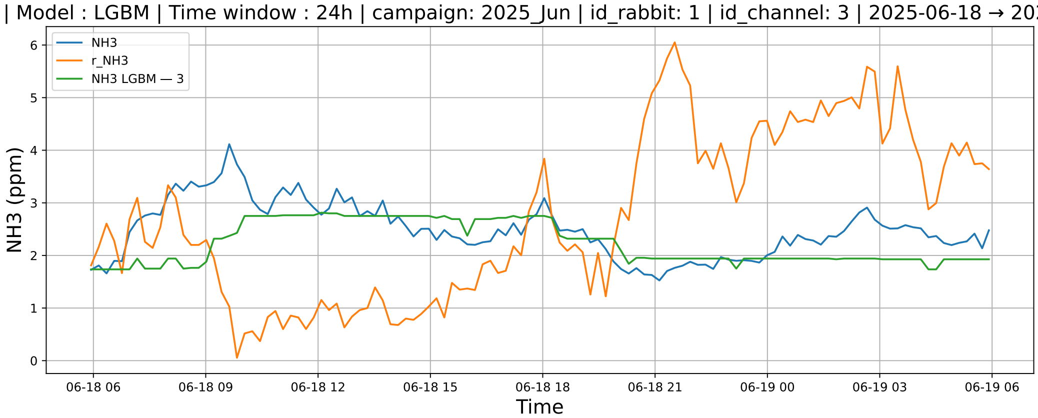
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

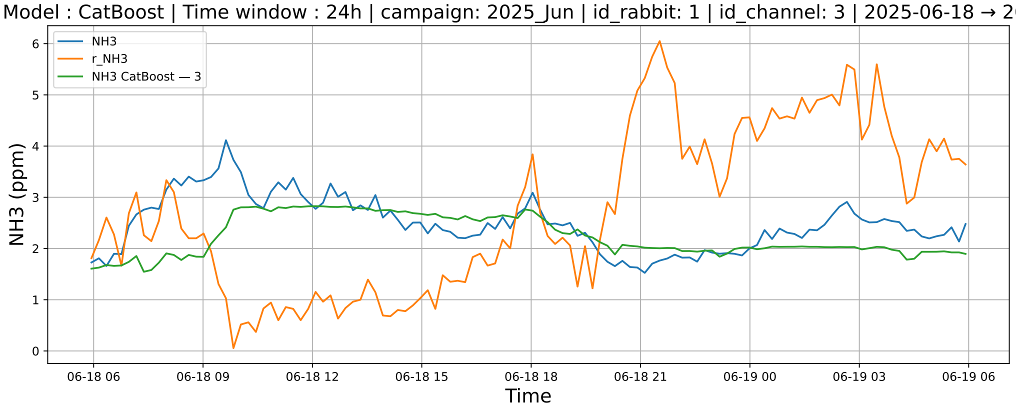
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

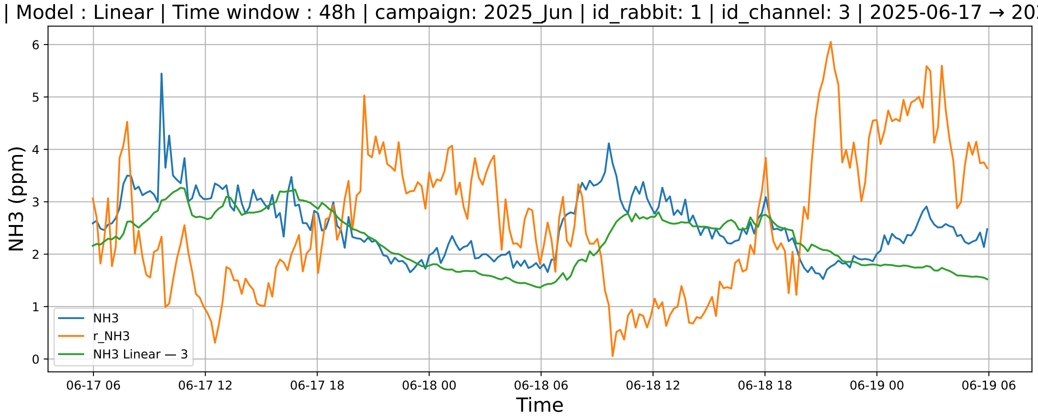


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

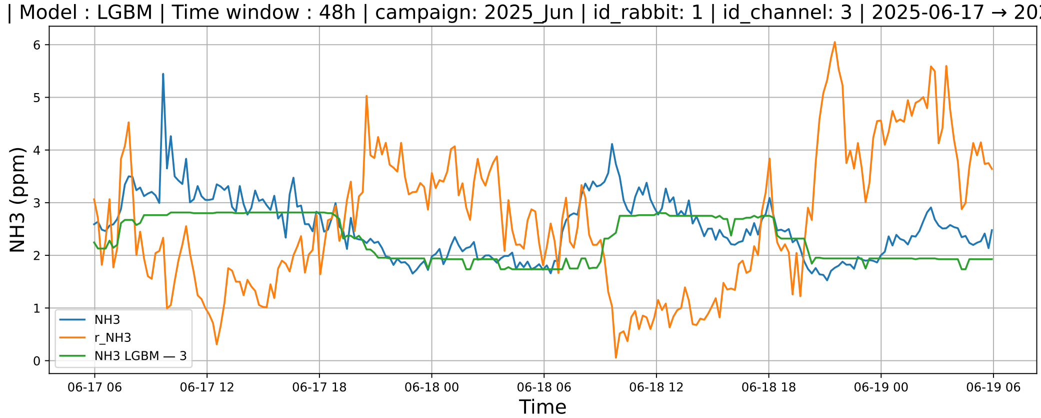


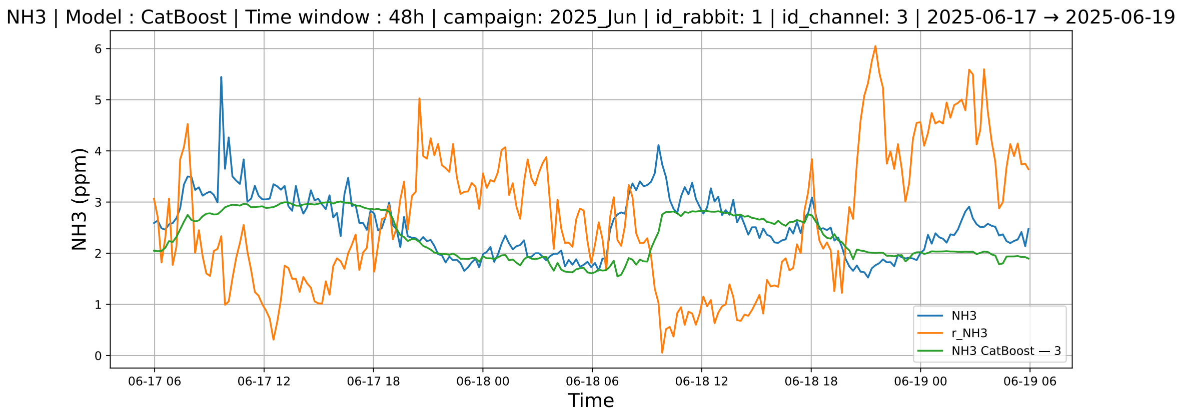
3.1.1.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

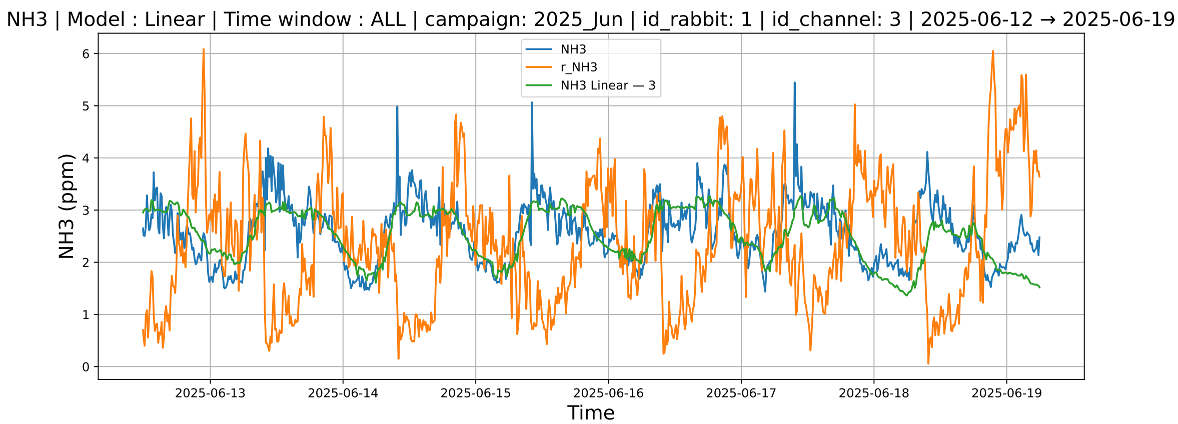


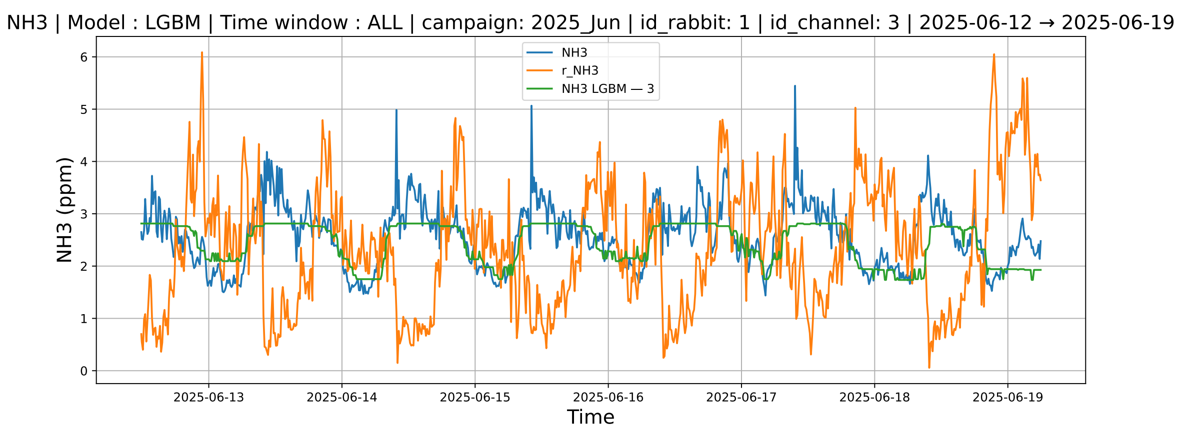
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



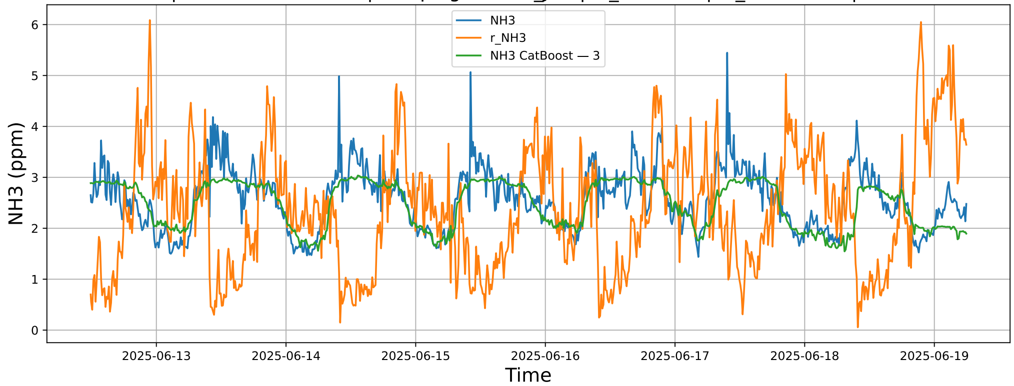


3.1.1.3 Time window : ALL





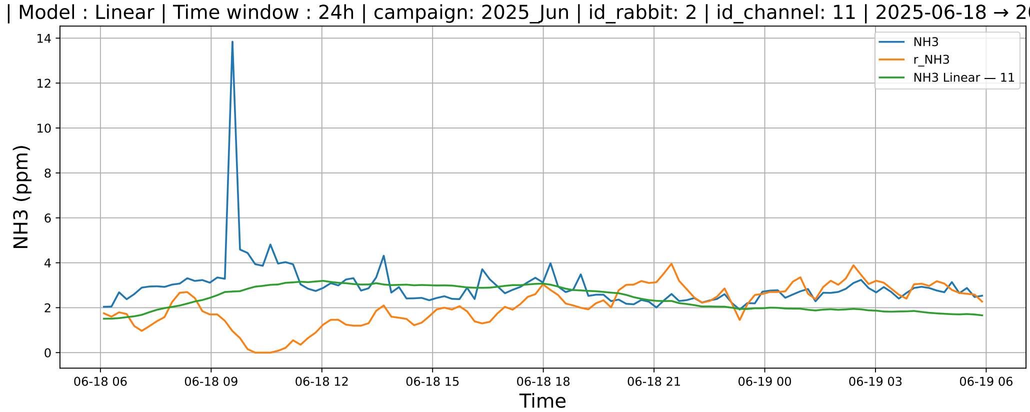
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



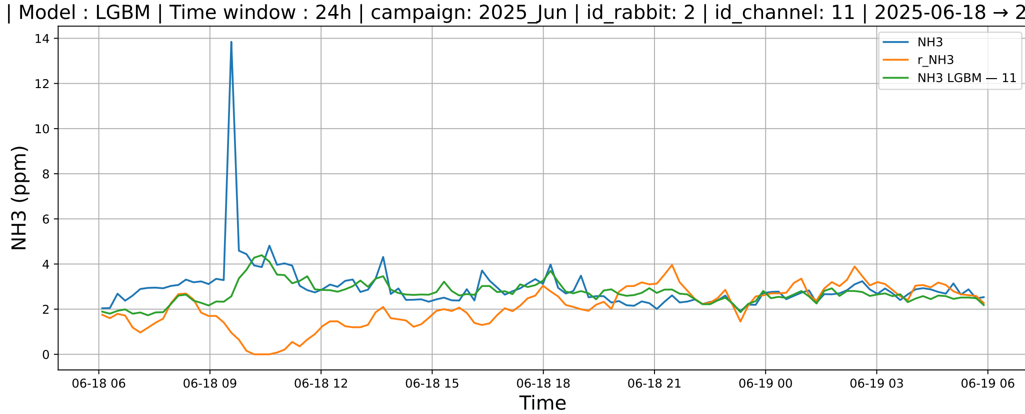
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

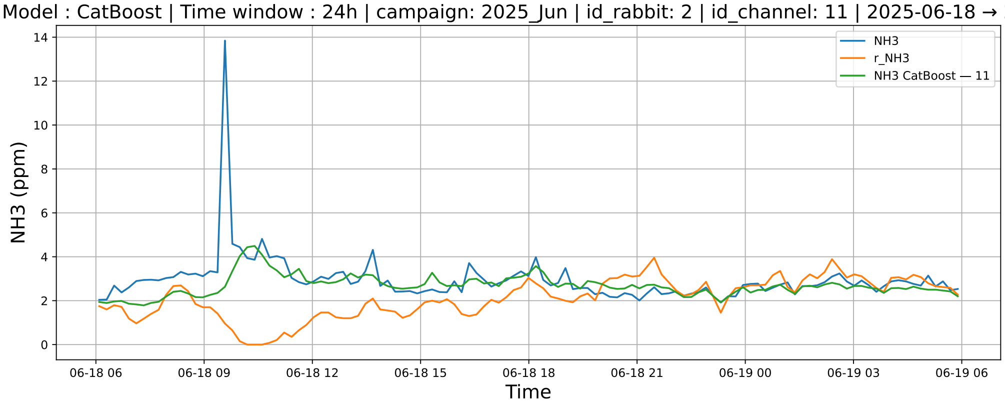
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

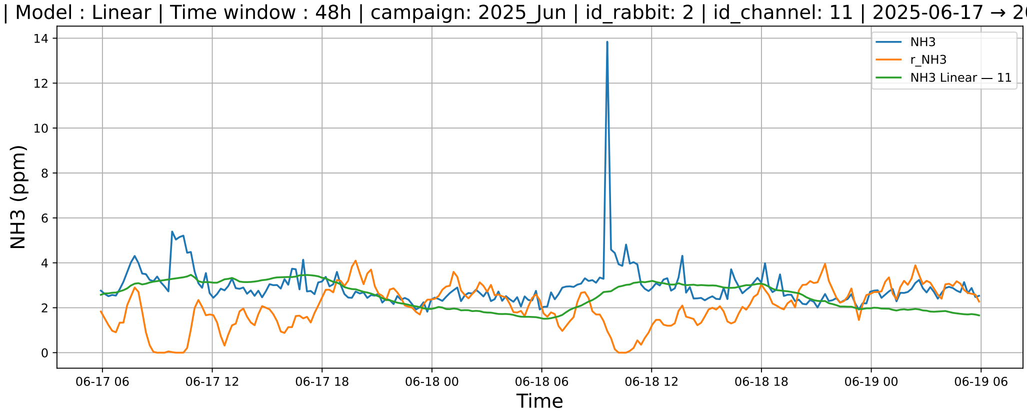


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

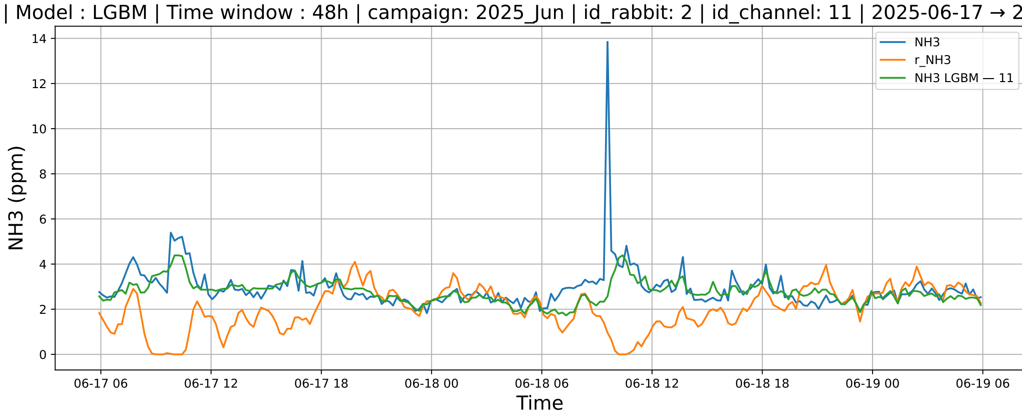


3.1.2.2 Time window : 48

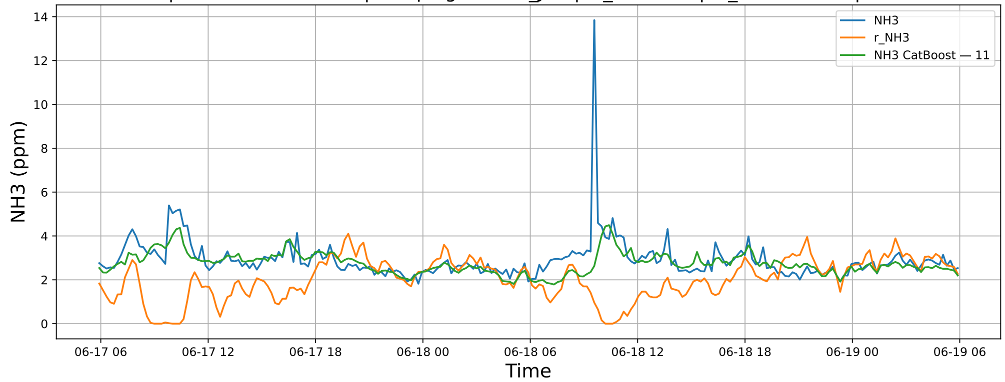
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

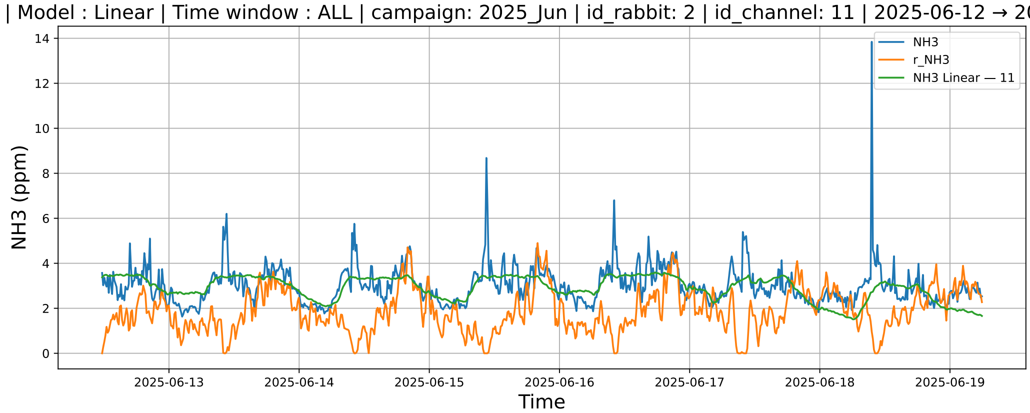


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

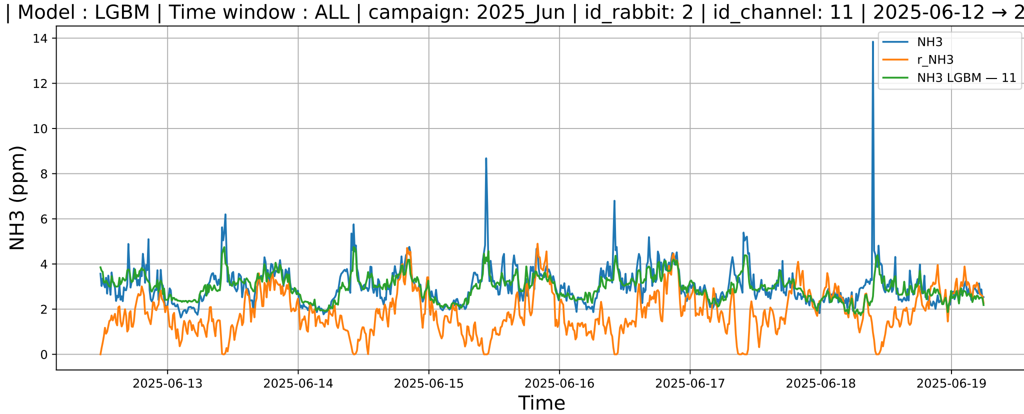


3.1.2.3 Time window : ALL

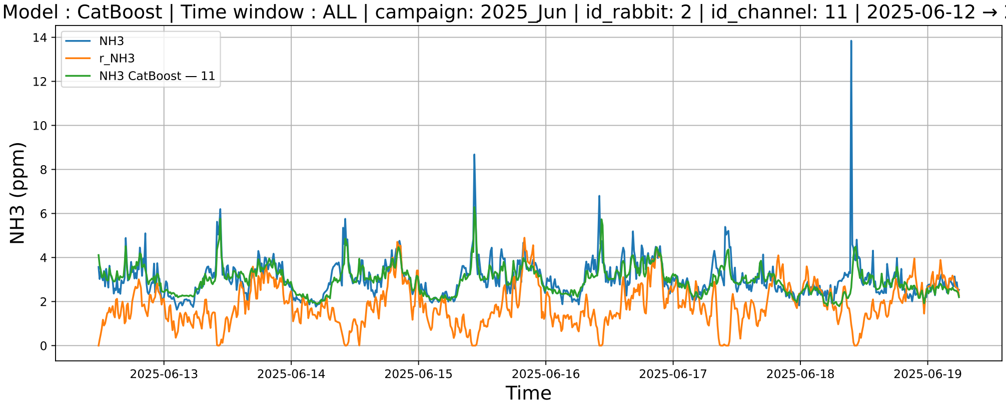
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



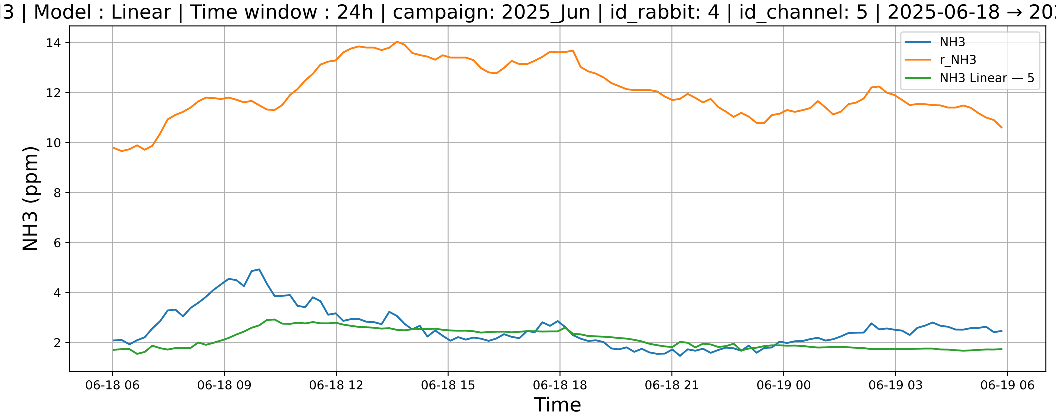
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



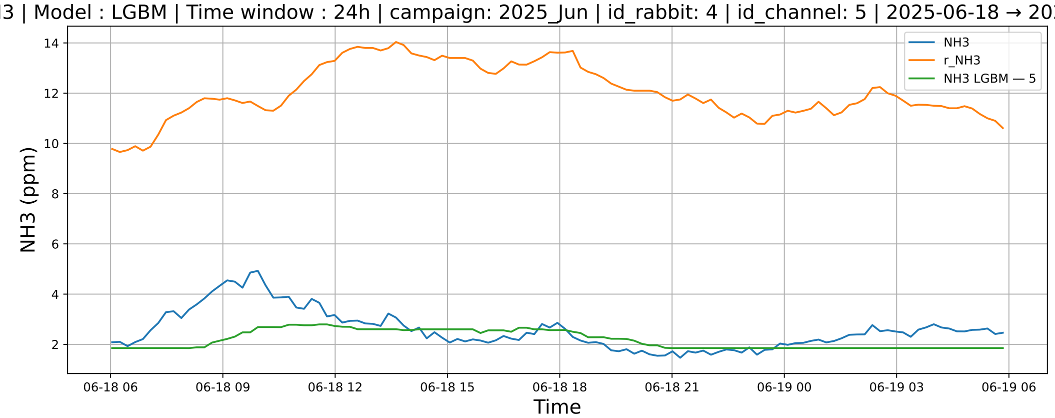
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

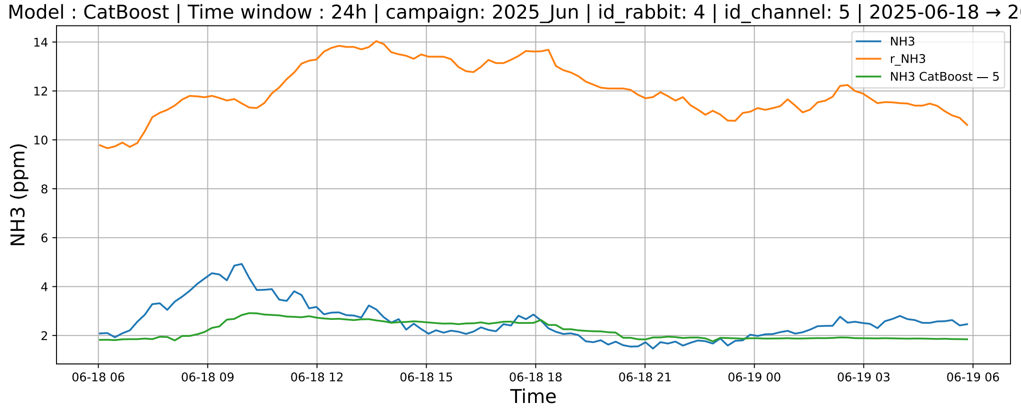
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

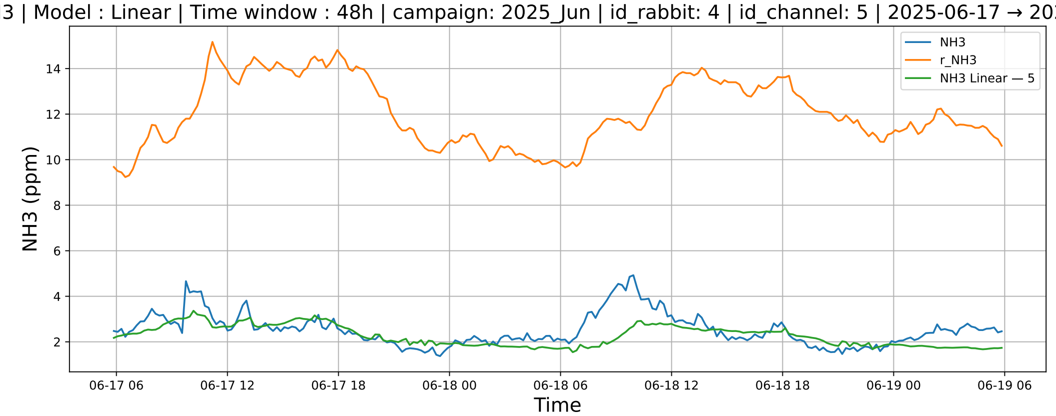


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

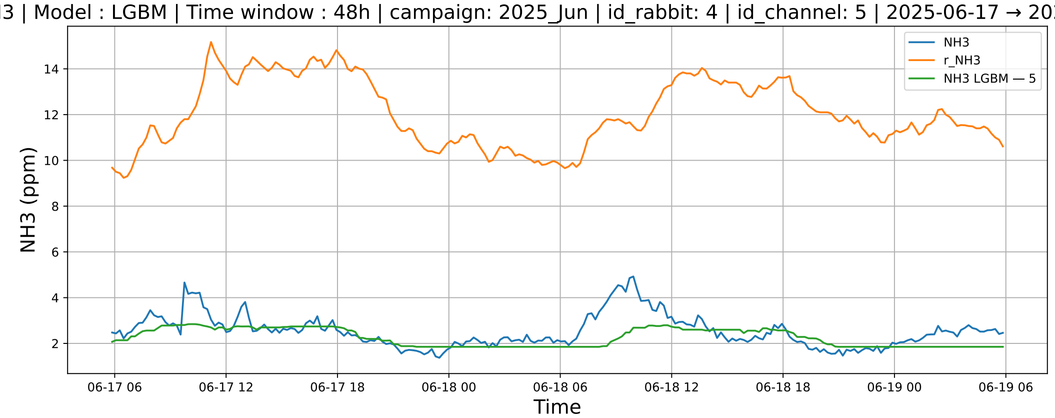


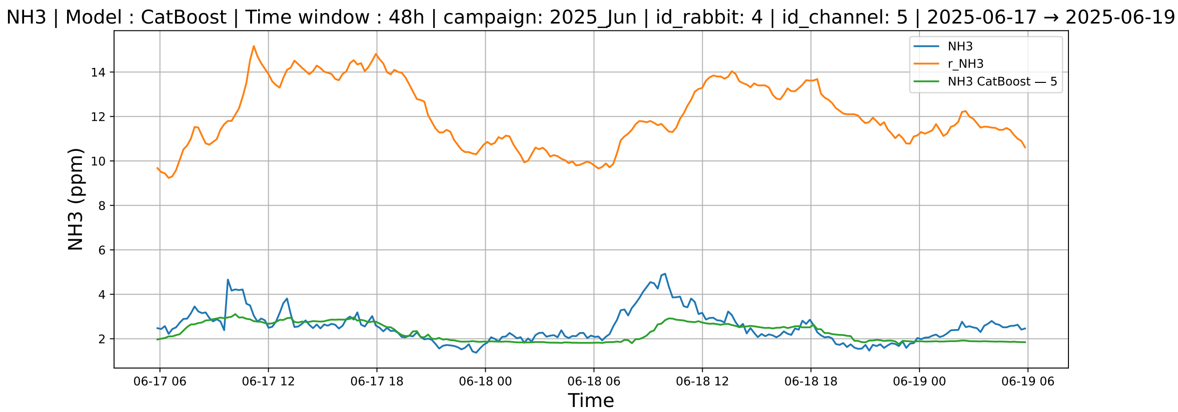
3.1.3.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

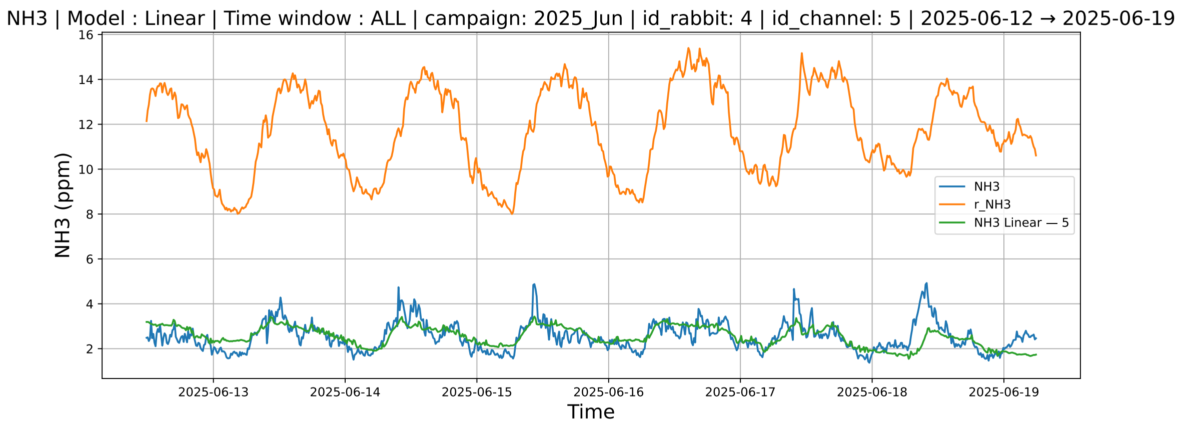


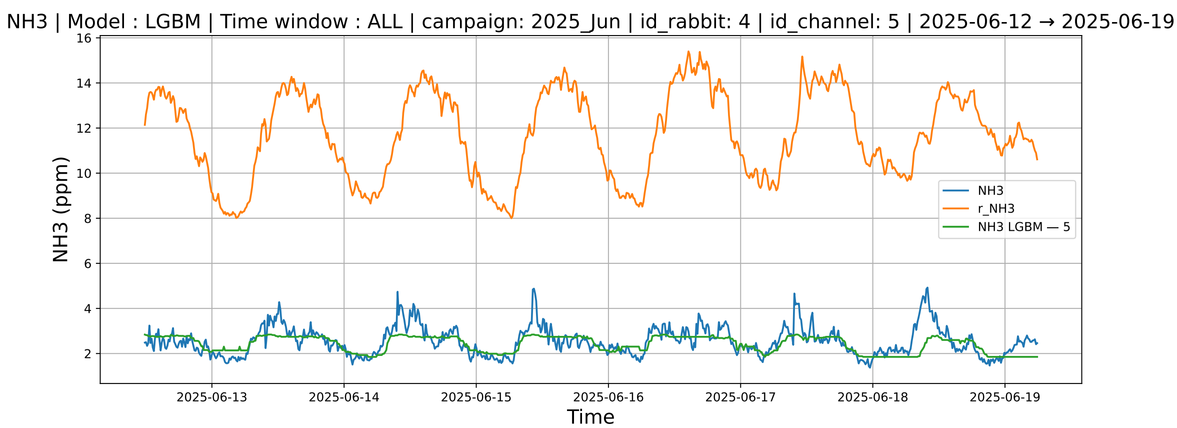
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



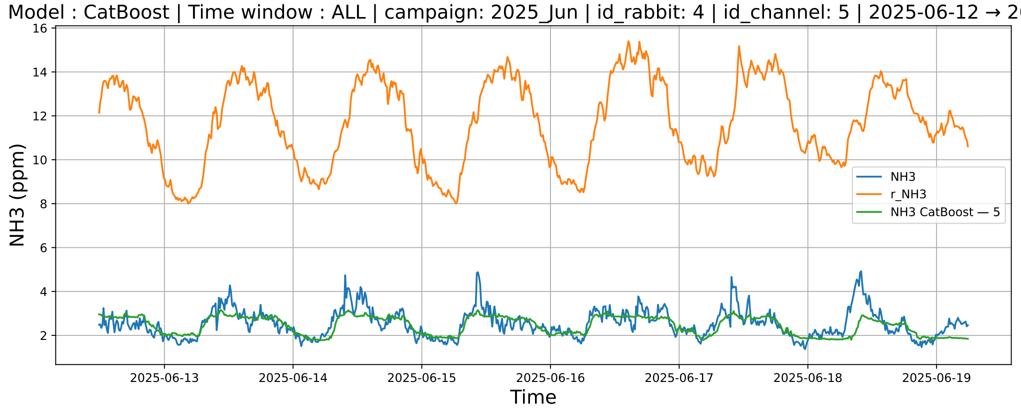


3.1.3.3 Time window : ALL





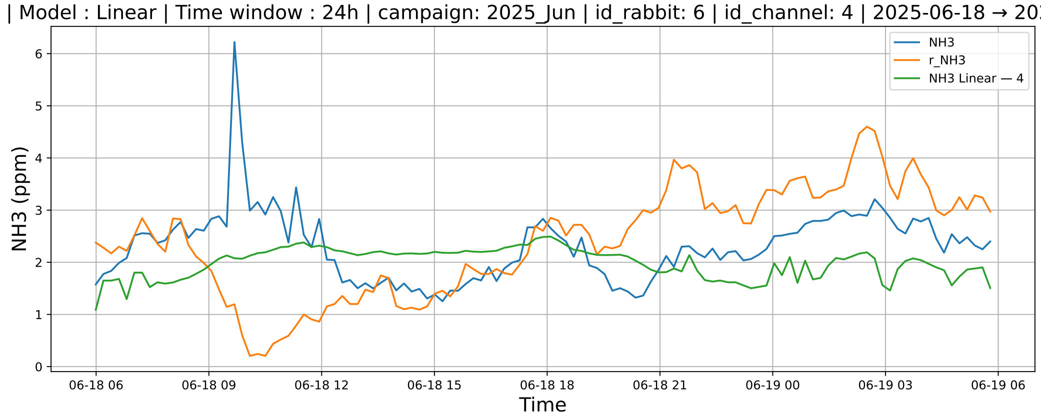
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



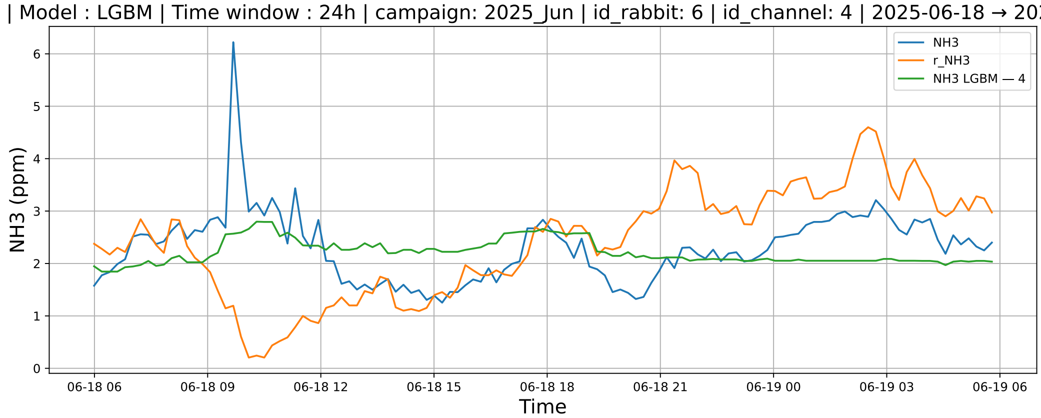
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

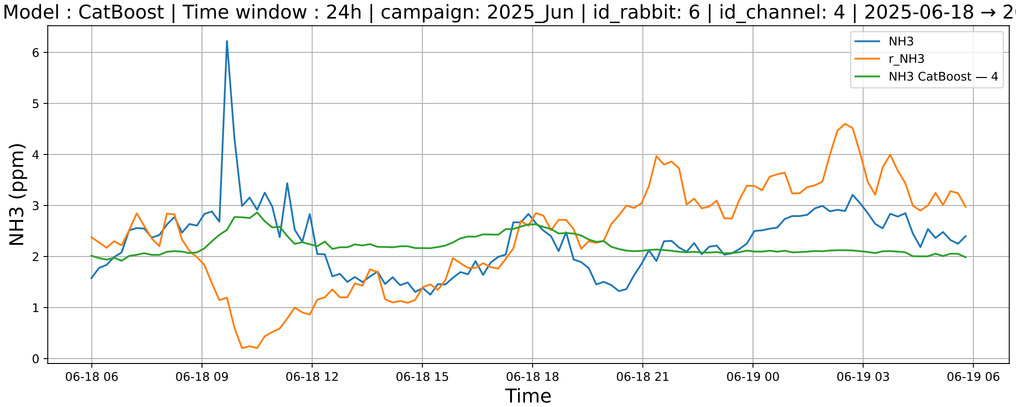
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

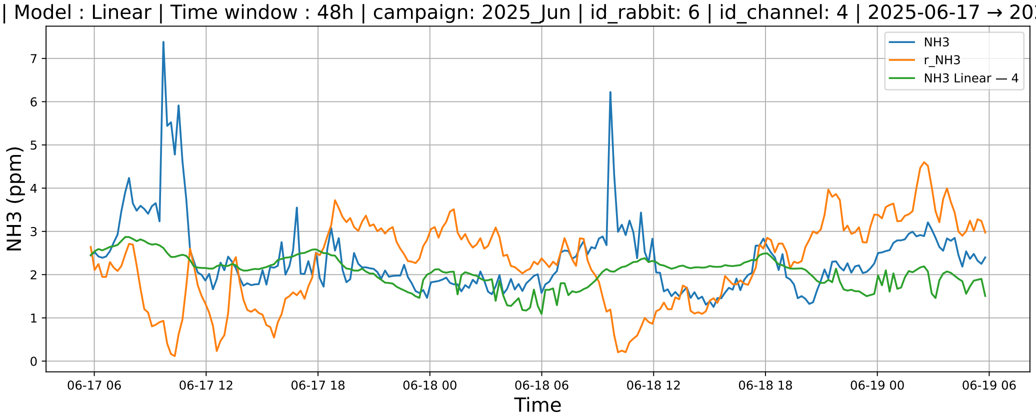


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

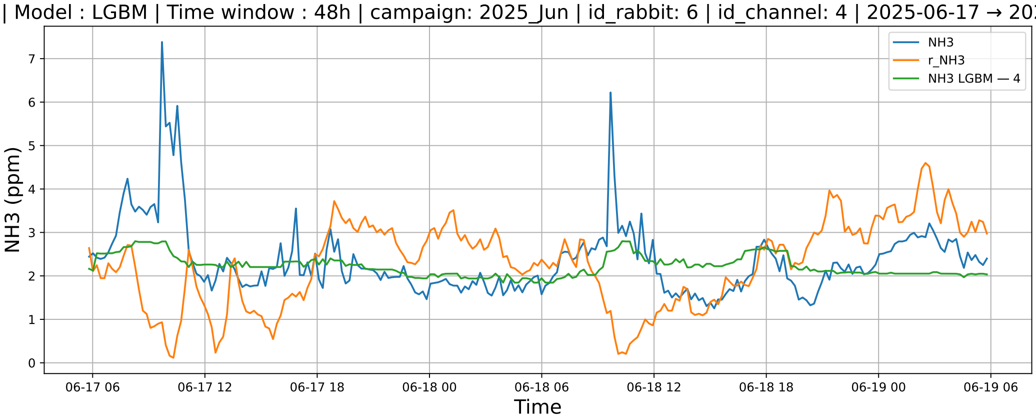


3.1.4.2 Time window : 48

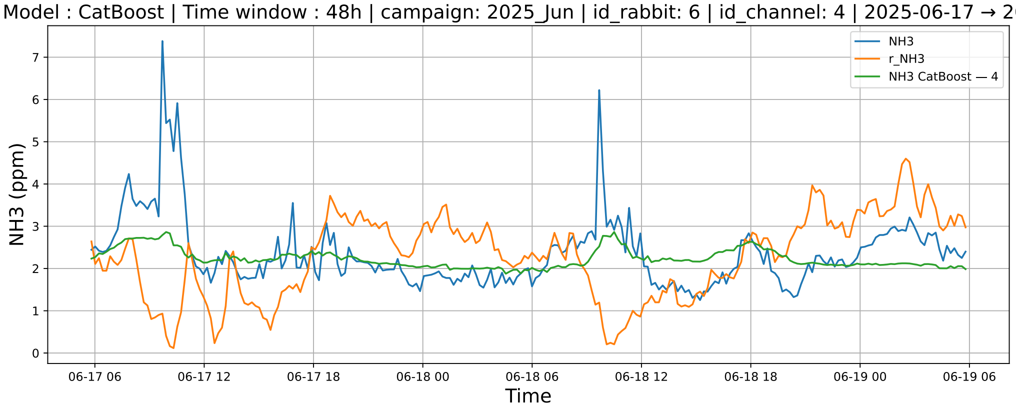
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

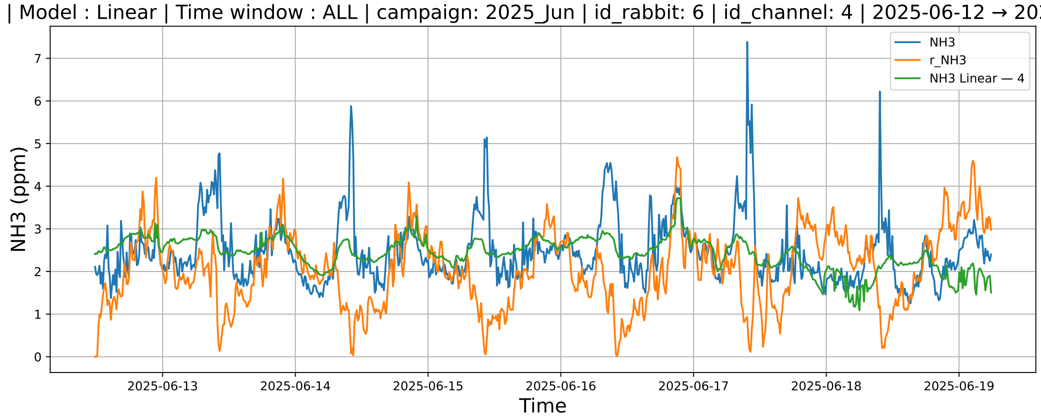


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

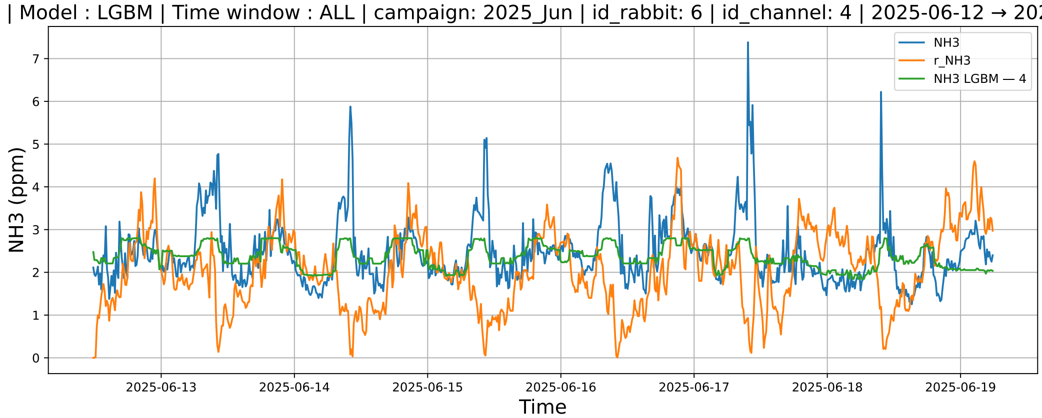


3.1.4.3 Time window : ALL

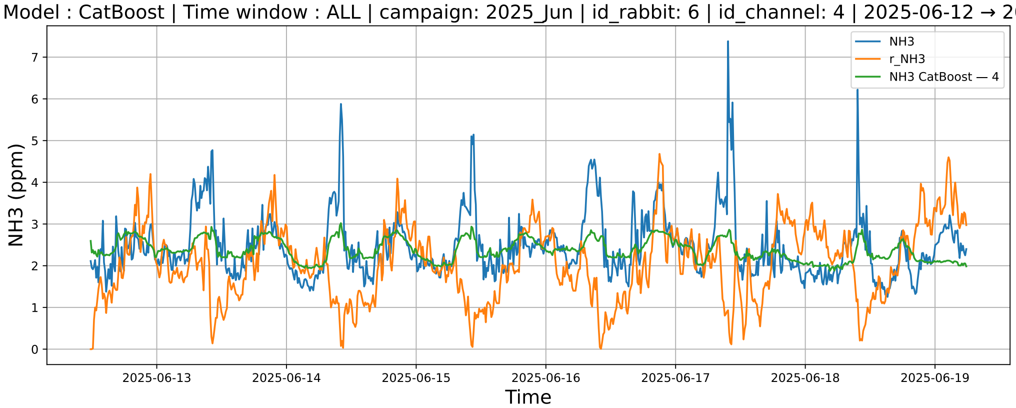
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



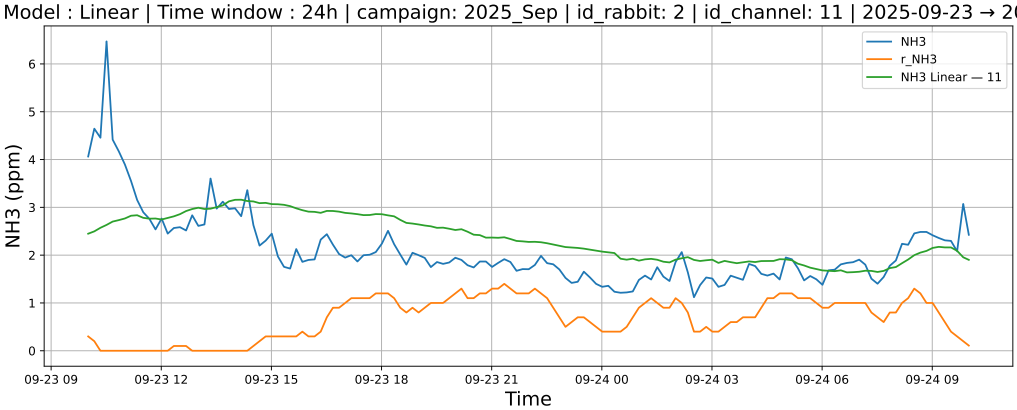
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



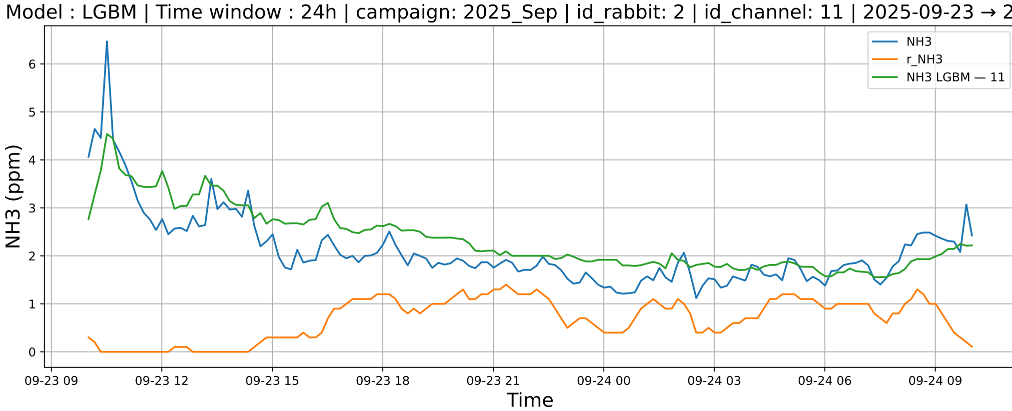
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

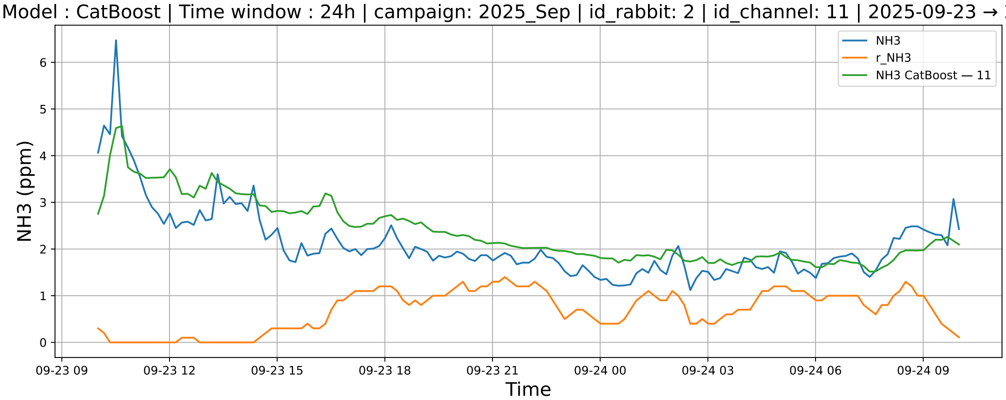
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

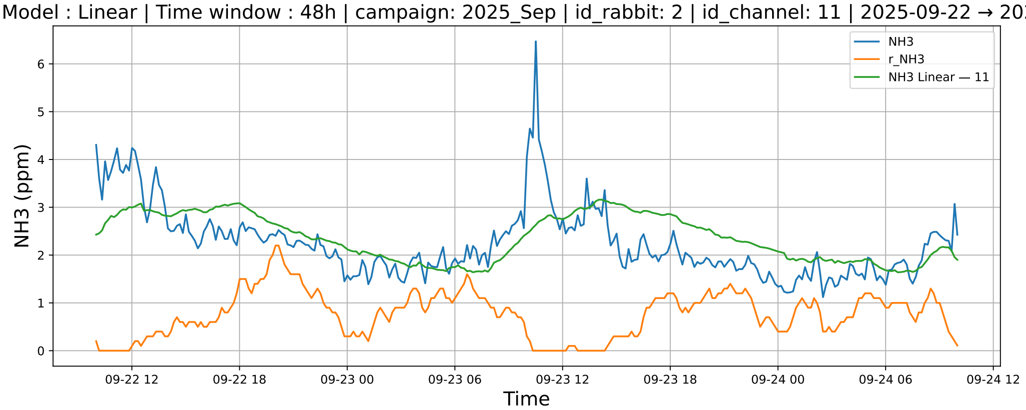


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

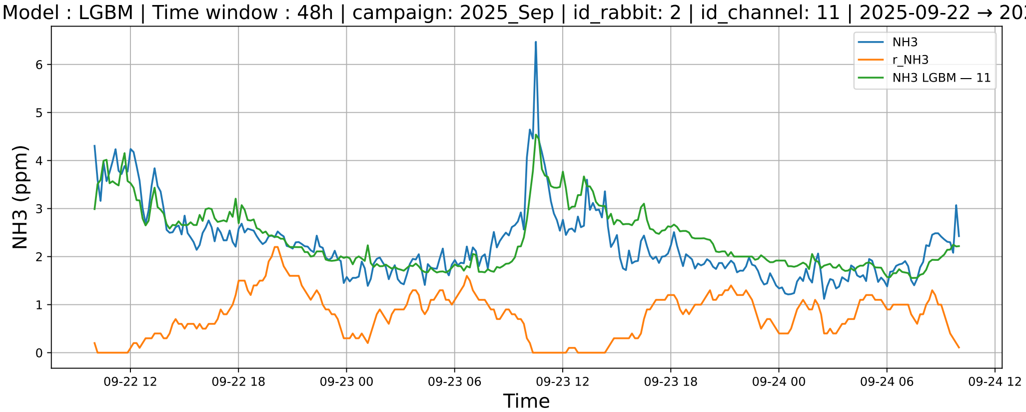


3.1.5.2 Time window : 48

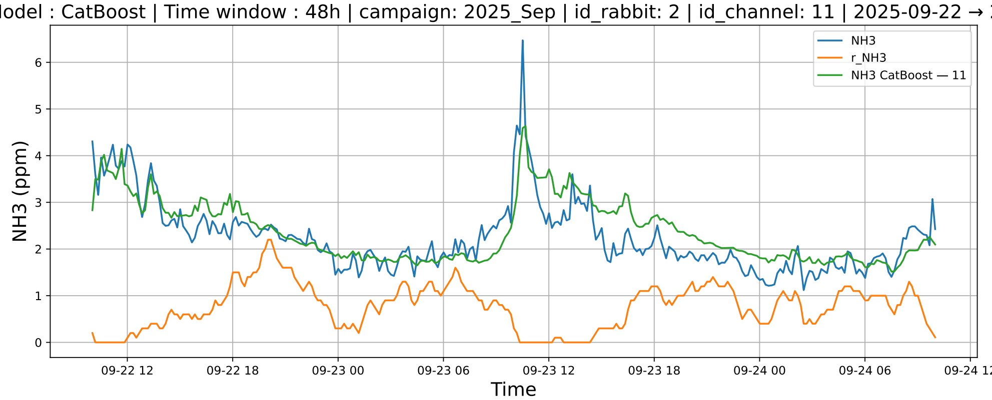
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

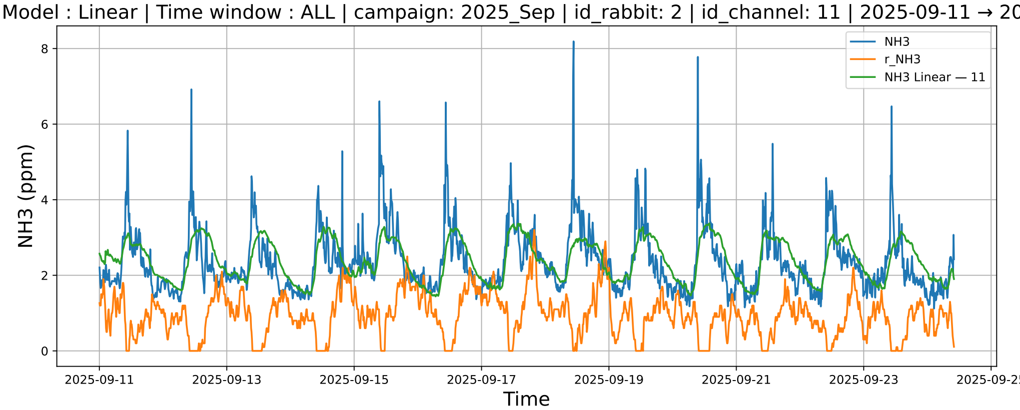


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

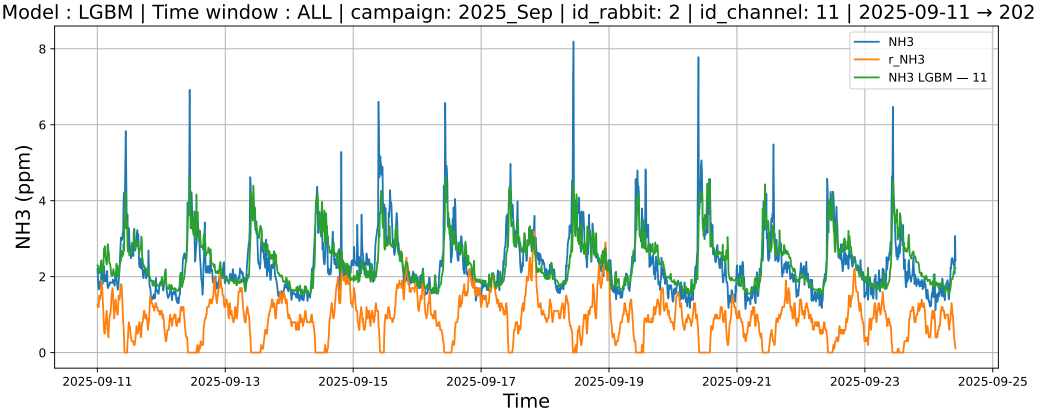


3.1.5.3 Time window : ALL

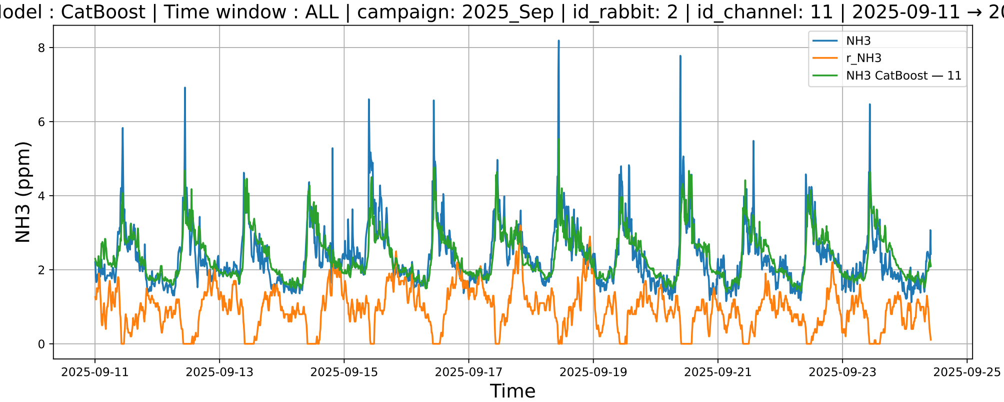
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



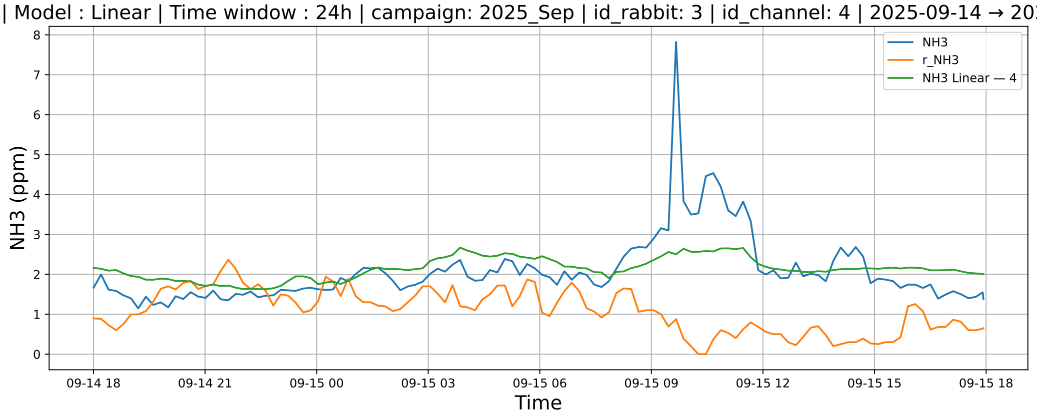
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



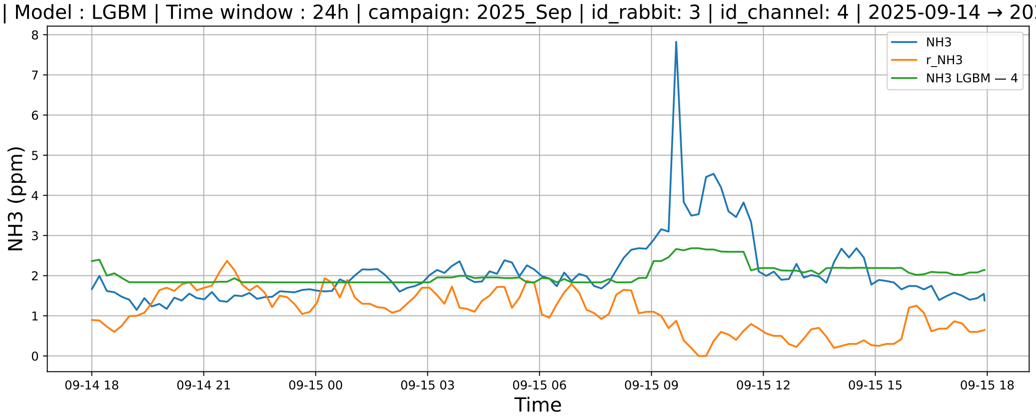
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

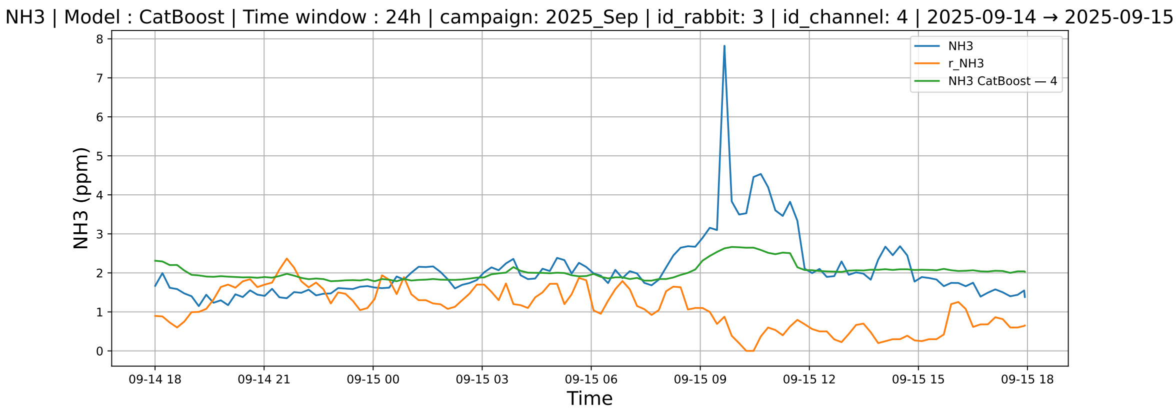
3.1.6.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

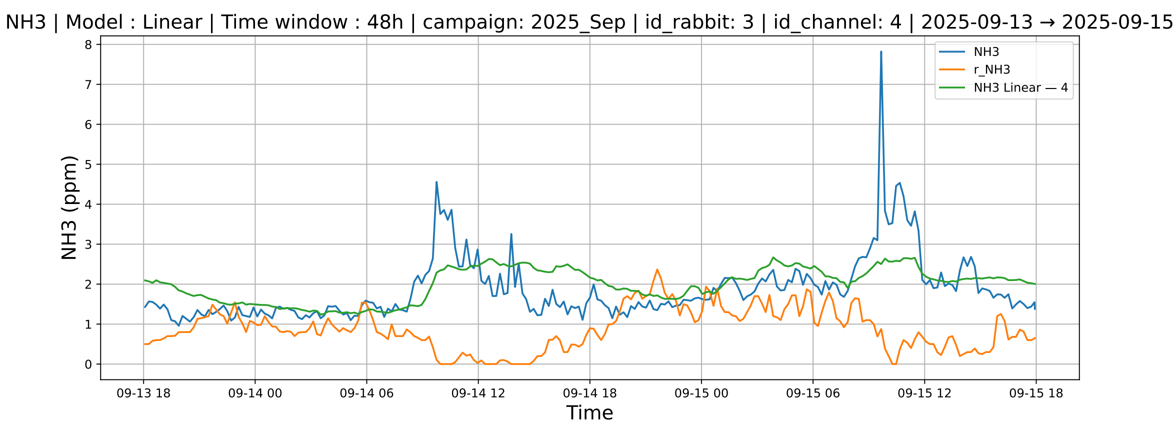


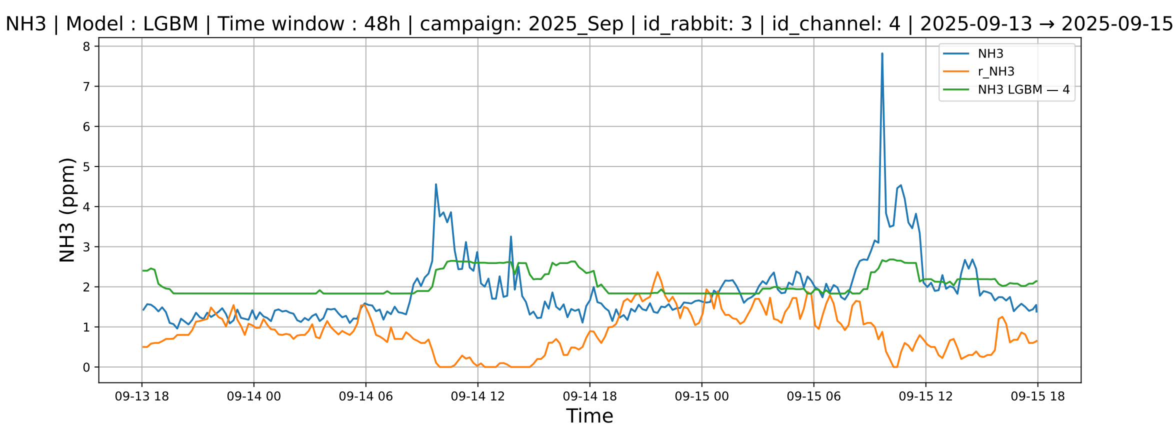
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

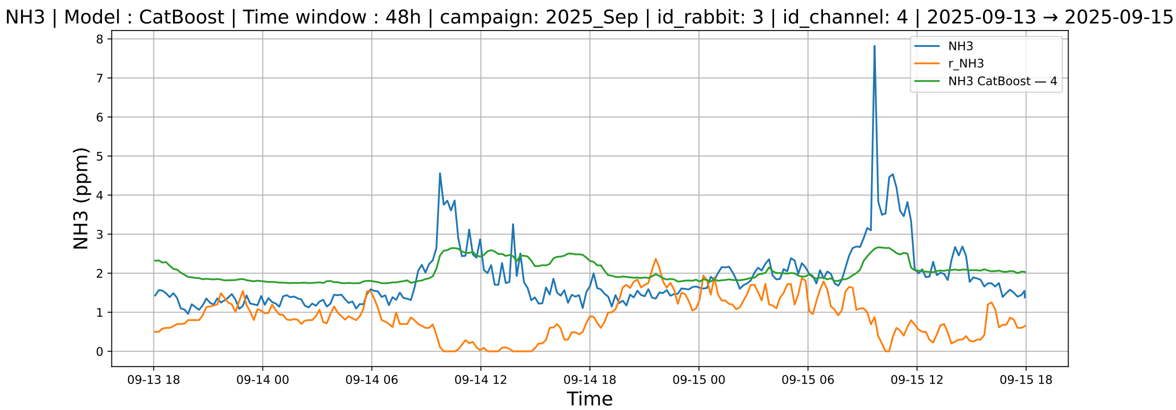




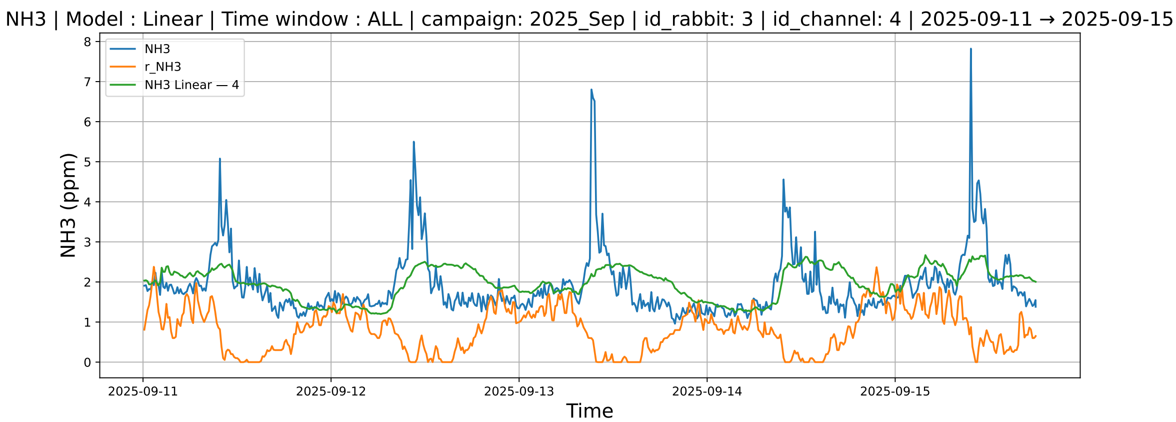
3.1.6.2 Time window : 48

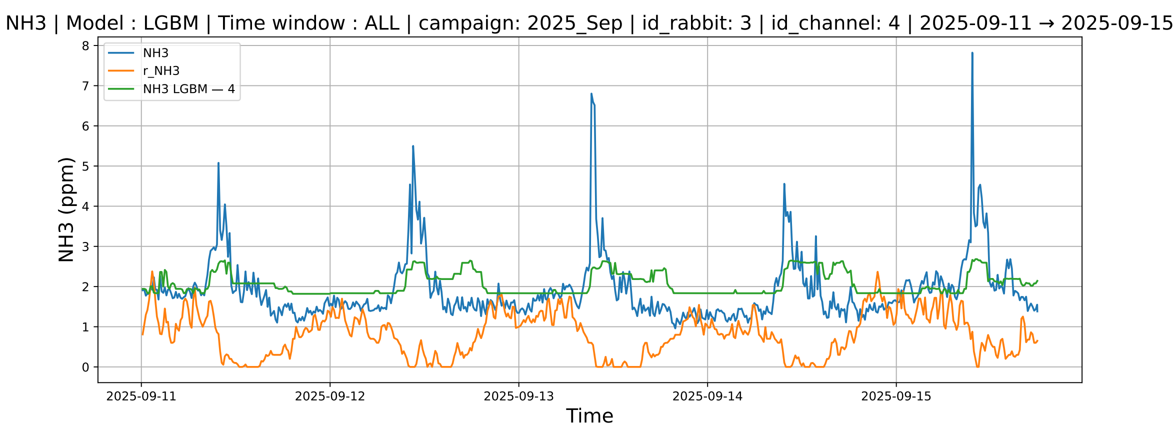


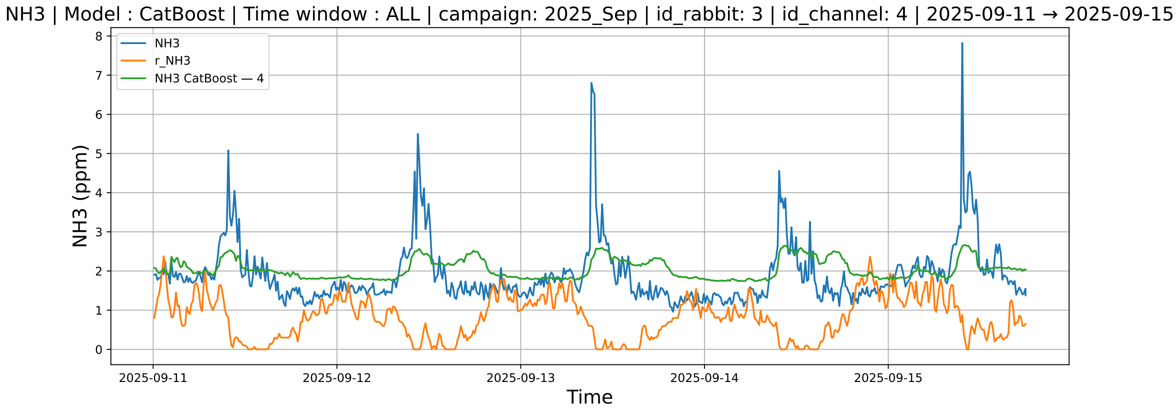




3.1.6.3 Time window : ALL

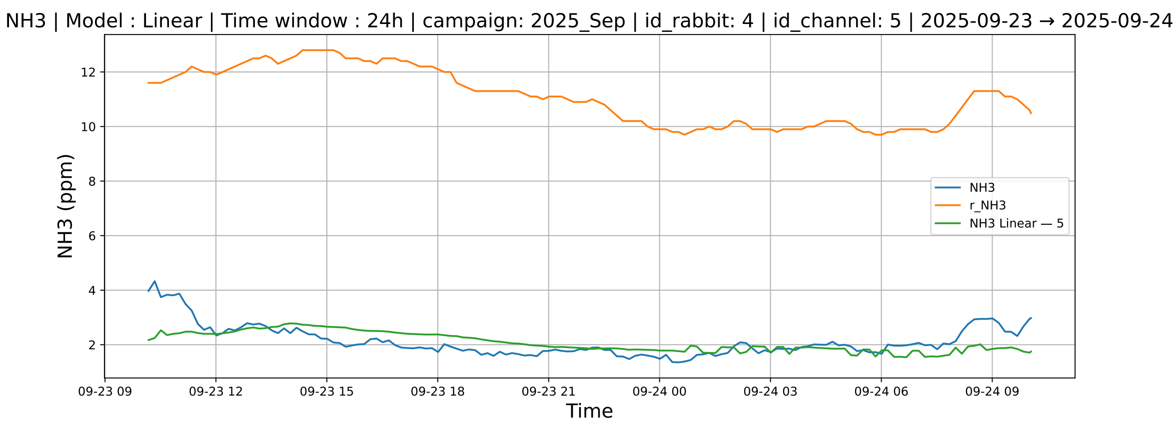


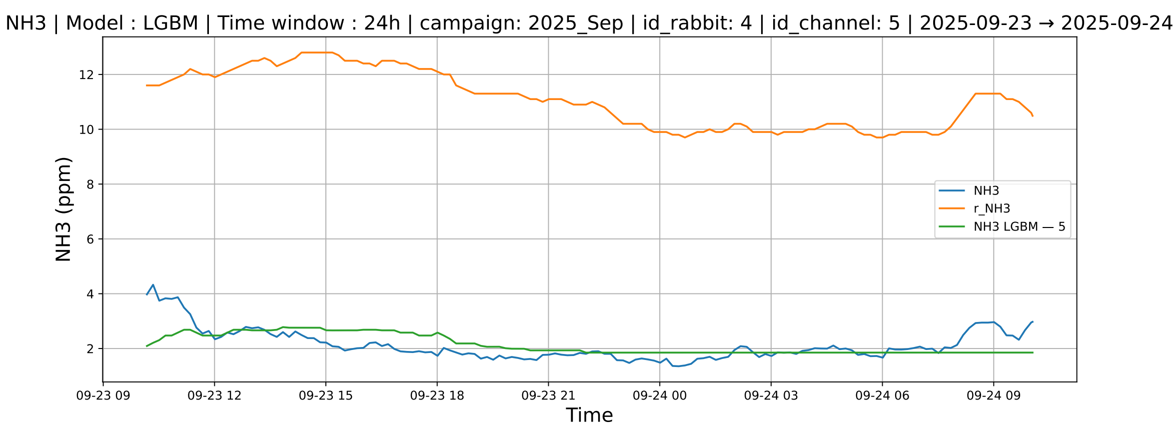




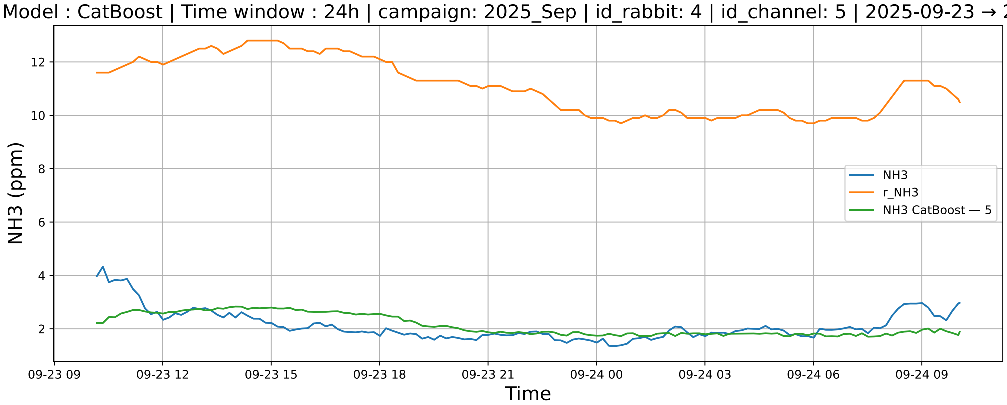
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24



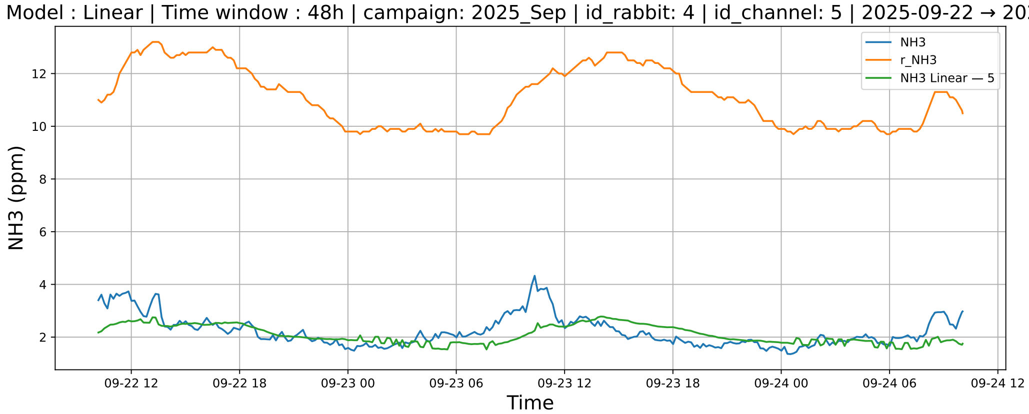


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

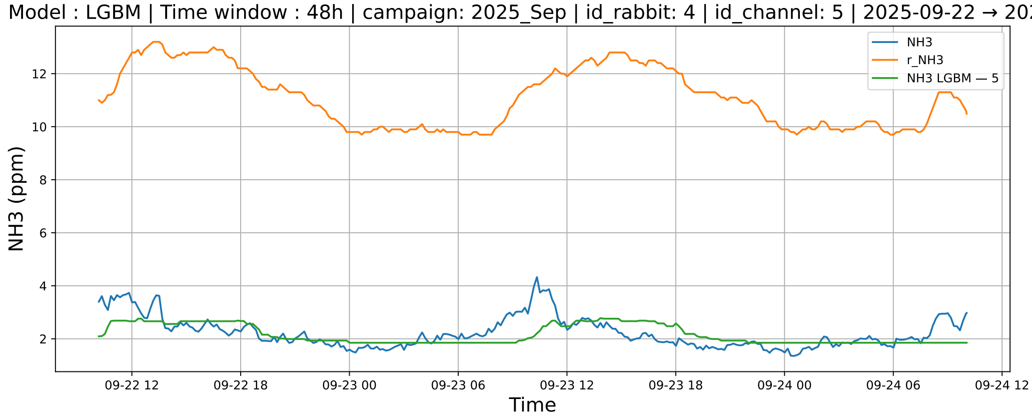


3.1.7.2 Time window : 48

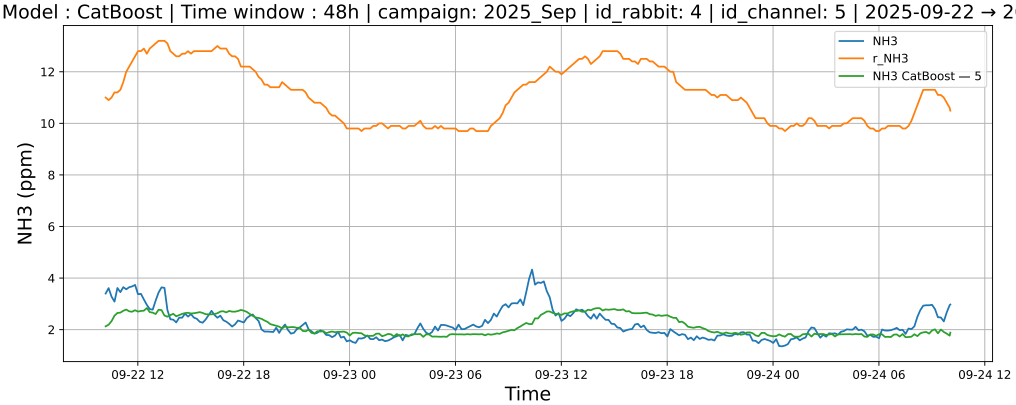
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

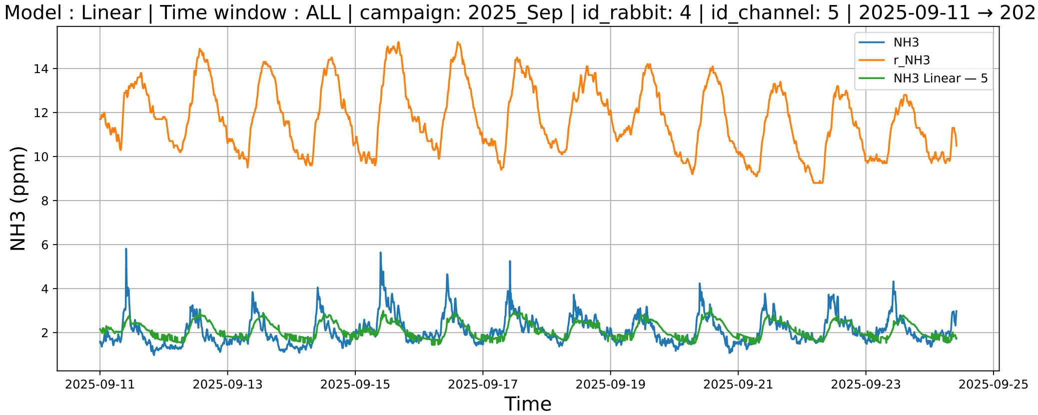


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

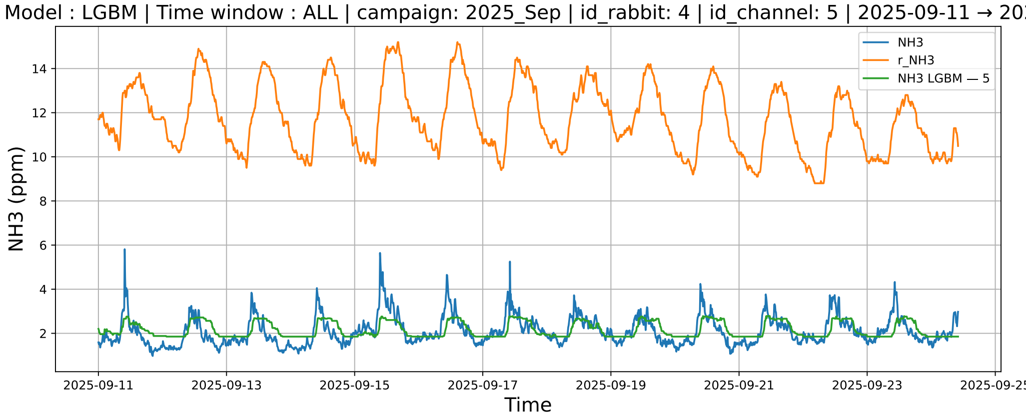


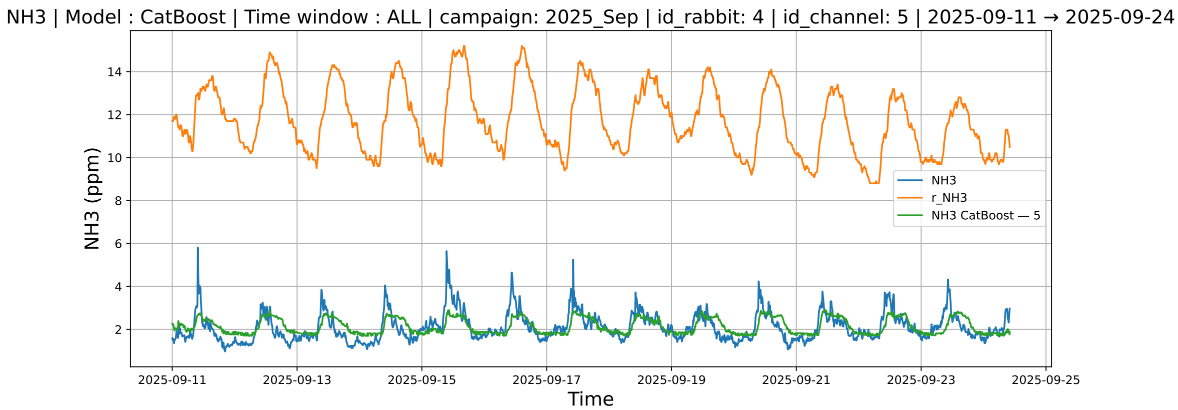
3.1.7.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



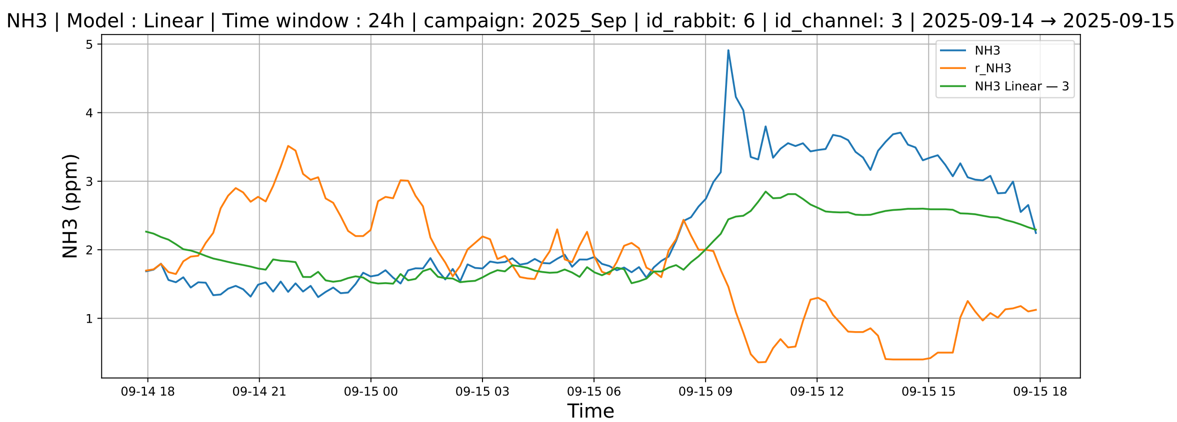
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

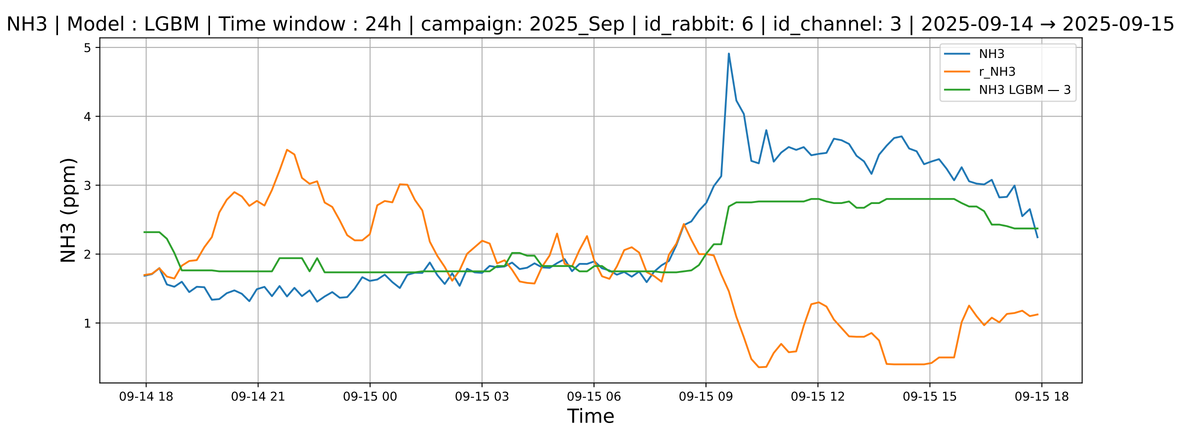




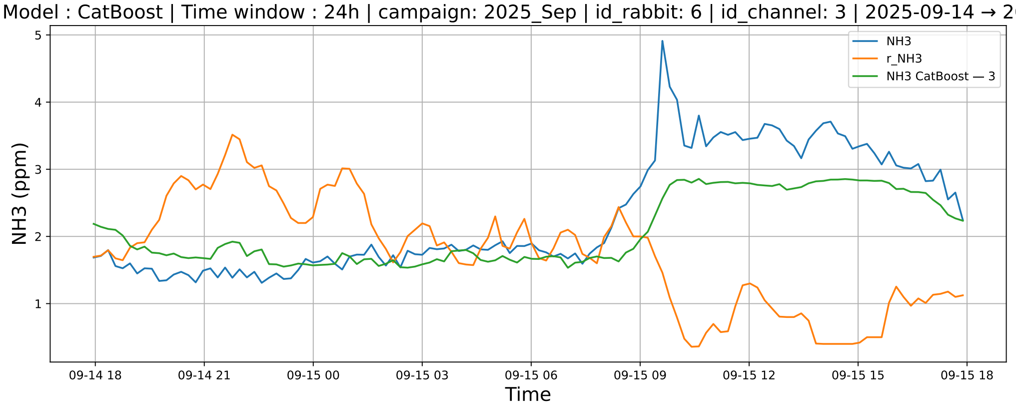
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24



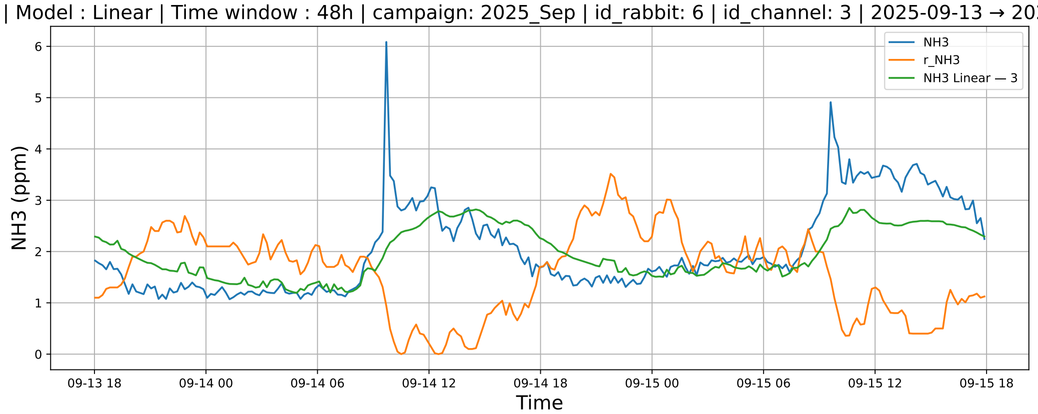


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

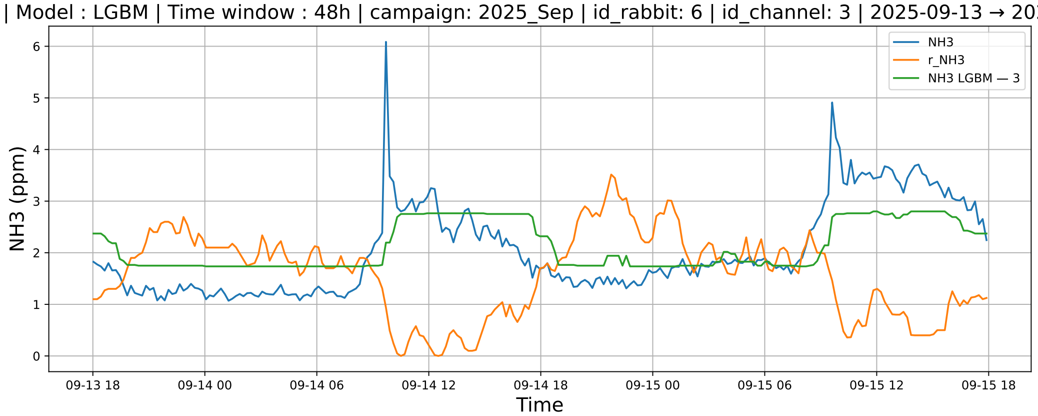


3.1.8.2 Time window : 48

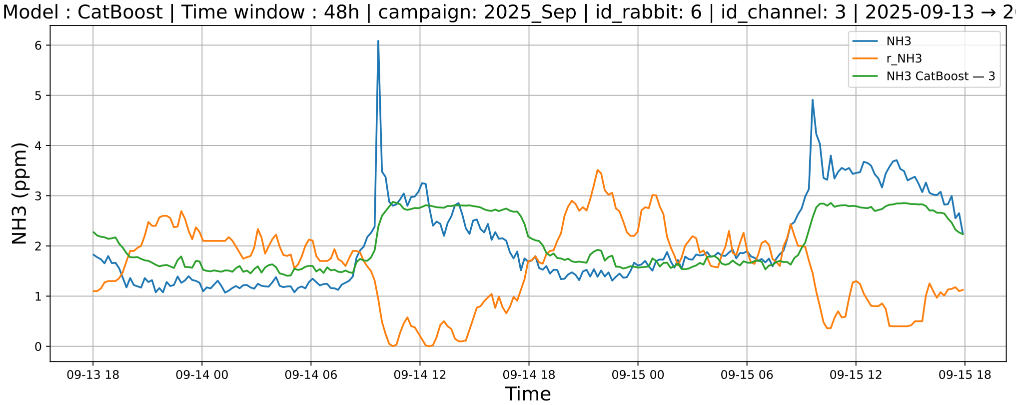
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

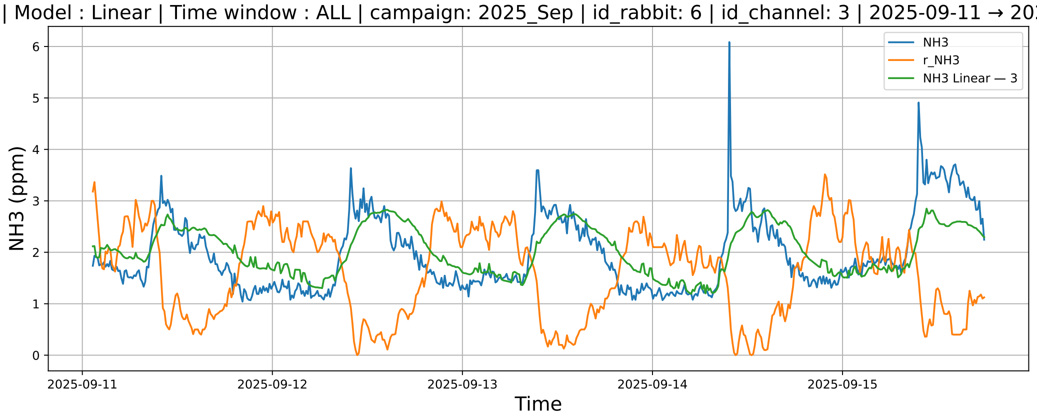


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

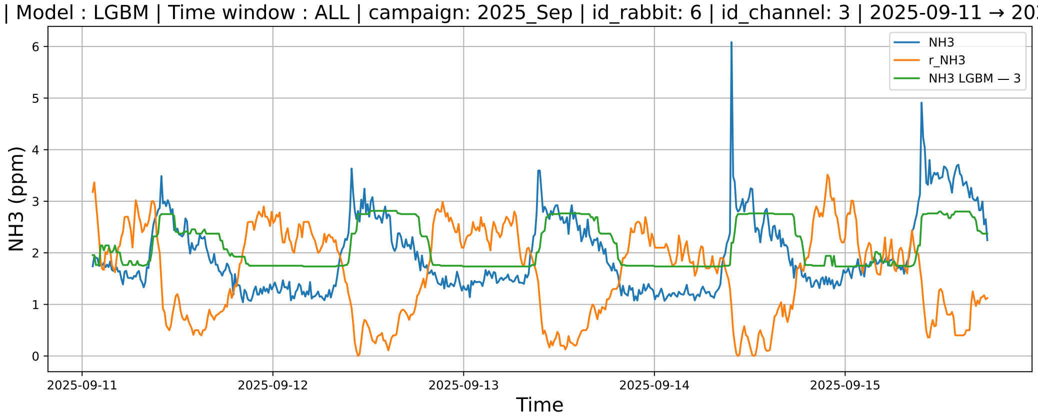


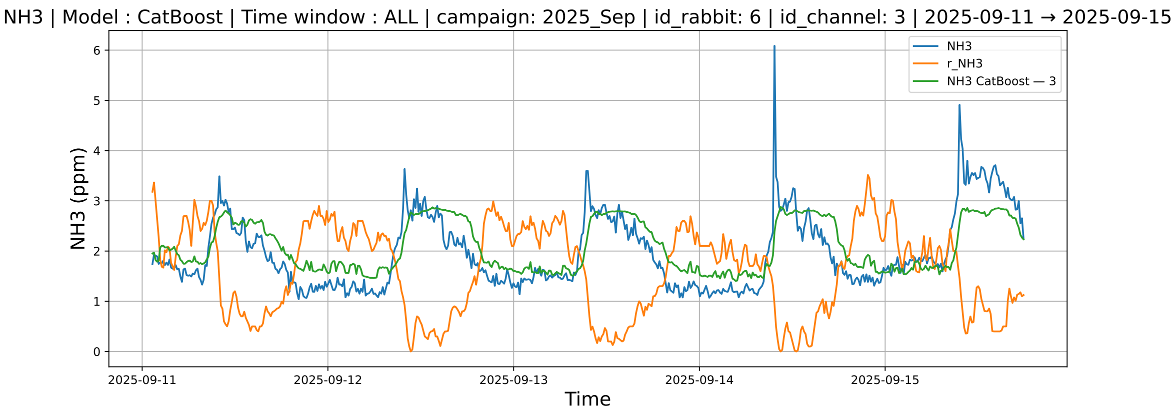
3.1.8.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

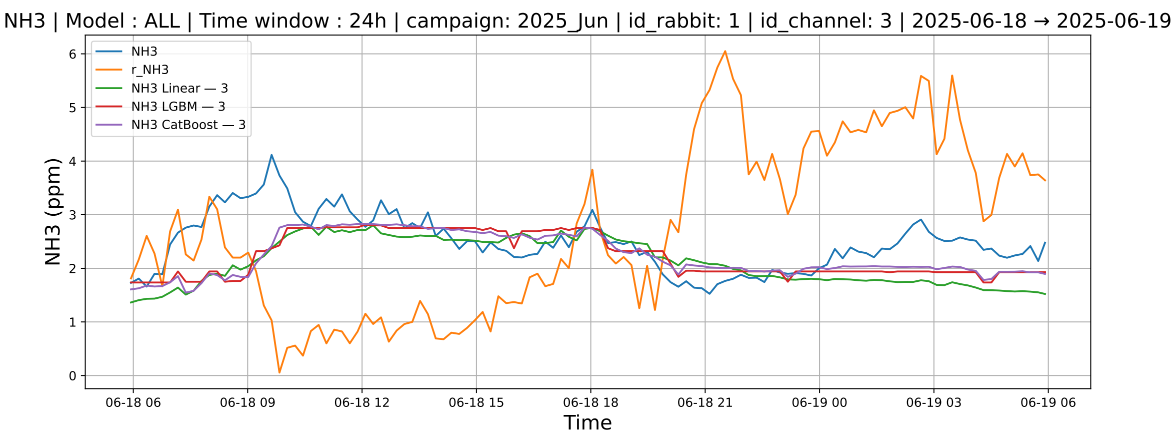




3.2 Compare plots

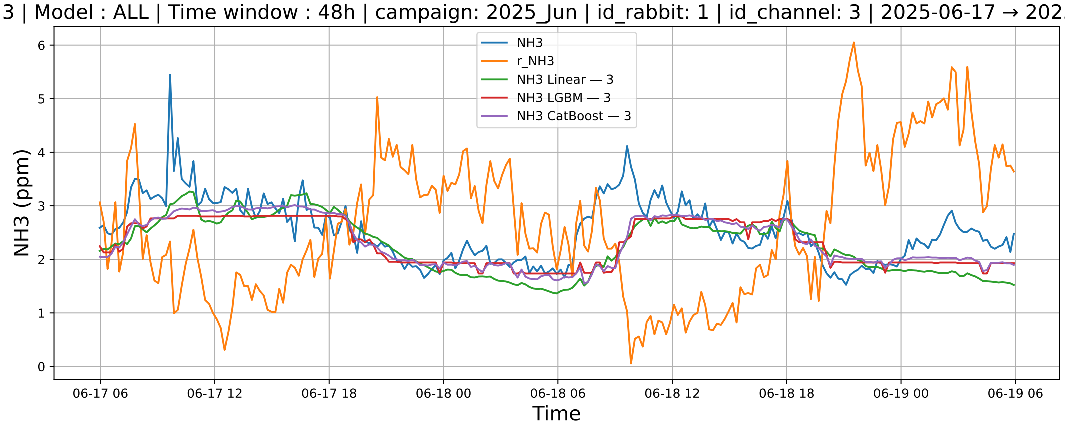
3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.2.1.1 Time window : 24



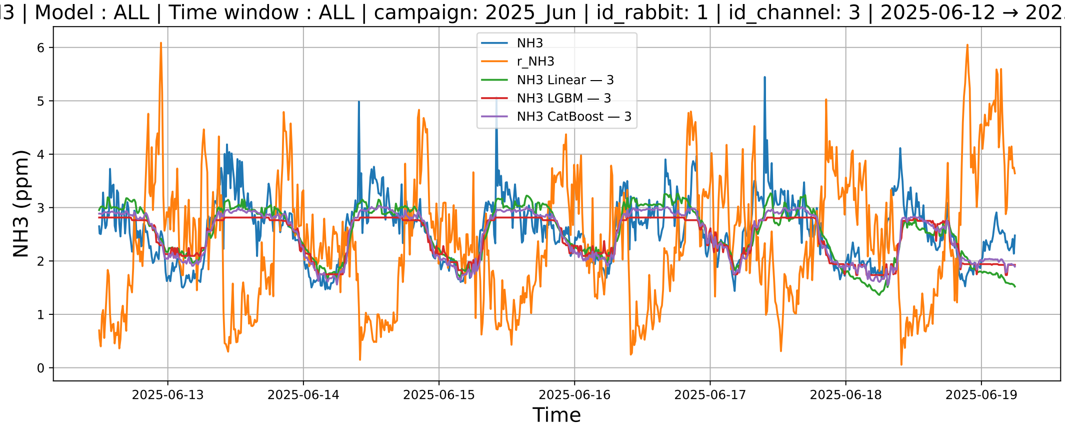
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

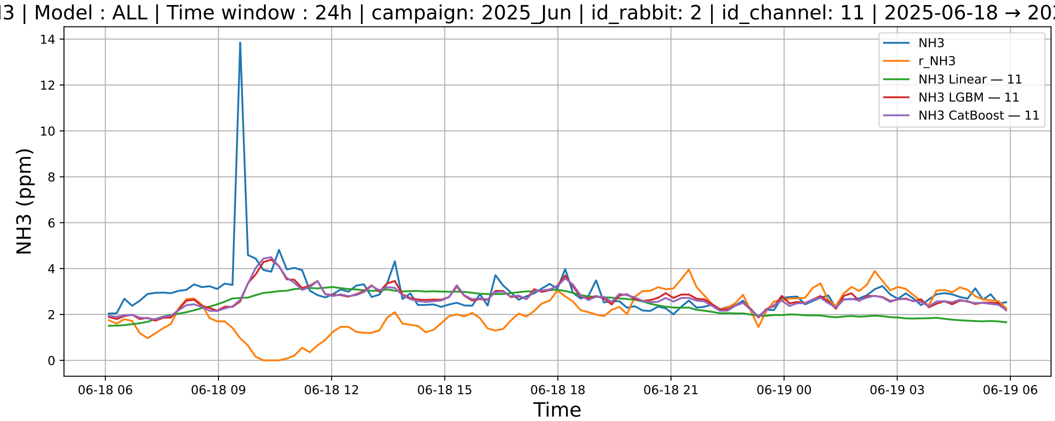
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

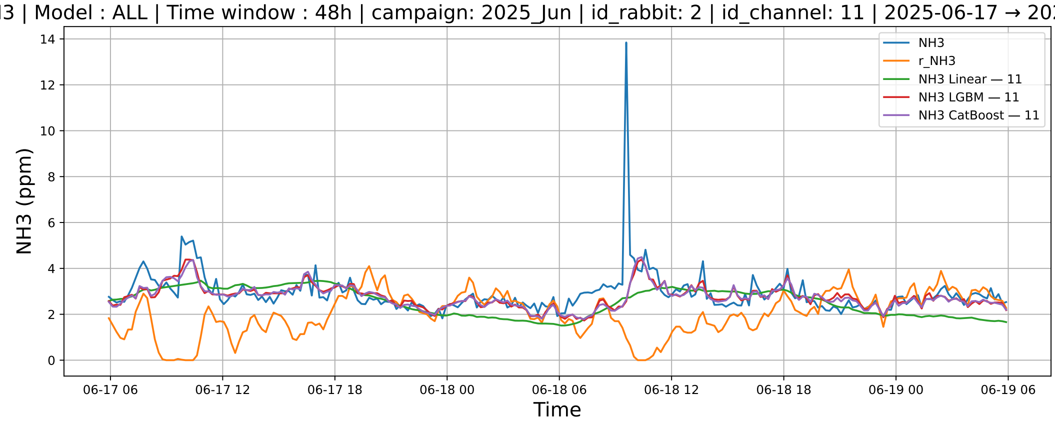
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



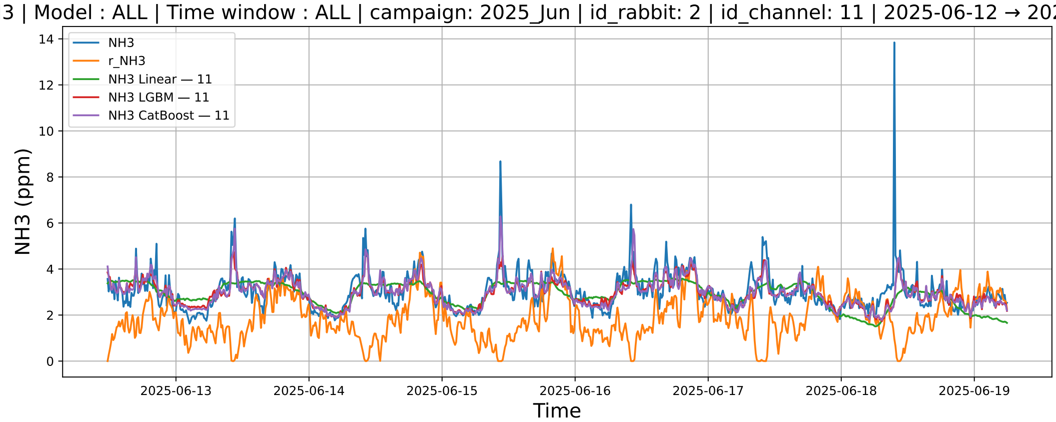
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



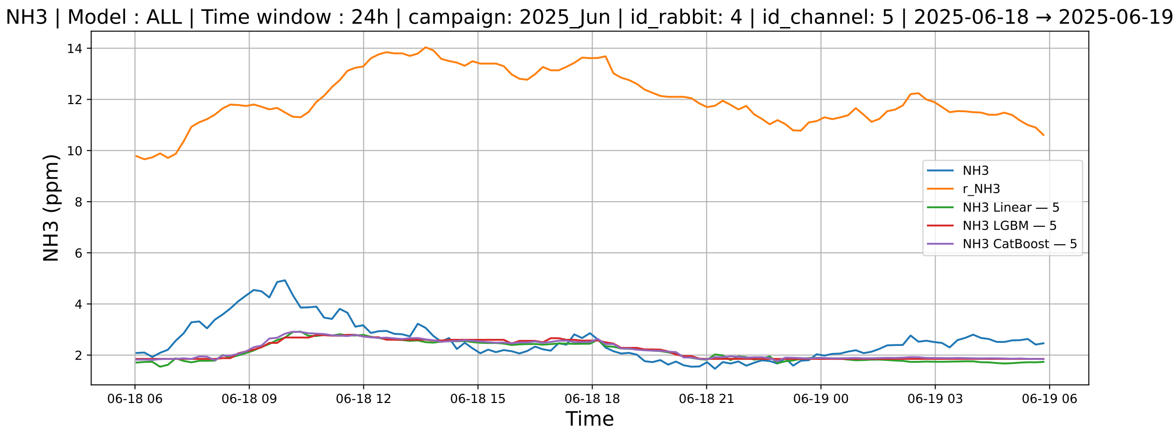
3.2.2.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

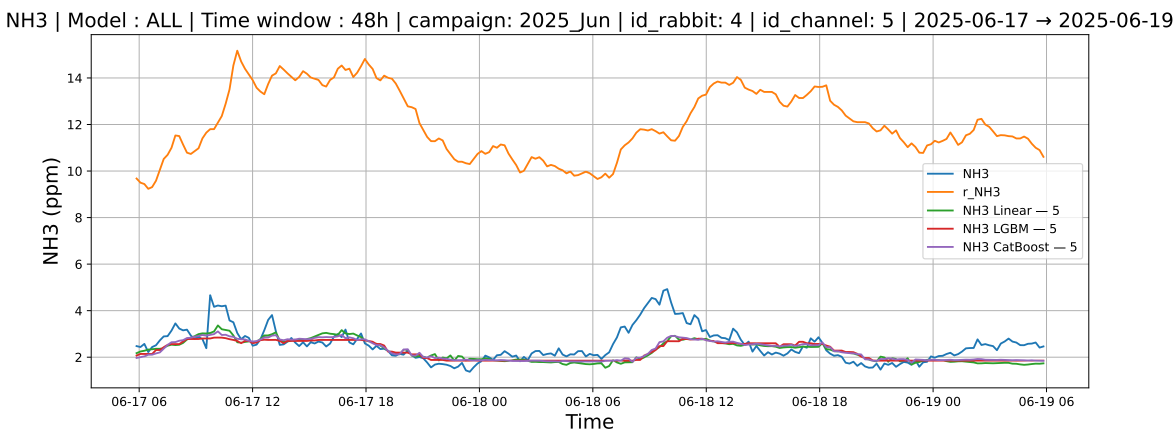


3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

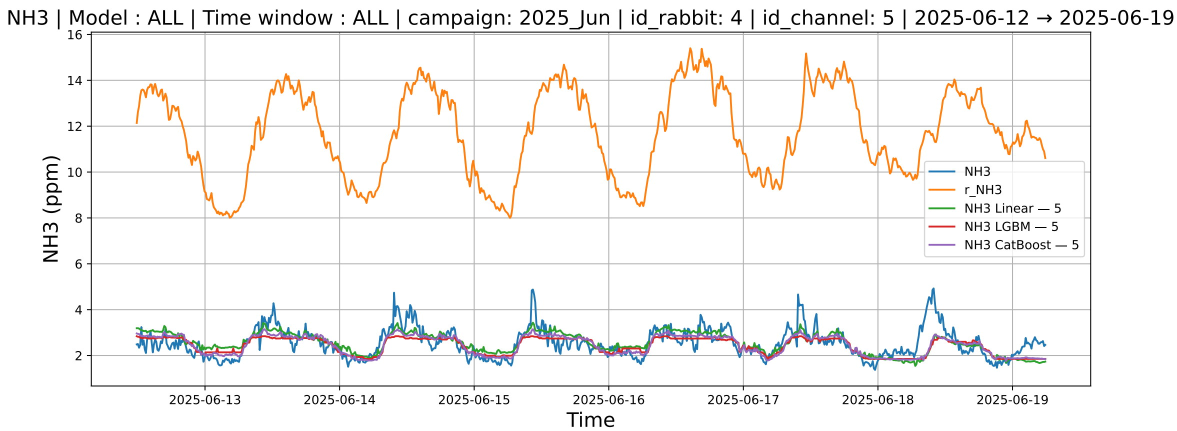
3.2.3.1 Time window : 24



3.2.3.2 Time window : 48

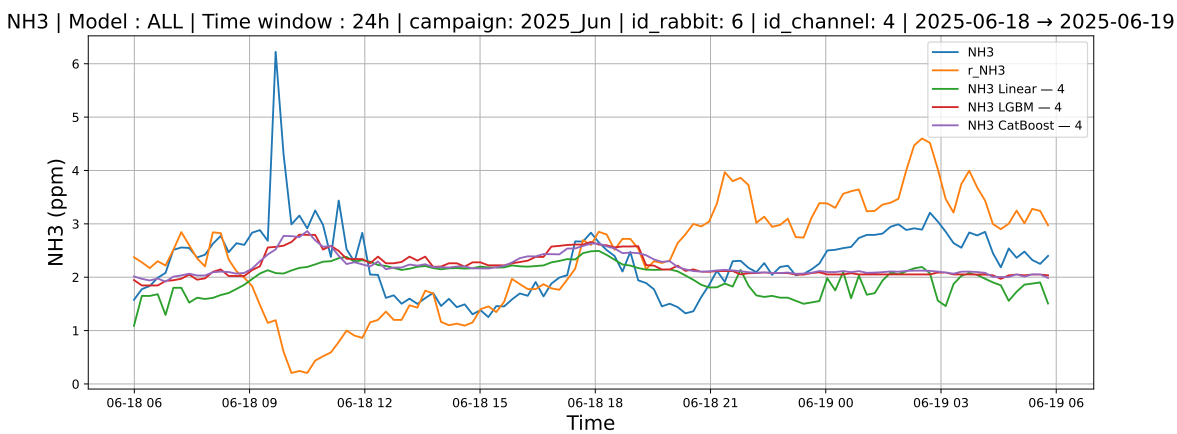


3.2.3.3 Time window : ALL

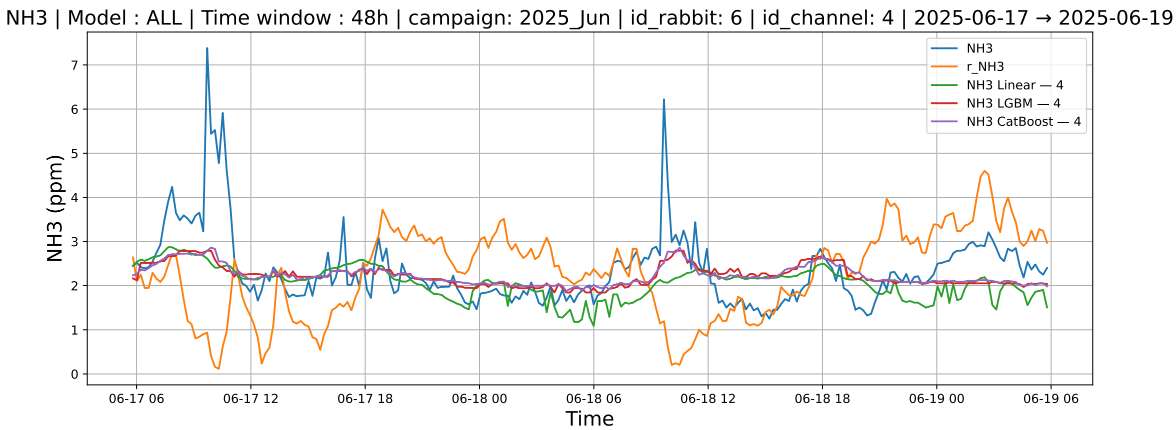


3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

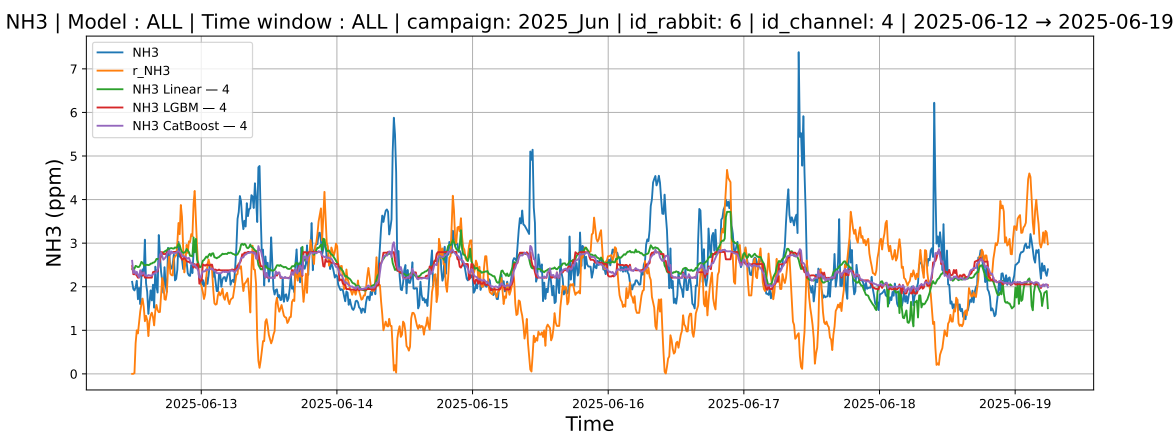
3.2.4.1 Time window : 24



3.2.4.2 Time window : 48

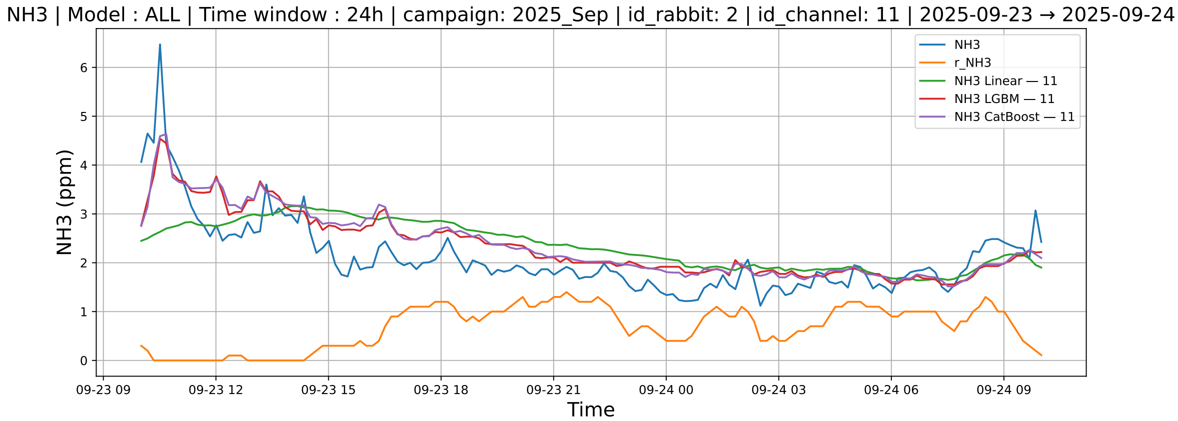


3.2.4.3 Time window : ALL

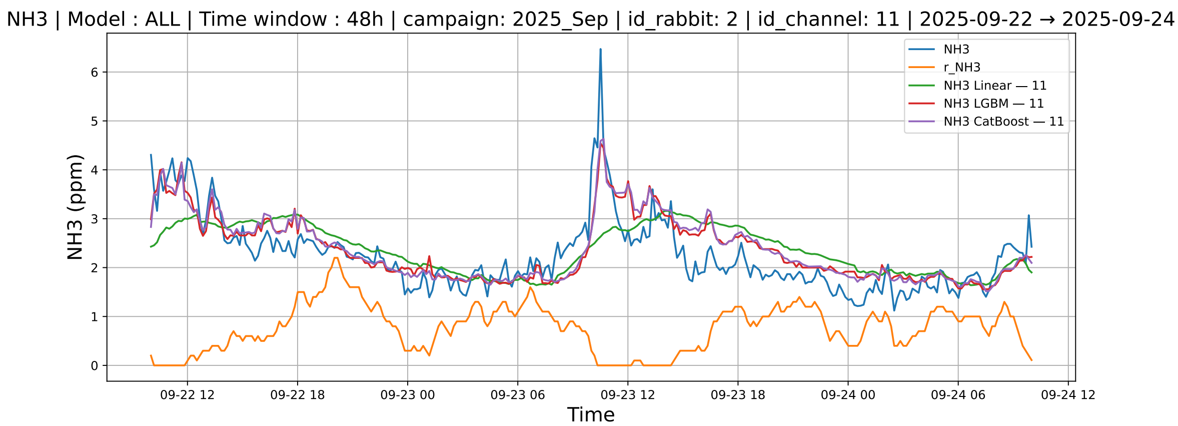


3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

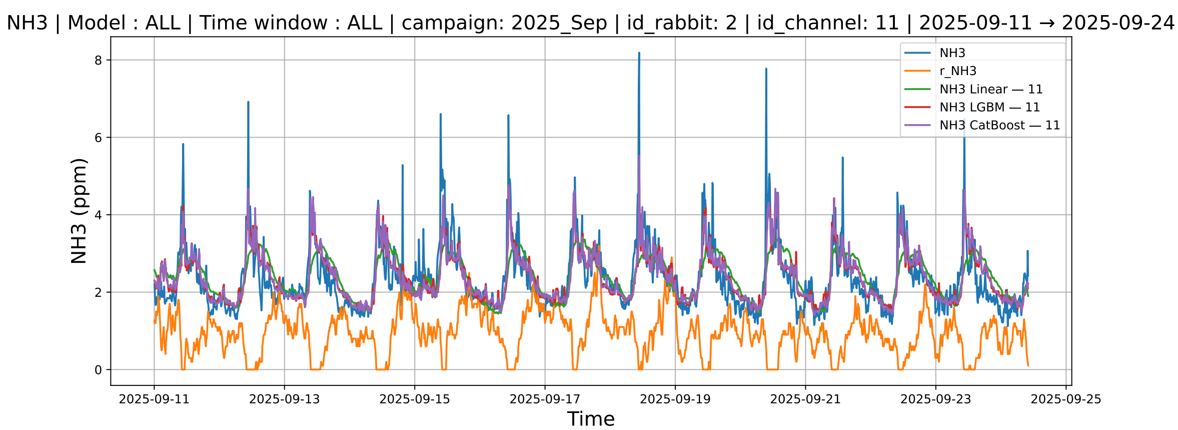
3.2.5.1 Time window : 24



3.2.5.2 Time window : 48

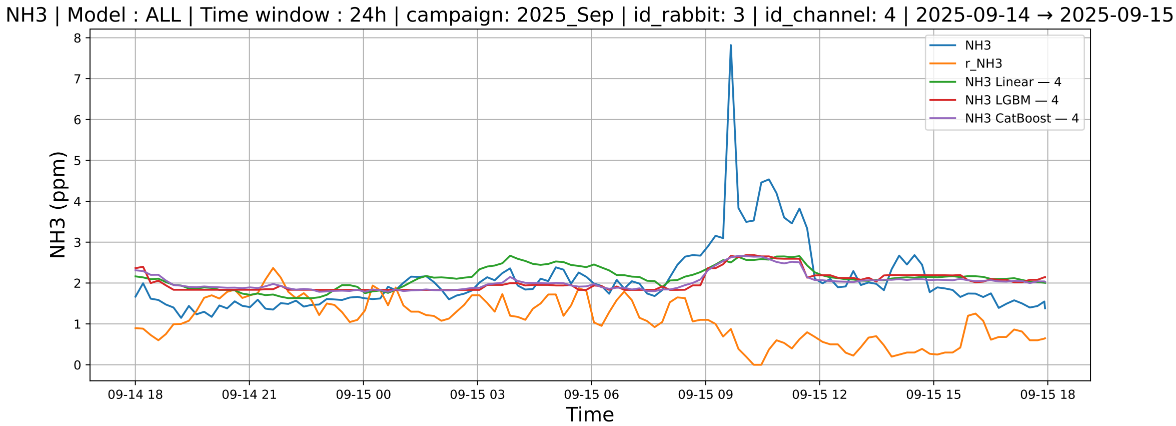


3.2.5.3 Time window : ALL

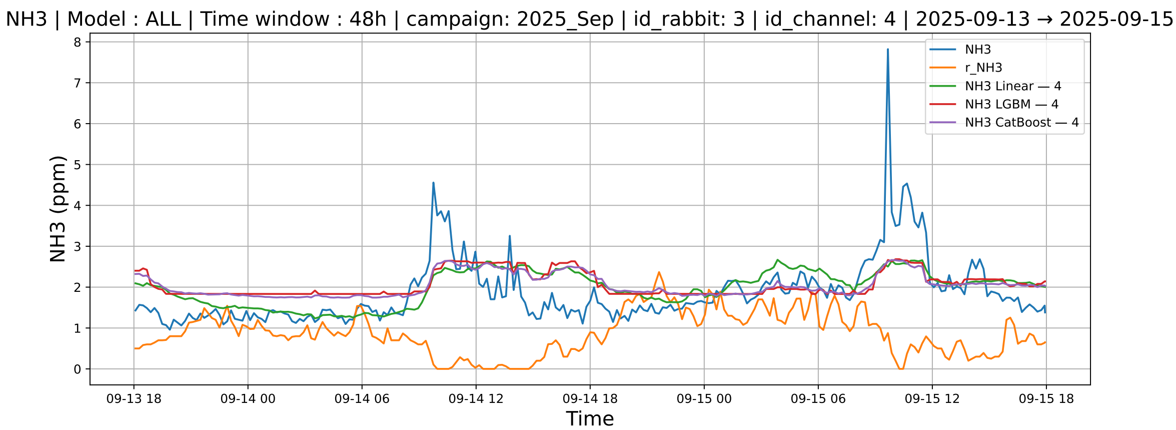


3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.2.6.1 Time window : 24

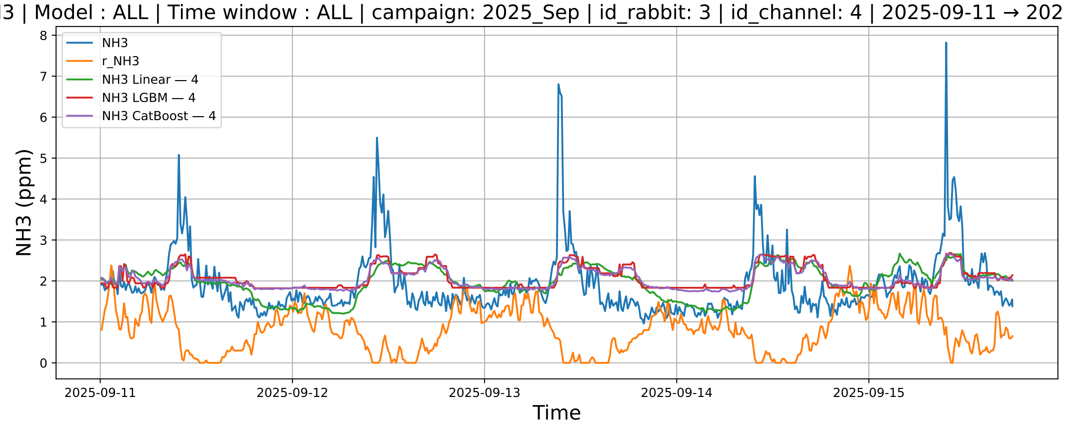


3.2.6.2 Time window : 48



3.2.6.3 Time window : ALL

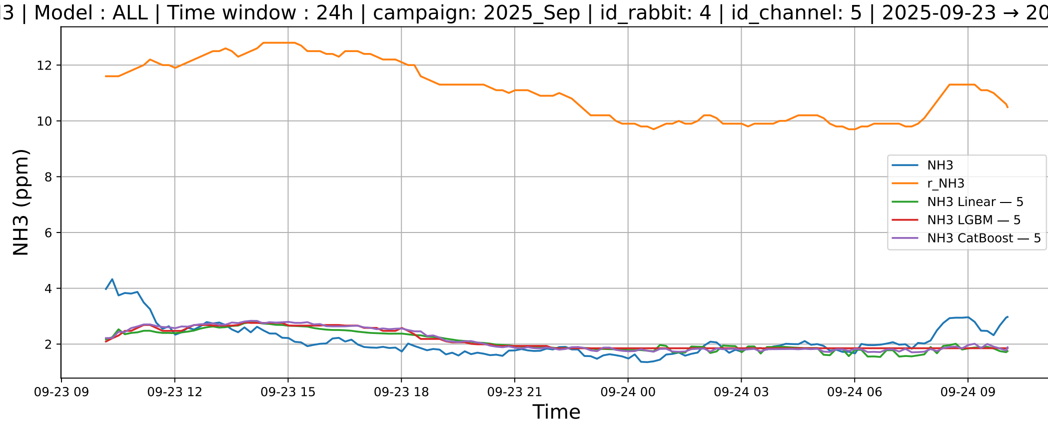
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



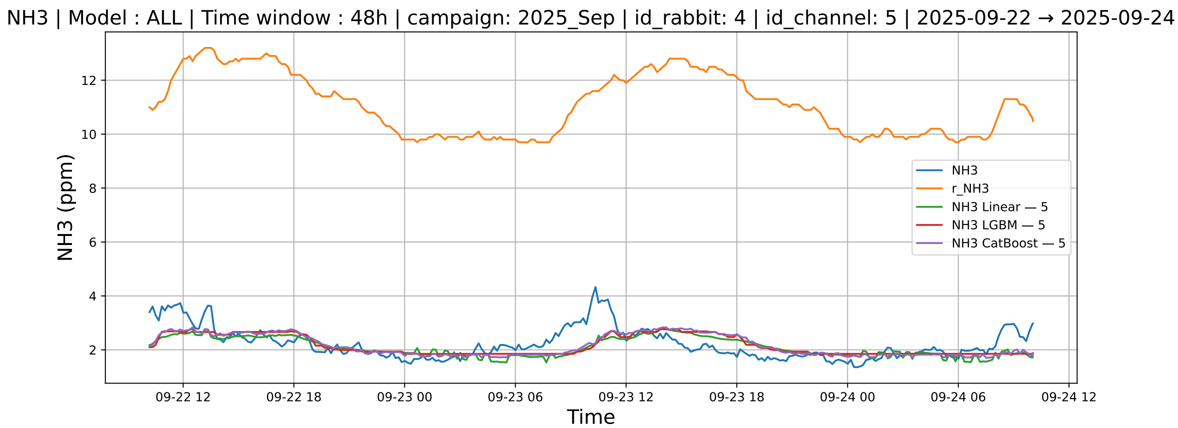
3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.2.7.1 Time window : 24

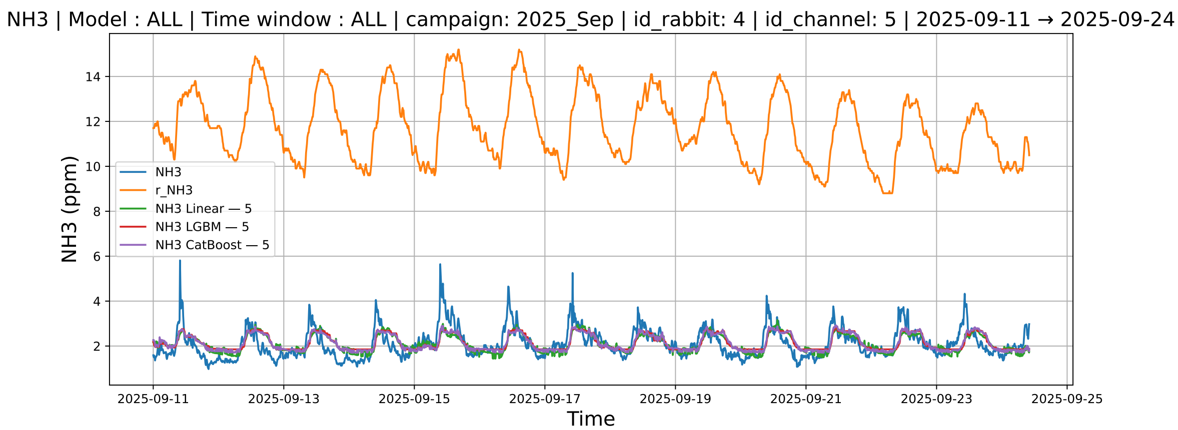
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



3.2.7.2 Time window : 48

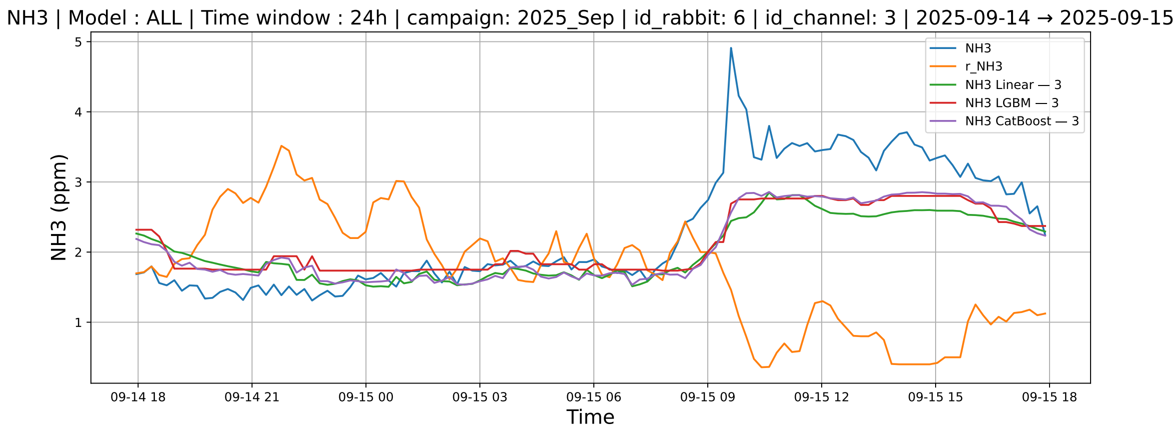


3.2.7.3 Time window : ALL

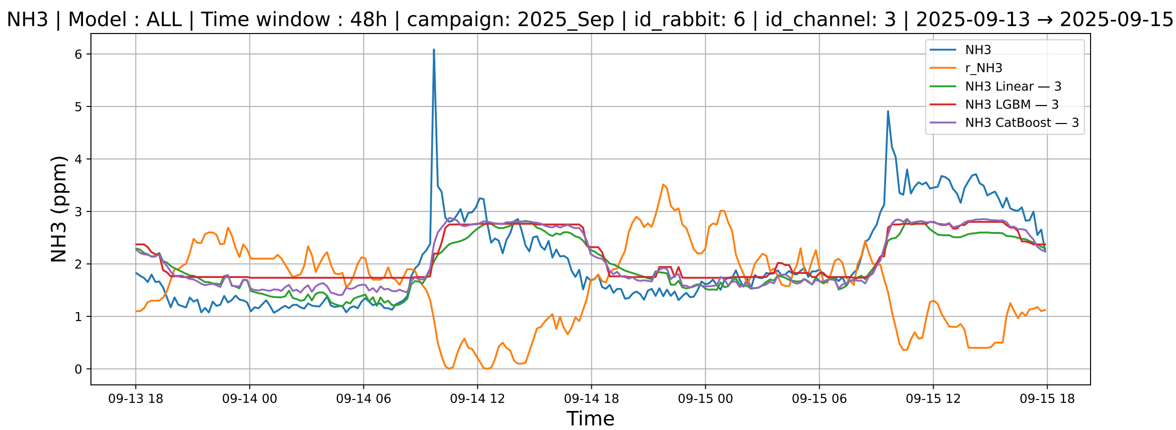


3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.2.8.1 Time window : 24



3.2.8.2 Time window : 48



3.2.8.3 Time window : ALL